

Techniques for computing and approximating large optimization problems in quantum information

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ABSTRACT

Many fundamental problems in quantum information theory can be formulated as optimization tasks over quantum states and measurements. While the convex structure of quantum theory often enables powerful semidefinite programming formulations, a major challenge persists: the exponential growth of Hilbert space dimension with system size. This dissertation discusses techniques to address these and related obstacles across several core problems in quantum information.

We first study optimization over separable quantum states. Such problems are typically NP-hard and standard semidefinite programming hierarchies typically suffer from the curse of dimensionality. We introduce a hybrid algorithm that leverages a quantum co-processor to reduce the effective dimension of the problem. Unlike several other approaches, our method is guaranteed to always produce a feasible separable state. We apply this framework to approximate the minimum separable energy of many-body Hamiltonians, and define an entanglement measure for their ground spaces, and demonstrate numerical scalability to systems of up to 28 qubits.

We next consider the general Bayes framework, as introduced by Helstrom, for state discrimination when, instead of a classical description of the candidate states, one has access to their *source code*: the quantum circuit that prepares them. We show that the typical semidefinite program for the discrimination problem can be reformulated in terms of the

Gram matrix of these states, reducing the variable dimensions from dL to NL , where d is the Hilbert space dimension, N is the number of candidate states, and L is the number of possible guesses. Importantly, we further introduce a quantum pre-processing procedure which efficiently constructs the reduced problem from the source code, enabling our method to operate directly on quantum data. We consider two applications. First, we characterize the optimal identifications for quantum changepoint problems under several reward structures, that were previously computationally inaccessible. Second, we consider a quantum error classification problem and show how our reduction makes it tractable for systems of hundreds of qubits.

We also analyze structural measurement strategies. As a counterpart to the well-known pretty good measurement for minimum error state discrimination, we introduce the pretty bad measurement for the exclusion variant, and prove that both are guaranteed to perform at least as well as blind guessing in their respective objectives. Finally, we consider evaluating Boolean functions under severe memory constraints and prove that a simple local “greedy” measurement strategy is globally optimal precisely for affine Boolean functions, while more generally achieving at least the square of the optimal global success probability.

Together, these results demonstrate that high-dimensional quantum optimization problems often admit hidden structure enabling dimension reduction, hybrid computation, and provable approximation guarantees.

Techniques for computing and approximating large optimization problems in quantum information

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GENERAL AUDIENCE ABSTRACT

Many fundamental problems in quantum information theory can be understood as questions of choosing the best possible quantum state or measurement for a given task. For instance, one may want to pick the quantum state that optimizes some desired property, or choose a measurement that extracts as much information as possible from a quantum system. These questions are naturally formulated as optimization problems. The mathematical structure of quantum theory often makes it possible to turn these questions into a powerful class of efficiently structured optimization problems called semidefinite programs. However, this approach faces a basic obstacle: as the number of quantum bits increases, the size of their mathematical description grows exponentially. As a result, even problems that are well structured can quickly become too large to solve directly. This dissertation discusses techniques to address these and related obstacles across several core problems in quantum information.

We begin by studying quantum systems that are said to be *separable*, i.e., systems whose parts can be described independently of one another. This is an important starting point because one of the most distinctive features of quantum theory is that systems that are not separable can exhibit correlations that have no classical analogue. Understanding which states do not have this special type of correlation, i.e., that are *entangled*, and how to optimize over them, is a basic problem in quantum information theory. These problems are believed to be computationally difficult and no efficient method is known yet for solving them exactly

in full generality. Although formulating these problems as a semidefinite program provides a systematic way to obtain an approximate solution, the resulting computations often become too large to handle as the number of quantum bits grow. We introduce a hybrid algorithm that combines classical computation with a quantum co-processor, a quantum device used alongside a classical computer. The role of the quantum co-processor is to encode part of the problem in a smaller effective space, making the resulting optimization task more manageable. Unlike several other approaches, our method is guaranteed to always produce a solution that is feasible with respect to the optimization problem. We apply this framework to study the lowest possible energy of large quantum systems under the restriction that the states are separable. This also leads to a way of quantifying how much entanglement is present in the lowest-energy states. Our numerical results demonstrate that the method can scale to systems with as many as 28 quantum bits.

We next consider *quantum state discrimination*: the task of identifying a given set of quantum bits, from a known list of possibilities. Building on the general Bayesian framework introduced by Helstrom, we study this problem in a setting where the candidate states are specified not by an explicit classical description, but by their preparation procedures, namely the quantum circuits that generate them. We show that the semidefinite program can be reformulated as a smaller problem. Instead of working directly with all of the quantum bits, whose description can be extremely large, our formulation uses only the matrix of inner products between the candidate states. This reformulation replaces a problem whose size depends on all of the quantum bits with one whose size depends mainly on the number of candidate states and possible guesses. An important part of our approach is a quantum pre-processing step that builds this smaller optimization problem directly from the circuits that prepare the quantum bits. This allows the method to work directly from quantum circuits, without first requiring a full classical description of the states. We demonstrate the versa-

tility of this framework through two applications. First, we consider quantum changepoint problems, where the goal is to identify when the quantum bits change from one behavior to another. Our approach allows us to handle more general reward structures, which were previously too large to compute. Second, we apply the framework to a problem where the goal is to identify the class of error affecting the quantum bits. Our reduction makes it possible to analyze hundreds of quantum bits, far beyond what direct methods can typically handle.

We also study measurement strategies that have restrictions on the available quantum memory, but still come with rigorous performance guarantees. We introduce a new strategy for ruling out incorrect quantum bits, analogous to a well-known strategy for identifying the correct them. We prove that these structured strategies are always at least as good as making a guess without using the quantum bits. Finally, we study how to make coarse grained predictions when only limited memory is available, showing exactly when a simple step-by-step strategy is optimal and proving that it remains provably close to optimal more generally.

Taken together, these results show that even very large quantum optimization problems can sometimes be made more manageable by uncovering the right underlying structure. This structure can allow us to reduce the size of the problem, combine classical and quantum computation effectively, and still obtain feasible solutions with rigorous performance guarantees.

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List of Abbreviations

NISQ Noisy intermediate-scale quantum

PBM Pretty bad measurement

PGM Pretty good measurement

POVM Positive operator-valued measure

PPT Positive partial transpose

QSD Quantum state discrimination

QSE Quantum state exclusion

SDP Semidefinite program

Chapter 1

Introduction

Quantum computing represents a paradigm shift in information processing by leveraging the fundamental principles of quantum mechanics, such as superposition and entanglement, to perform tasks that may be beyond the reach of classical computation. A primary objective has been to characterize the resources required to achieve such a speedup or advantage over classical computing. Shor’s algorithm [[Sho97](#)] for integer factorization, which achieves an exponential speedup over the best known classical algorithm, and Grover’s algorithm [[Gro96](#)] for unstructured search, which achieves a quadratic speedup over the best possible classical strategy, are two foundational examples. Beyond computation, many central questions in quantum information theory can be naturally formulated as optimization problems. Given a quantum system, one may ask how much entanglement it contains, which measurement best extracts the information encoded in it, or what resources are required to learn a desired property from this system.

The mathematical structure of quantum theory makes semidefinite programming [[Tun16](#)] particularly well suited to addressing such questions. Semidefinite programming is a class of convex optimization problems, whose individual instances are called semidefinite programs,

in which a linear objective function is optimized subject to linear constraints and positive semidefiniteness conditions. Quantum states are represented by positive semidefinite operators, while measurements are described by collections of positive semidefinite operators that sum to the identity. Moreover, many quantities of interest can be expressed as linear functions of these operators. Consequently, a broad range of optimization problems in quantum information theory admit natural formulations as semidefinite programs. This perspective has played a central role in the field, both as a computational framework and as a tool for establishing structural results [SC23].

Jain, Ji, Upadhyay, and Watrous [JJUW11] used the matrix multiplicative weights update method to approximate the optimal value of a semidefinite program, leading to the proof that QIP, the class of problems having quantum interactive proof systems, is equivalent to PSPACE, the class of problems decidable using polynomial space. More recently, Ji, Natarajan, Vidick, Wright, and Yuen [JNV⁺21] used the NPA hierarchy to show that MIP*, the class of multi-prover interactive proofs with entangled quantum provers, contains all recursively enumerable languages. Together, these works demonstrate how semidefinite programming and related hierarchies provide a powerful language for resolving deep structural questions in quantum information.

Despite its power, the semidefinite programming perspective faces two major obstacles. The first is computational hardness: some natural structural constraints, most notably separability, are NP-hard to test in general. The second is dimensionality: the Hilbert space of an n -qubit system has dimension 2^n , so even optimization problems that are efficiently computable in principle may become computationally infeasible at scale. This thesis studies several problems where these barriers arise and develops methods that exploit the underlying structure to reduce the effective dimension, enable hybrid quantum-classical computation, or provide provable guarantees for simpler measurement strategies. In particular, we employ

semidefinite programming as both a theoretical framework as well as a practical computational tool to investigate two areas of quantum information: the characterization of entanglement and the optimal discrimination of quantum states. The remainder of this section outlines specific problems addressed in this work, beginning with the challenge of identifying entanglement in bipartite systems.

Separability problem The first part of this thesis concerns the *separability problem*. Entanglement is one of the defining features of quantum theory and a central resource in quantum information processing [HHHH09]. Given a bipartite quantum state, one asks whether it can be expressed as a convex combination of product states. States admitting such a decomposition are called *separable*, while those that do not are said to be entangled. Determining whether a given state is separable is a fundamental problem in quantum information theory, but it is NP-hard in general [Gur04, Gha10]. Although one may consider separability or entanglement in multipartite systems, this thesis focuses on the bipartite setting.

A variety of semidefinite programming relaxations have been developed to test separability. One of the most prominent is the Peres-Horodecki criterion, also known as the *positive partial transpose* (PPT) condition, which states that every separable state remains positive semidefinite under partial transposition [Per96, HHH96]. The PPT condition is both necessary and sufficient for separability of 2×2 and 2×3 dimensional systems [HHH96], but in higher dimensions it is only a necessary condition. Other criteria, such as the realignment criterion [CW03, Rud05], and more systematic approaches, such as the Doherty-Parrilo-Spedalieri (DPS) hierarchy of symmetric extensions [DPS02], provide increasingly refined tests. In particular, the DPS hierarchy gives a sequence of semidefinite programming relaxations that converge to the set of separable states in the limit.

Despite its many applications [Ioa07, BCY11, GSU13, CS12, HMW13, LWFLJ14, CAW22, JW23], the separability problem remains difficult for two main reasons. First, even verifying that a proposed optimal solution is separable can be computationally hard. Second, the problem suffers from the curse of dimensionality: when the two subsystems are n -qubit spaces, the relevant Hilbert space has dimension 2^{2n} , rendering direct numerical methods infeasible even for moderately large systems.

In Chapter 2, we address these obstacles by combining semidefinite programming techniques with quantum heuristics. By using semidefinite programming-based algorithms, one can solve small instances (i.e., when the dimensions of the two systems are small) with reasonable levels of success. We introduce a heuristic algorithm to reduce the dimension of the optimization problem, and discuss when this pre-processing step can be run on a quantum co-processor thereby alleviating the curse of dimensionality. In a sense, we do not need to perform any vector-represented calculations, the interactions are done on a quantum mechanical level and the desired quantities can be inferred from measurement statistics. Overall, this quantum-assisted pre-processing with semidefinite programming allows us to optimize over separable states in dimensions that would otherwise be inaccessible.

A key feature of our approach is that, although the algorithm is heuristic in nature, it always returns a feasible separable state.¹ Thus, even in cases where the algorithm does not find an optimal solution,² its output is still an explicit valid candidate solution, rather than merely a numerical bound.

As an application, we approximate the separable state that optimizes the energy with respect to a fixed Hamiltonian. We refer to this value as the *separable ground energy*.³ This

¹This remains true even when the quantum processing component is noisy.

²Due to standard complexity-theoretic barriers $P \neq NP$, one should not expect any efficient method to solve all instances of the separability problem.

³Since we formulate the relevant optimization problems as maximization problems, we regard the largest eigenvalue as the ground energy. This convention is without loss of generality, since one can replace H by

approximation provides a way to quantify the amount of entanglement present in the ground space of the Hamiltonian. We demonstrate the method numerically for the one-dimensional Ising Hamiltonian on systems of up to 28 qubits.

Quantum state discrimination The second part of this thesis studies several aspects of *quantum state discrimination* [BK15a], one of the oldest and most fundamental tasks in quantum information processing. In its basic form, the task is to identify which state from a known set of candidate states was prepared. More formally, Alice has a finite set of quantum states $\{\rho_1, \dots, \rho_N\}$, prepared with corresponding prior probabilities $\{p_1, \dots, p_N\}$, and Bob is given a single state drawn from this set. His goal is to perform a measurement and guess the index of the state Alice sent. This simple two-party game, illustrated in Fig. 1.1, underlies a wide range of problems in quantum information, including quantum key distribution [KCCY18], two-party cryptography [ATSVY00, VE02, Cab00, Amb01, Sik17], the hidden subgroup problem [HKK08], quantum algorithms [RRS09, HKK08], and dimension witnessing [SB13].

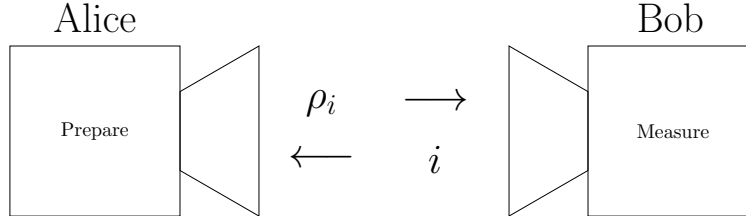


Figure 1.1: Alice prepares $\rho_i \in \{\rho_1, \dots, \rho_N\}$ and sends it to Bob. Bob wins the game if he correctly identifies the index i .

It is well known that Bob can identify the state perfectly if and only if the set of states are pairwise orthogonal. Since this condition is often not satisfied, Bob must instead choose a figure of merit and optimize over all possible measurements. Two standard examples are minimum-error discrimination, in which Bob always outputs a guess and seeks to minimize $-H$ if one prefers a minimization convention.

the probability of error [Hel69, Bae13], and unambiguous discrimination, in which incorrect guesses are forbidden at the cost of allowing inconclusive outcomes [Iva87, Die88, Per88]. Other objectives are also possible, depending on the application, including state exclusion [BJOP14] and reward functions that assign partial credit to approximate or nearly correct answers. We refer the reader to the surveys [CB98, Che00, BHH04, Ber07, BC09, Ber10, BK15b] for broader background.

A convenient way to unify these different tasks is through Helstrom's general Bayes framework [Hel03]. In this formulation, Bob constructs measurements $\{M_1, \dots, M_L\}$, where the L correspond to the number of his possible guesses. A reward or penalty is assigned to each pair consisting of an input state and a measurement outcome. Thus, if R_{ij} denotes the reward associated when i is guessed given that the received state is ρ_j , then Bob seeks to optimize an objective of the form

$$\sum_{i=1}^L \sum_{j=1}^N p_j R_{ij} \langle M_i, \rho_j \rangle,$$

subject to the constraint that $\{M_1, \dots, M_L\}$ form a set of valid measurements. Different choices of the rewards R_{ij} recover many familiar discrimination tasks. For an arbitrary ensemble, such optimization problems can be formulated as semidefinite programs [EMV03, Eld03b]. However, closed-form solutions are known only in special cases, such as discrimination of two states [Hel69], arbitrary qubit states with equal prior probabilities [DT10], geometrically uniform states [EMV04], and mirror-symmetric states [ABGH02].

In Chapter 3, we study this Bayes framework in a setting motivated by quantum computation. Rather than assuming that the candidate states are given as exponentially large classical vectors or matrices, we suppose that one has access to their *source code*: quantum circuits that prepare the states. If the states have dimension d and there are L possible

outcomes, the standard semidefinite programming formulation involves L measurement operators acting on a d -dimensional Hilbert space, leading to a problem whose size scales with dL . This quickly becomes infeasible when the states are generated by many-qubit circuits.

We show that the corresponding semidefinite programs can be reformulated entirely in terms of the Gram matrix of the candidate states. This reduces the dimension of the optimization problem from one depending on the ambient Hilbert space dimension to one depending only on the number of candidate states and the number of possible outcomes. In particular, for N candidate states and L possible guesses, the reduced formulation has dimension NL , rather than dL . Moreover, the Gram matrix can be constructed by quantum pre-processing procedures that estimate inner products directly from the preparation circuits. This yields a hybrid quantum-classical method for Bayes-optimal quantum state discrimination that can operate directly on quantum data.

We apply this framework to two problems. The first is quantum changepoint identification in which a quantum system changes from one state to another at one or more unknown locations, and the task is to identify the changepoint positions. Our framework allows us to study multiple-changepoint regimes and reward functions that account for approximate localization, thereby capturing settings in which a nearly correct changepoint estimate should receive partial credit. The second application is quantum error type classification where the goal is to determine which error, from a known family of possible errors, has occurred. In this setting, the Gram matrix reduction makes the associated discrimination problems tractable for systems with hundreds of qubits.

Chapter 4 focuses on a different aspect of state discrimination: the construction of simple measurements with provable performance guarantees. In the minimum-error setting, Bob uses the measurements $\{M_1, \dots, M_N\}$, where outcome i corresponds to guessing that the

state was ρ_i . His success probability is $\sum_{i=1}^N p_i \langle M_i, \rho_i \rangle$, and the optimal value is

$$\mathcal{P}_{\text{Best}} = \max \sum_{i=1}^N p_i \langle M_i, \rho_i \rangle. \quad (1.1)$$

Although this optimization can be solved numerically by semidefinite programming, the optimal measurement is difficult to characterize explicitly except in special cases. This motivates the study of measurements with a closed form that approximate the optimal one.

One such measurement is the *pretty good measurement* (PGM), first discussed in [Bel75a, Bel75b, HJW93]. The PGM has a simple closed form and is known to be optimal, or near optimal, in many structured settings. Examples include the hidden subgroup problem [BCvD06, MR07], quantum changepoint detection [SBC⁺16], and port-based teleportation [Led22]. More generally, the PGM often performs well for ensembles with a high degree of symmetry.

We then turn to the exclusion variant of state discrimination. Whereas discrimination asks Bob to identify which state was sent, exclusion asks him to name a state that was *not* sent. Thus, if Alice sends ρ_j , Bob wins by outputting any index $i \neq j$. This task, illustrated in Fig. 1.2, has been studied in connection with the Pusey-Barrett-Rudolph game [PBR12] and related foundational questions [BJOP14]. Perfect exclusion, also known as antidistinguishability, asks whether Bob can exclude a state without error; characterizing when this is possible remains an active area of research [HB20, RS23, LD20, MNW24, UBK⁺20, JRS25].

In the exclusion setting, Bob seeks to minimize the probability of incorrectly excluding the state that was actually sent. If he constructs the measurements $\{M_1, \dots, M_N\}$, where outcome j corresponds to excluding ρ_j , then the probability of an incorrect exclusion is

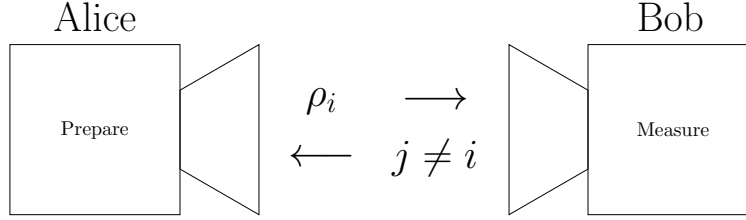


Figure 1.2: Alice prepares $\rho_i \in \{\rho_1, \dots, \rho_n\}$ and sends it to Bob. Bob wins the game if he correctly *excludes* an index $j \neq i$.

$\sum_{i=1}^N p_i \langle M_i, \rho_i \rangle$. Thus, the optimal success probability is

$$\mathcal{P}_{\text{Worst}} = \min \sum_{i=1}^N p_i \langle M_i, \rho_i \rangle. \quad (1.2)$$

This is the opposite objective from minimum-error discrimination: the best exclusion measurement is, in this precise sense, the worst measurement for discrimination. Motivated by the role of the PGM as a well-studied approximation to the optimal discrimination measurement, we introduce the *pretty bad measurement* as its analogue for state exclusion. We prove that the pretty good and pretty bad measurements each satisfy a natural baseline guarantee: they perform at least as well as blind guessing in their respective tasks.

Finally, in Chapter 5, we study state discrimination under severe memory constraints. Up to this point, the most general measurements have been joint measurements, in which Bob may store all received quantum systems coherently and measure them together. Such measurements are mathematically optimal but may be experimentally demanding, especially for long sequences of states. This motivates the question of when simpler local or memoryless strategies can match, or approximate, the performance of optimal joint measurements.

Many quantum information-processing protocols naturally involve sequences of quantum states. For example, in the BB84 quantum key distribution protocol, Alice sends Bob qubits chosen from the set $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$ [BB14]. From Bob's perspective, the received signal

is a quantum sequence whose components are drawn from a known finite set. The resources required to optimally discriminate such sequences can vary significantly depending on the structure of the states and on the goal of the task. For two pure states given in finitely many copies, adaptive local strategies using limited classical memory can match the performance of optimal joint measurements, whereas memoryless strategies generally achieve this only asymptotically [ABB⁺05]. For two mixed states, joint measurements can strictly outperform all local strategies even in the asymptotic limit [CDVMTB10]. In quantum changepoint detection, joint measurements provide an advantage in the minimum-error setting [SBC⁺16], while in the unambiguous setting the optimality of local versus global strategies depends on the overlap between the states [SCMT17]. For symmetric pure-state identification, local strategies using only limited classical memory have also been shown to perform optimally in several cases [SMVMT22].

Recent work has shown that, when the task is to identify an entire sequence of independently prepared states, memoryless strategies can achieve the same performance as joint measurements for both minimum-error and unambiguous discrimination [GMB24, GMRB24]. This naturally raises a further question: does such an equivalence persist when Bob does not need to identify the entire sequence, but only a coarse-grained property of it?

Motivated by this question, we initiate a study of learning Boolean properties of quantum sequences. Suppose Bob receives a sequence

$$|\psi_{x_1}\rangle \otimes |\psi_{x_2}\rangle \otimes \cdots \otimes |\psi_{x_n}\rangle,$$

where each $x_i \in \{0, 1\}$, and where $|\psi_0\rangle$ and $|\psi_1\rangle$ are known non-orthogonal states. Rather than learning the entire string $x = x_1 \cdots x_n$, Bob may only wish to determine a Boolean function $f(x)$. For example, he may want to decide whether the number of occurrences of one

state exceeds the number of occurrences of the other, or whether the number of occurrences is even. These tasks are trivial when the classical string x is directly available, but become non-trivial when x is encoded into non-orthogonal quantum states.

We compare two resource regimes, illustrated in Fig. 1.3. In the *full-memory* setting, Bob may store the entire sequence and perform an optimal joint measurement after all systems have been received. In the *memoryless* setting, Bob has neither quantum nor classical memory: each state must be measured immediately using predetermined local measurements, and the final guess for $f(x)$ is computed from the resulting measurement outcomes. We characterize which Boolean functions can still be learned optimally under these stringent constraints. For functions outside this class, we give tight bounds showing how well such memoryless strategies can perform even when they are not optimal.

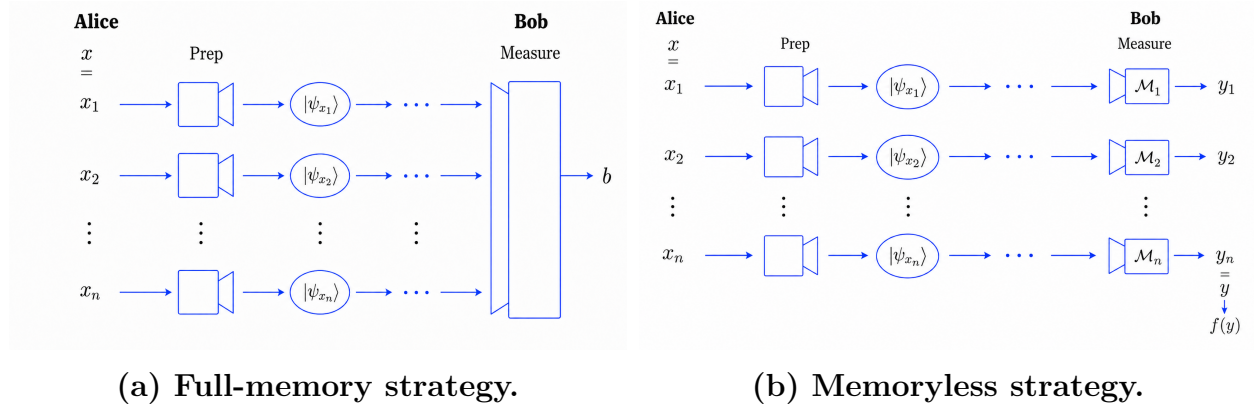


Figure 1.3: Alice prepares the state $|\psi_x\rangle = |\psi_{x_1}\rangle \otimes \dots \otimes |\psi_{x_n}\rangle$ and sends it to Bob. Bob's goal is to compute a function $f(x)$ with binary properties. Bob can either store all his received states and then perform a joint measurement (Fig. ??) to obtain a guess b for $f(x)$, or he can measure each of his received states as soon as it is received using predetermined local measurements to obtain a guess y_i for x_i which can be collectively used to guess $f(x)$ (Fig. ??).

Together, we show that high-dimensional quantum optimization problems can sometimes be tackled using structure in their naive formulations. By identifying and exploiting this structure, one can sometimes reduce the dimension, incorporate quantum subroutines into

classical optimization pipelines, and obtain rigorous guarantees for simple measurement solutions. Next is a brief overview of the main topics, followed by the relevant background relevant for this thesis.

1.1 Background

In this section, we establish the relevant notations and background for this thesis. We first review concepts in linear algebra and convex analysis. Then, we discuss topics in quantum information and semidefinite programming.

1.1.1 Notations

A *complex Euclidean space*, denoted by $\mathbb{C}^n$, is an n -dimensional vector space over the field of complex numbers $\mathbb{C}$ equipped with the inner product $\langle u, v \rangle = \sum_{i=1}^n \bar{u}_i v_i$ for $u, v \in \mathbb{C}^n$, where $\bar{u}_i$ denotes the complex conjugate of u_i . We denote complex Euclidean spaces by letters in calligraphic font, e.g., $\mathcal{A}, \mathcal{B}, \mathcal{X}$ and $\mathcal{Y}$, and use the shorthand $[n]$ for $\{1, \dots, n\}$. The set of all n -bit strings is denoted by $\{0, 1\}^n$.

A *Boolean function* on n variables is a function $f : \{0, 1\}^n \rightarrow \{0, 1\}$. In Chapter 5, we consider the AND function and the MAJ function. The AND function on n bits is defined by

$$\text{AND}(x_1, \dots, x_n) = x_1 \wedge x_2 \wedge \dots \wedge x_n = \begin{cases} 1 & \text{if } x_i = 1 \text{ for all } i, \\ 0 & \text{otherwise.} \end{cases} \quad (1.3)$$

For odd $n = 2k + 1$, the majority function is defined by

$$\text{MAJ}(x_1, \dots, x_{2k+1}) = \begin{cases} 1 & \text{if } \sum_{i=1}^{2k+1} x_i > k, \\ 0 & \text{if } \sum_{i=1}^{2k+1} x_i \leq k. \end{cases} \quad (1.4)$$

A Boolean function is said to be *affine* if it can be written as $f(x) = \langle a, x \rangle \oplus b$, where $a \in \{0, 1\}^n$, $b \in \{0, 1\}$, $\oplus$ denotes addition modulo 2. For strings $x, y \in \{0, 1\}^n$, the *Hamming distance* between x and y , denoted by $d(x, y)$, is the number of bit positions in which they differ. The *Hamming weight* of a string $x \in \{0, 1\}^n$, denoted by $|x|$, is the number of ones in x , equivalently $|x| = \sum_{i=1}^n x_i$.

For integers n and k satisfying $0 \leq k \leq n$, the *binomial coefficient* is defined as $\binom{n}{k} = \frac{n!}{k!(n-k)!}$. It counts the number of ways to choose k elements from a set of size n . Equivalently, it is the number of strings in $\{0, 1\}^n$ with Hamming weight k .

A random variable X is said to have a *Bernoulli distribution* with parameter $p \in [0, 1]$, written $X \sim \text{Bernoulli}(p)$, if X takes values in $\{0, 1\}$ with

$$\Pr[X = 1] = p \quad \text{and} \quad \Pr[X = 0] = 1 - p. \quad (1.5)$$

A random variable Y is said to have a *binomial distribution* with parameters n and p , written $Y \sim \text{Binomial}(n, p)$, if Y is the sum of n independent Bernoulli random variables, each with parameter p . Equivalently,

$$\Pr[Y = k] = \binom{n}{k} p^k (1 - p)^{n-k} \quad (1.6)$$

for each $k \in \{0, \dots, n\}$.

1.1.2 Linear algebra

The p -norm for a vector $u \in \mathbb{C}^n$ is defined as

$$\|u\|_p = \left(\sum_{i=1}^n |u_i|^p \right)^{\frac{1}{p}} \quad (1.7)$$

for every positive integer p and

$$\|u\|_\infty = \max\{|u_i| : i \in [n]\}. \quad (1.8)$$

For a matrix A , we define its *adjoint* as the matrix A^* which is computed as $A^* = \overline{A}^\top$ where $A^\top$ is the transpose of A and $\bar{A}$ is the entry-wise complex conjugate of A .

For matrices

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \in \mathbb{C}^{m \times n} \quad \text{and} \quad B = \begin{pmatrix} b_{11} & \cdots & b_{1q} \\ \vdots & \ddots & \vdots \\ b_{p1} & \cdots & b_{pq} \end{pmatrix} \in \mathbb{C}^{p \times q},$$

we define their *tensor product* as

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix} \in \mathbb{C}^{mp \times nq}. \quad (1.9)$$

The *trace* of $A \in \mathbb{C}^{n \times n}$ is defined as $\text{tr}(A) = \sum_{i=1}^n A(i, i)$. This function satisfies $\text{tr}(AB) = \text{tr}(BA)$ for all $A \in \mathbb{C}^{m \times n}$ and $B \in \mathbb{C}^{n \times m}$, which is referred to as the *cyclic property* of trace. The trace can be used to define an inner product as $\langle A, B \rangle = \text{tr}(A^*B)$ for all $A \in \mathbb{C}^{m \times n}$ and

$B \in \mathbb{C}^{m \times p}$, called the *Hilbert-Schmidt inner product*.

We denote the largest and smallest eigenvalue of a matrix A by $\lambda_{\max}(A)$ and $\lambda_{\min}(A)$ respectively. A matrix $A \in \mathbb{C}^{n \times n}$ is said to be *Hermitian* if it satisfies $A = A^*$. We denote the set of all Hermitian matrices on $\mathcal{X}$ as $\text{Herm}(\mathcal{X})$. The eigenvalues of a Hermitian matrix are real and thus can be ordered in either increasing or decreasing order.

A Hermitian matrix $B \in \mathbb{C}^{n \times n}$ is said to be *positive semidefinite* if it admits the decomposition $B = A^*A$ for some $A \in \mathbb{C}^{n \times n}$. The set of all positive semidefinite matrices on $\mathcal{X}$ is denoted as $\text{Pos}(\mathcal{X})$. The eigenvalues of a positive semidefinite matrix are non-negative, and we write $B \succcurlyeq 0$ to indicate that a matrix B is positive semidefinite. More generally, for matrices $A, B \in \text{Herm}(\mathcal{X})$, the notation $A \succcurlyeq B$ means that $A - B$ is positive semidefinite. For a positive semidefinite matrix B , its *square root* is the unique matrix $Y \succcurlyeq 0$ satisfying $Y^2 = B$, and we denote it as $B^{1/2}$.

A matrix $X \in \mathbb{C}^{n \times n}$ is *invertible* if there exists a matrix $Y \in \mathbb{C}^{n \times n}$ such that $YX = \mathbb{1}$. When such a Y exists it is denoted by X^{-1} . Since X is square, we also have that $XX^{-1} = \mathbb{1}$. The *Moore-Penrose pseudoinverse* of X is the operator X^+ satisfying:

1. $X X^+ X = X$,
2. $X^+ X X^+ = X^+$, and
3. $X X^+$ and $X^+ X$ are both Hermitian.

When X is invertible, $X^+ = X^{-1}$.

A matrix $U \in \mathbb{C}^{n \times n}$ is a *unitary* operator if it preserves the Euclidean norm, i.e., $\|Ux\| = \|x\|$ for all $x \in \mathbb{C}^n$. These matrices preserve inner products as well, i.e., $\langle Ux, Uy \rangle = \langle x, y \rangle$ for all $x, y \in \mathbb{C}^n$. They satisfy the equation $U^*U = UU^* = \mathbb{1}$ and are invertible.

For a matrix $A \in \mathbb{C}^{m \times n}$ and a real number $p \geq 1$, we define the *Schatten p -norm* as

$$\|A\|_p = \left(\text{Tr} \left((A^* A)^{\frac{p}{2}} \right) \right)^{\frac{1}{p}}. \quad (1.10)$$

The Schatten 1-norm is also referred to as the *trace* norm and the Schatten 2-norm is referred to as the *Frobenius* norm.

The Cauchy-Schwarz inequality for Hermitian matrices is

$$|\langle X, Y \rangle| \leq \|X\|_F \|Y\|_F \quad (1.11)$$

with equality if and only if X and Y are linearly dependent. Therefore, if *nonzero* Hermitian X and Y satisfy

$$\langle X, Y \rangle = \|X\|_F \|Y\|_F \quad (1.12)$$

then there must exist $\lambda \neq 0$ such that $X = \lambda Y$.

Let $\mathcal{A}$ and $\mathcal{B}$ be complex Euclidean spaces. The *adjoint* of a linear map $\Phi : \mathcal{A} \rightarrow \mathcal{B}$ is the unique linear map $\Phi^* : \mathcal{B} \rightarrow \mathcal{A}$ that satisfies $\langle \Phi(A), B \rangle = \langle A, \Phi^*(B) \rangle$ for all A in $\mathcal{A}$ and B in $\mathcal{B}$. The *identity map* on operators on $\mathcal{A}$, denoted by $\mathcal{I}_{\mathcal{A}}$, maps every operator on $\mathcal{A}$ to itself. The *partial trace* over $\mathcal{A}$, denoted by $\text{Tr}_{\mathcal{A}}$, is defined as $\text{Tr}_{\mathcal{A}}(A \otimes B) = \text{Tr}(A) B$ for every operator A on $\mathcal{A}$ and every operator B on $\mathcal{B}$. Finally, a linear map Ξ from operators on $\mathcal{A}$ to operators on $\mathcal{B}$ is *Hermiticity-preserving* if $\Xi(A)$ is Hermitian for every Hermitian operator A on $\mathcal{A}$.

The *Gram matrix* G of a set of vectors $\{v_1, \dots, v_N\}$ is the Hermitian positive semidefinite matrix whose entries are given by $G_{ij} = \langle v_i, v_j \rangle$.

An $N \times N$ *Toeplitz matrix* T is a matrix with entries satisfying

$$T_{i,j} = T_{i+1,j+1} \text{ for all } i, j \in [N-1]. \quad (1.13)$$

A Toeplitz matrix can be written as

$$T = \begin{pmatrix} t_0 & t_{-1} & t_{-2} & \cdots & t_{-(N-1)} \\ t_1 & t_0 & t_{-1} & \cdots & t_{-(N-2)} \\ t_2 & t_1 & t_0 & \cdots & t_{-(N-3)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ t_{N-1} & t_{N-2} & t_{N-3} & \cdots & t_0 \end{pmatrix},$$

noting that every superdiagonal has constant entries. Since each diagonal of the Toeplitz matrix has the same value, we can alternately express T as

$$T = \sum_{k=-(N-1)}^{N-1} t_k \Theta_k, \quad (1.14)$$

where Θ_k is the $N \times N$ Toeplitz matrix with ones on the k -th diagonal and zeros elsewhere. Here, $k = 0$ indicates the principal diagonal, $k > 0$ denotes the sub-diagonals, and $k < 0$ corresponds to the super-diagonals. Note that a Toeplitz matrix is not necessarily square.

A set $\mathcal{C}$ is *convex* if we have $\lambda x + (1 - \lambda)y \in \mathcal{C}$, for all $x, y \in \mathcal{C}$ and $\lambda \in [0, 1]$. A set $\mathcal{K}$ is a *cone* if $\lambda x \in \mathcal{K}$ for all $x \in \mathcal{K}$ and $\lambda \in [0, 1]$. A cone that is also convex is referred to as a *convex cone*. *Cone programming* is the study of optimizing a linear function over a variable in a convex cone $\mathcal{K}$ subject to affine constraints. In standard form, we can write a cone program as

$$\text{maximize } \left\{ \langle X, A \rangle : \Xi(X) = B, X \in \mathcal{K} \right\} \quad (1.15)$$

where Ξ is a linear map. The *convex hull* of $\mathcal{A}$, denoted $\text{conv}(\mathcal{A})$, is the intersection of all convex sets containing $\mathcal{A}$.

An order- r *Krylov subspace* generated by an $n \times n$ matrix A and an n -dimensional vector b is the set of vectors

$$\mathcal{K}_r(A, b) = \text{span} \{b, Ab, A^2b, \dots, A^{r-1}b\}.$$

Let $r_0 = \dim \text{span} \{b, Ab, A^2b, \dots\}$. We have that $\{b, Ab, A^2b, \dots, A^{r-1}b\}$ are linearly independent unless $r > r_0$.

1.1.3 Quantum information

A *density operator* is a positive semidefinite operator with unit trace. We denote the set of all density operators on $\mathcal{X}$ by $D(\mathcal{X})$. A *quantum state*, often referred to simply as a *state*, is represented by a density operator.

We use Dirac notation, where $|\psi\rangle$ denotes a vector in a complex Euclidean space and $\langle\psi|$ denotes its adjoint. For vectors $|\psi\rangle$ and $|\phi\rangle$, the inner product is written as $\langle\psi|\phi\rangle$, and their outer product is expressed as $|\psi\rangle\langle\phi|$.

Consider states $\{\rho_1, \dots, \rho_n\}$ with associated probabilities $\{p_1, \dots, p_n\}$. The collection $\{(p_i, \rho_i) : i \in [n]\}$ is called an *ensemble* of states, and the state $\rho = \sum_{i=1}^n p_i \rho_i$ is called the corresponding *mixture*. A rank-one state is called a *pure state*.

A state $\rho \in D(\mathcal{X}_1 \otimes \dots \otimes \mathcal{X}_n)$ is said to be a *product state* if it can be expressed as $\rho = \sigma_1 \otimes \dots \otimes \sigma_n$, where $\sigma_i \in D(\mathcal{X}_i)$ for each $i \in [n]$. A state is called *separable* if it can be expressed as a convex combination of product states; otherwise, it is called *entangled*.

The *maximally mixed state* on a complex Euclidean space $\mathcal{X}$ is the state $\frac{1}{\dim(\mathcal{X})} \mathbf{1}_{\mathcal{X}}$. The *uniform superposition state* in $\mathcal{X} = \mathbb{C}^n$ is defined as $\frac{1}{\sqrt{n}} \sum_{i=1}^n |i\rangle$.

A *measurement* is described by a positive operator-valued measure, or POVM, which is a collection of operators M_i for $i \in [n]$ satisfying $M_i \succcurlyeq 0$ for each $i \in [n]$, and $\sum_{i=1}^n M_i = \mathbb{1}$. The operators M_i are referred to as the measurement operators, and each $i \in [n]$ indexes the possible measurement outcomes. If the system is in state $\rho \in \mathcal{D}(\mathcal{X})$, then the probability of obtaining the outcome i upon measuring with M_i is given by $\langle M_i, \rho \rangle$. This expression for the outcome probabilities is known as *Born's rule*.

1.1.4 Similarity in quantum computing

Swap test. Given two (possibly unknown) pure states $|\psi\rangle$ and $|\phi\rangle$, the swap test outputs 0 with probability $\frac{1}{2} + \frac{1}{2} |\langle \psi | \phi \rangle|^2$. Figure 1.4 depicts the corresponding circuit. One can learn $|\langle \psi | \phi \rangle|^2$ to additive accuracy ϵ with failure probability at most δ using $\mathcal{O}\left(\frac{1}{\epsilon^2} \log\left(\frac{1}{\delta}\right)\right)$ copies of the states and $\tilde{\mathcal{O}}\left(\frac{1}{\epsilon^2} \log\left(\frac{1}{\delta}\right)\right)$ operations [BCWDW01, HBR21].

Hadamard test. Given access to a unitary U that maps the state $|\psi\rangle$ to the state $|\phi\rangle$, this method creates a random variable whose expected value is the real part of $\langle \psi | \phi \rangle$. Figure 1.5 illustrates the circuit for this procedure. The imaginary part of $\langle \psi | \phi \rangle$ can be computed by applying a phase gate S on the first qubit before the controlled unitary operation. We can learn $|\langle \psi | \phi \rangle|^2$ to additive accuracy ϵ with failure probability at most δ using $\mathcal{O}\left(\frac{1}{\epsilon^2} \log\left(\frac{1}{\delta}\right)\right)$ copies of the states and $\tilde{\mathcal{O}}\left(\frac{1}{\epsilon^2} \log\left(\frac{1}{\delta}\right)\right)$ operations [AJL06, HBR21].

1.1.5 Semidefinite programming

Semidefinite programming is an area of convex optimization where the goal is to optimize a linear function of a positive semidefinite matrix X over affine constraints. A semidefinite

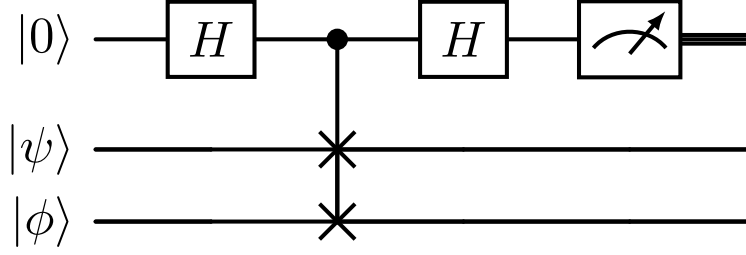


Figure 1.4: Circuit for the swap test which can learn the inner product between two (perhaps unknown) states $|\psi\rangle$ and $|\phi\rangle$, by approximating $|\langle\psi|\phi\rangle|$ from many measured samples.

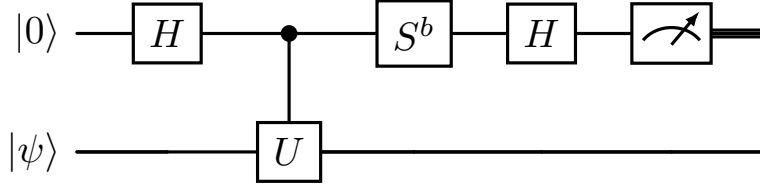


Figure 1.5: Circuit for the Hadamard test to learn the inner product between two (perhaps unknown) states $|\psi\rangle$ and $|\phi\rangle$. When $b = 0$, the Hadamard test approximates from many samples $\text{Re}(\langle\psi|\phi\rangle)$, and when $b = 1$, $\text{Im}(\langle\psi|\phi\rangle)$ can be approximated instead. The unitary U maps the state $|\psi\rangle$ to the state $|\phi\rangle$.

program (SDP) can be written in standard form

$$\begin{aligned} \alpha = \inf \quad & \langle C, X \rangle \\ \text{s.t.} \quad & \langle A_i, X \rangle = b_i, \quad i \in [m] \\ & X \succcurlyeq 0, \end{aligned}$$

where the matrices C and A_i are Hermitian and b_i are real, and $\inf$ is short for infimum. Every SDP has a dual which is an SDP itself and is defined as

$$\begin{aligned} \beta = \sup \quad & \langle b, y \rangle \\ \text{s.t.} \quad & \sum_{i=1}^m y_i A_i \preceq C \\ & y \in \mathbb{R}^m, \end{aligned} \tag{1.16}$$

where $\sup$ is short for supremum. We have that $\alpha \leq \beta$, a fact known as weak duality. If both α and β are finite and one has a feasible solution where the inequalities are strict (known as strict feasibility), then $\alpha = \beta$. This condition is known as strong duality.

The above optimization in this case is called *semidefinite programming* and is of great interest in many areas including combinatorial optimization [Tun16] and quantum information [Wat18] and we refer to a specific instance as a semidefinite program. While optimizing over separable operators is indeed intractable, one can often solve SDPs efficiently. Current numerical solvers for solving SDPs exist such as CVX [GB14], SeDuMi [Stu99], SDPT3 [TTT99] and Mosek [AA00]. Since SDPs can often be solved efficiently, they are a popular choice for approximating convex sets which are difficult to deal with.

Chapter 2

Quantum heuristics for optimizing large separable operators

The contents of this chapter is based on work with Tobias Haug, Kishor Bharti and Jamie Sikora in [\[MHBS25\]](#).

In this chapter, we develop a dimension-reduction framework for optimizing over separable operators that preserves feasibility. The central idea is to restrict the search to carefully chosen low-dimensional ansatz subspaces, solve a reduced separable optimization problem, and then lift the resulting solution back to the original space. This produces an inner approximation of the original problem and therefore guarantees valid separable solutions. To construct the reduced problem in regimes where the full operator cannot be explicitly represented, we leverage a quantum co-processor to estimate the data required for the reduction.

We demonstrate the effectiveness of this approach by applying it to the problem of approximating separable ground energies of Ising Hamiltonians. This application allows us to introduce a normalized measure quantifying how close separable states can approach the

true ground energy and to study how entanglement manifests across system sizes. Numerical experiments illustrate both the accuracy of the proposed heuristic and its ability to scale beyond the limits of direct classical computation.

This chapter is organized as follows.

- Section 2.1: We study optimization-based algorithms for solving (small) separability problems and examine numerically how well they perform.
- Section 2.2: We focus to large separability problems whose classical descriptions could involve exponentially large vectors and matrices. To combat this curse of dimensionality, we introduce our quantum heuristics for reducing the dimension to optimize over both separable operators and separable states. We also discuss how a quantum co-processor is used to compute the data for the reduced optimization problem.
- Section 2.3: As a concrete application, we apply the heuristic to find separable ground states of Hamiltonians. We first introduce a measure of entanglement in the groundspace of a Hamiltonian and then discuss how the ansatz states are chosen and evaluate the performance of our heuristic as we modify this number. For the Ising Hamiltonian, we compare the ground energy, separable ground energy, and their approximations.
- Section 2.4: We discuss related work that deals with optimizing over separable states and those specific to Hamiltonians. This is followed with a brief discussion of how our algorithms compare to these works.

2.1 Semidefinite programming-based algorithms for the separability problem, a quick discussion

For a density operator acting on the bipartite space $\mathcal{A} \otimes \mathcal{B}$, we wish to decide whether it belongs to the set of separable states, defined as

$$\text{SepD}(\mathcal{A} : \mathcal{B}) = \text{conv} \{ \rho \otimes \sigma \} \quad (2.1)$$

where ρ ranges over density operators on $\mathcal{A}$, and σ ranges over density operators on $\mathcal{B}$. In this work, we also look at the set of separable *operators* which is defined as

$$\text{Sep}(\mathcal{A} : \mathcal{B}) = \text{conv} \{ X \otimes Y \} \quad (2.2)$$

where X is a positive semidefinite operator acting on $\mathcal{A}$ and Y is a positive semidefinite operator acting on $\mathcal{B}$. We note that $\text{Sep}(\mathcal{A} : \mathcal{B})$ and $\text{SepD}(\mathcal{A} : \mathcal{B})$ differ only in the fact that the latter has a unit trace constraint.

A general linear optimization over the set of separable operators takes the form

$$\text{maximize} \left\{ \langle X, C \rangle : \Xi(X) = B, X \in \text{Sep}(\mathcal{A} : \mathcal{B}) \right\} \quad (2.3)$$

where C and B are Hermitian matrices, and Ξ is a hermiticity-preserving linear map. While we consider the general problem above in this work, we often focus on the special case of linear optimization over the set of separable *states* taking the form

$$\text{maximize} \left\{ \langle \tau, C \rangle : \tau \in \text{SepD}(\mathcal{A} : \mathcal{B}) \right\}. \quad (2.4)$$

In this work, we are concerned primarily with the cone of *separable operators*, i.e., when $\mathcal{K} = \text{Sep}$, but also when $\mathcal{K}$ is the set of positive semidefinite operators.

Notice that if we have the optimal solution τ , which always exists by compactness of the set SepD , then a simple convexity argument¹ shows that there exists pure states $|\psi\rangle$ and $|\phi\rangle$ such that $|\psi\rangle\langle\psi| \otimes |\phi\rangle\langle\phi|$ is an optimal solution. Therefore, we can optimize over product states instead of all of SepD . Under this restriction, we can rewrite the problem (2.4) as

$$\text{maximize } \left\{ \langle \rho \otimes \sigma, C \rangle : \rho \in D(\mathcal{A}), \sigma \in D(\mathcal{B}) \right\}, \quad (2.5)$$

where we have denoted the set of *density matrices* acting on $\mathcal{X}$ as $D(\mathcal{X})$. (Technically, we can assume ρ and σ to be pure if it helps, but we do not need to assume this.) Since the optimization problem (2.5) is quadratic, it still remains hard in general. However, there are folklore algorithms called *see-saw* which often perform well in practice (and are a special case of coordinate descent algorithms [Wri15]). (Note that no algorithm will always perform well due to NP complexity barriers.) The see-saw pseudo-algorithm is stated below for this setting.

Algorithm 1 See-saw

- 1: **Input:** $\hat{\rho} \in D(\mathcal{A}), C \in \text{Herm}(\mathcal{A} \otimes \mathcal{B})$
 - 2: **repeat**
 - 3: Solve for optimal $\hat{\sigma} \in D(\mathcal{B})$: maximize $\langle \hat{\rho} \otimes \sigma, C \rangle$.
 - 4: Solve for optimal $\hat{\rho} \in D(\mathcal{A})$: maximize $\langle \rho \otimes \hat{\sigma}, C \rangle$.
 - 5: **until** Progress is not being made.
 - 6: **return** $\hat{\rho}$ and $\hat{\sigma}$.
-

This approach generally works well, but it could depend on how the initial state $\hat{\rho}$ is chosen. Of course, if one starts with $\hat{\rho}$ from an optimal solution $\rho \otimes \sigma$, then this algorithm terminates immediately with an optimal solution. However, due to the curvature of the function, it can

¹If an optimal solution is a convex combination of feasible solutions, then one of those feasible solutions has objective function value at least as good as the original optimal solution.

easily get stuck in a local maximum, rendering it difficult to find a global maximum without a restart. We consider three choices of initial states in this work, the maximally mixed state $\hat{\rho} = \frac{1}{\dim(\mathcal{A})} \mathbb{1}_{\mathcal{A}}$, the uniform superposition state $\hat{\rho} = |\psi\rangle\langle\psi|$ where $|\psi\rangle = \frac{1}{\sqrt{\dim(\mathcal{A})}} \sum_i |i\rangle$, and 100 randomly chosen states.

An important aspect of the see-saw algorithm is that it gives a feasible solution, in the sense that $\hat{\rho} \otimes \hat{\sigma}$ is always separable. Thus, this gives us a *lower bound* on the optimal value and a corresponding feasible solution. We note that many SDP-based algorithms for approximating the separable set are *outer approximations* which do not always have this feature of yielding a feasible solution (such as the PPT criterion [HHH96, Per96], symmetric extensions [DPS02, CJK⁺14, LHRZ19], and the realignment criterion [CW03] (as discussed in Chapter 1)). Therefore, this is a nice feature of the see-saw algorithm.

2.1.1 A bottleneck in the see-saw algorithm

Recall that in see-saw we wish to alternately optimize over ρ and σ . Concentrating on optimizing over σ for now, we can rewrite $\langle \hat{\rho} \otimes \sigma, C \rangle$ as $\langle \sigma, C_{\mathcal{B}} \rangle$, where

$$C_{\mathcal{B}} = \text{Tr}_{\mathcal{A}}[(\hat{\rho} \otimes \mathbb{1})C] \in \text{Herm}(\mathcal{B}).$$

Thus, we are effectively solving the optimization problem

$$\text{maximize } \left\{ \langle \sigma, C_{\mathcal{B}} \rangle : \sigma \in D(\mathcal{B}) \right\} \quad (2.6)$$

whose solution corresponds to the principal eigenvector of $C_{\mathcal{B}}$. Similarly, $\langle \rho \otimes \hat{\sigma}, C \rangle = \langle \rho, C_{\mathcal{A}} \rangle$ where $C_{\mathcal{A}} = \text{Tr}_{\mathcal{B}}[(\mathbb{1} \otimes \hat{\sigma})C] \in \text{Herm}(\mathcal{A})$ whose solution is the principal eigenvector of $C_{\mathcal{A}}$. Thus, the see-saw algorithm is simply a sequence of principal eigenvector computations.

2.1.2 Special-case instances

If $\dim(\mathcal{A}) = \dim(\mathcal{B}) = d$, then $C \in \text{Herm}(\mathcal{A} : \mathcal{B})$ is a $d^2 \times d^2$ matrix. Therefore solving $\langle \rho \otimes \sigma, C \rangle$ for either ρ or σ requires dealing with a $d^2 \times d^2$ matrix. Calculating $C_{\mathcal{A}}$ and $C_{\mathcal{B}}$ using a brute force method involves matrix computations on a $d^2 \times d^2$ matrix, which can grow expensive rather quickly. Before discussing our dimension-reduction technique, we discuss a special case that can speed up computations at this level.

We now consider instances with a particular structure that circumvents the pre-computation issue raised above while still capturing many applications of interest. Suppose C can be written in the following way

$$C = \sum_{m=1}^N K_m \otimes L_m, \quad (2.7)$$

where $K_m \in \text{Herm}(\mathcal{A})$ and $L_m \in \text{Herm}(\mathcal{B})$, for all m . Observe that

$$C_{\mathcal{B}} = \text{Tr}_{\mathcal{A}}[(\hat{\rho} \otimes \mathbb{1})C] = \text{Tr}_{\mathcal{A}} \left[(\hat{\rho} \otimes \mathbb{1}) \left(\sum_m K_m \otimes L_m \right) \right] = \sum_m \langle \hat{\rho}, K_m \rangle L_m \quad (2.8)$$

and similarly $C_{\mathcal{A}} = \text{Tr}_{\mathcal{B}}[(\mathbb{1} \otimes \hat{\sigma})C] = \sum_m \langle \hat{\sigma}, L_m \rangle K_m$. Therefore, the see-saw pre-computation involves a series of matrix computations on matrices of size $d \times d$ now, offering a quadratic speed up.

2.1.3 Numerical tests

Efficiency. Using the approach for general C , we can perform see-saw up to local dimension of 30 (which for our numerical experiments take roughly 30 hours). However for instances with the structure in Eq. (2.7), we could solve for much larger dimensions. For this, we generated random C matrices of the form $C = \sum_{m=1}^{10} K_m \otimes L_m$ and with this we observe that our experiments for a local dimension of 30 takes less than 0.008 seconds. Moreover we

are able to perform see-saw for instances with this specific structure up to a local dimension of 2500. At this point, our numerical experiments required about 35 days of runtime. (The hardware specifications are indicated in the computational platform section (Section 2.3.3).)

Accuracy. In our numerical experiments, we compare the lower bounds against some outer approximations obtained by relaxing the set SepD into a larger set that can be represented as an SDP (positive partial transpose criterion [Per96, HHH96] and symmetric extensions [DPS02]). To quantify the performance, we track the difference $\beta - \gamma$ where β is the least upper bound (e.g., from all of the outer approximations calculated) and γ is the greatest lower bound (e.g., by performing see-saw over the various starting points). If the difference is small, then we have a good approximation and the states outputted from the best see-saw algorithm is an approximately optimal solution. A large difference indicates that we have found a tricky instance over which at least one of these algorithms failed to produce a good approximation.

For varying dimensions, we ran this over 100 random instances. Our numerical experiments showed that this difference was at most 10^{-3} for values of local dimension up to 13. Beyond this value of the local dimension, the median time required to compute the upper bounds over 100 random instances of C exceeded 30 minutes and thus had to be terminated. Since the see-saw algorithm is seen to be effective both in terms of performance as well as scalability, we rely on see-saw for approximations for larger dimensions in the rest of this work.

Data. The data collected from these numerical experiments is hosted on an interactive web application https://ankith-mohan.shinyapps.io/SEP_app.

2.2 Reducing the dimension of separability problems via a quantum co-processor

Now that we have argued that the see-saw algorithm provides good inner approximations on small separability problems, what can we do with large separability problems? An even tougher problem:

What if the problem is so big we cannot even write it down?

Indeed, many quantum problems suffer from the curse of dimensionality meaning that classical descriptions could involve exponentially large vectors and matrices. We now consider heuristics for reducing the dimension by generalizing the NISQ SDP solver idea in [BHVK22] to optimize over both separable operators and separable states.

2.2.1 General linear optimization over separable operators

Consider the optimization problem

$$\mu = \text{maximize } \left\{ \langle X, C \rangle : \Xi(X) = B, X \in \text{Sep}(\mathcal{A} : \mathcal{B}) \right\}. \quad (2.9)$$

Suppose we are given *two* sets of ansatz states

$$\{|\psi_1\rangle, \dots, |\psi_L\rangle\} \subset \mathcal{A} \quad \text{and} \quad \{|\phi_1\rangle, \dots, |\phi_M\rangle\} \subset \mathcal{B}.$$

From these states, we define the two matrices

$$\Psi := \sum_{i=1}^L |\psi_i\rangle\langle i| \quad \text{and} \quad \Phi := \sum_{k=1}^M |\phi_k\rangle\langle k|. \quad (2.10)$$

Then we can *guess*² that an optimal solution takes the form

$$X = (\Psi \otimes \Phi)Y(\Psi \otimes \Phi)^* \quad (2.11)$$

for some $Y \in \text{Sep}(\mathbb{C}^L : \mathbb{C}^M)$. Now consider the following separability problem

$$\mu' = \text{maximize } \left\{ \langle Y, D \rangle : \Xi'(Y) = B, Y \in \text{Sep}(\mathbb{C}^L : \mathbb{C}^M) \right\}, \quad (2.12)$$

where $D = (\Psi \otimes \Phi)^*C(\Psi \otimes \Phi)$ and $\Xi'(\cdot) = \Xi((\Psi \otimes \Phi) \cdot (\Psi \otimes \Phi)^*)$. It can be verified that

- $\langle X, C \rangle = \langle Y, D \rangle$,
- $\Xi(X) = \Xi'(Y)$, and
- $X \in \text{Sep}(\mathcal{A} : \mathcal{B})$ when $Y \in \text{Sep}(\mathbb{C}^L : \mathbb{C}^M)$.

To see the last claim, if we take a separable decomposition of $Y = \sum_i q_i \rho_i \otimes \sigma_i$, we have that

$$X = (\Psi \otimes \Phi)Y(\Psi \otimes \Phi)^* = \sum_i q_i \Psi \rho_i \Psi^* \otimes \Phi \sigma_i \Phi^* \in \text{Sep}(\mathcal{A} : \mathcal{B}). \quad (2.13)$$

Moreover, this gives an *inner approximation* for the original separability problem; if Y is feasible with objective function value $\langle Y, D \rangle$ in Eq. (2.12), then X is feasible in the original problem with objective function value $\langle X, C \rangle = \langle Y, D \rangle$. Thus, $\mu' \leq \mu$, yielding a bound on the optimal value of the original problem as well. We call the problem (2.12) the *reduced* problem since, typically, its size is much smaller than the original problem.

If we apply the see-saw algorithm to the reduced problem, this yields a feasible solution to the reduced problem. As we have previously discussed, the see-saw algorithm performs

²An optimal solution may not have the following form in general, however this does yield a feasible solution. Intuitively, the better the ansatz, the better feasible solution one would expect to obtain.

well on smaller separability problems. Moreover, this feasible solution can be turned into a feasible solution for the original problem. Thus, overall this process yields a feasible solution to the original problem, and this provides a lower bound on its value.

2.2.2 Optimizing over separable states

We now consider the special case when the constraint $\Xi(X) = B$ is written as $\text{Tr}(X) = 1$, i.e., we wish to optimize over separable quantum states such as in the following problem:

$$\text{maximize } \left\{ \langle \rho, C \rangle : \rho \in \text{SepD}(\mathcal{A} : \mathcal{B}) \right\}. \quad (2.14)$$

From the previous discussion, the reduced problem takes the form

$$\text{maximize } \left\{ \langle Y, D \rangle : \langle G_{\mathcal{A}} \otimes G_{\mathcal{B}}, Y \rangle = 1, Y \in \text{Sep}(\mathbb{C}^L : \mathbb{C}^M) \right\}, \quad (2.15)$$

where $G_{\mathcal{A}}$ is the Gram matrix of the first set of ansatz states and $G_{\mathcal{B}}$ is the Gram matrix of the second set of ansatz states:

$$G_{\mathcal{A}} = \sum_{i=1}^L \sum_{j=1}^L \langle \psi_i | \psi_j \rangle |i\rangle\langle j| \quad \text{and} \quad G_{\mathcal{B}} = \sum_{k=1}^M \sum_{l=1}^M \langle \phi_k | \phi_l \rangle |k\rangle\langle l|. \quad (2.16)$$

If we suppose that each set of ansatz states are linearly independent, then both $G_{\mathcal{A}}$ and $G_{\mathcal{B}}$ are invertible. Now, if we define the matrix

$$Z := \left(G_{\mathcal{A}}^{1/2} \otimes G_{\mathcal{B}}^{1/2} \right) Y \left(G_{\mathcal{A}}^{1/2} \otimes G_{\mathcal{B}}^{1/2} \right) \quad (2.17)$$

we can rewrite the reduced problem as the following:

$$\text{maximize } \left\{ \langle Z, \tilde{D} \rangle : Z \in \text{SepD}(\mathbb{C}^L : \mathbb{C}^M) \right\}, \quad (2.18)$$

where

$$\tilde{D} = (G_{\mathcal{A}}^{-1/2} \Psi^* \otimes G_{\mathcal{B}}^{-1/2} \Phi^*) C (\Psi G_{\mathcal{A}}^{-1/2} \otimes \Phi G_{\mathcal{B}}^{-1/2}). \quad (2.19)$$

To verify this, we can see that:

- Y is separable if and only if Z is separable,
- $\langle Z, \tilde{D} \rangle = \langle Y, D \rangle$, and
- $\text{Tr}(Z) = 1$ if and only if $\langle G_{\mathcal{A}} \otimes G_{\mathcal{B}}, Y \rangle = 1$.

Let us summarize what we have accomplished so far. We have taken the optimization over separable states given in Eq. (2.14) defined by the matrix $C \in \text{SepD}(\mathcal{A} : \mathcal{B})$ and replaced it with a reduced version which is itself an optimization over separable states given in Eq. (2.18) defined by the matrix $\tilde{D} \in \text{SepD}(\mathbb{C}^L : \mathbb{C}^M)$. This can be a considerably smaller optimization over separable states and, again, given a feasible solution to the reduced problem we can find a feasible solution to the original problem.

2.2.3 Calculating the data of the reduced problems with a quantum co-processor

One immediate concern arises when one wishes to compute $\tilde{D}$. This involves computing matrix operations on possibly huge vectors, which may not be possible on a classical computer. This is where the quantum co-processor comes into the picture. To this end, we make the assumption that we have physical access to the ansatz states $|\psi_i\rangle$ and $|\phi_j\rangle$. From

an algorithmic standpoint, we assume that they are efficiently preparable and that we have access to:

- circuits U_i that create $|\psi_i\rangle$ and their inverses, and
- circuits V_j that create $|\phi_j\rangle$ and their inverses.

By combining them, we thus have a circuit U_{ij} which maps $|\psi_i\rangle$ to $|\psi_j\rangle$ for any choice of i and j , and similarly for the $|\phi_i\rangle$ states. This way we can compute the Gram matrix $G_{\mathcal{A}}$ by performing the *Hadamard test* to learn $\text{Re}(\langle\psi_i|\psi_j\rangle)$ and $\text{Im}(\langle\psi_i|\psi_j\rangle)$ for each pair of i and j . We can similarly compute the Gram matrix $G_{\mathcal{B}}$.

In the event that all of the inner products are non-negative then we could alternatively use the *SWAP test* to compute each of the inner products in the Gram matrix. In this setting, we do not require any of the preparation circuits, just access to copies of the states.

Lastly, as long as the ansatz states are linearly independent and there are not too many of them, we can relatively easily compute the inverses of their Gram matrices.

With this said, all that remains is to compute $\tilde{D}$, which we now discuss. Recall that we have $\tilde{D} = \left(G_{\mathcal{A}}^{-1/2} \otimes G_{\mathcal{B}}^{-1/2}\right) D \left(G_{\mathcal{A}}^{-1/2} \otimes G_{\mathcal{B}}^{-1/2}\right)$, and we can write

$$D := \sum_{i=1}^L \sum_{j=1}^L \sum_{k=1}^M \sum_{l=1}^M \langle\psi_i| \langle\phi_k| C |\psi_j\rangle |\phi_l\rangle |i\rangle\langle j| \otimes |k\rangle\langle l|. \quad (2.20)$$

Thus, if we can compute D via a quantum co-processor, then we are done.

If we were to write $C = \sum_m P_m$ as a linear combination of Pauli matrices, and we also have access to Pauli gates, then we can compute each entry of D in a manner similar to that described above. That is, we can apply each P_m to each of the states $|\psi_j\rangle |\phi_l\rangle$ and calculate the required inner products via the Hadamard or SWAP test, say.

As a demonstration, a particularly nice case is when each of the ansatz states $|\psi_i\rangle$ can be prepared as $P_i^A |\psi\rangle$ where $|\psi\rangle$ is a fixed reference state, and $|\phi_k\rangle$ can be prepared as $P_k^B |\phi\rangle$ where $|\phi\rangle$ is a fixed reference state, where each $P \in \{\mathbb{1}, X, Y, Z\}^{\otimes m}$ is a Pauli string. Then, each entry of the matrix D can be expressed as

$$D_{i,j,k,l} = \langle \psi | \langle \phi | \left(\sum_m (P_i^A \otimes P_k^B) P_m (P_j^A \otimes P_l^B) \right) | \psi \rangle | \phi \rangle = \langle \psi | \langle \phi | P | \psi \rangle | \phi \rangle, \quad (2.21)$$

where P is simply a sum of Pauli strings (with $\{\pm 1, \pm i\}$ coefficients). So calculating $D_{i,j,k,l}$ is equivalent to computing the expectation value of a Pauli string with respect to the state $|\psi\rangle |\phi\rangle$, which can be done efficiently as long as $|\psi\rangle |\phi\rangle$ are efficiently preparable as discussed above and we have access to Pauli gates.

2.2.4 A special case which speeds up the see-saw algorithm when applied to the reduced problem

We note that when C can be expressed as $C = \sum_m K_m \otimes L_m$, we can rewrite the reduced problem (2.18) as

$$\text{maximize} \left\{ \left\langle Z, \sum_{m=1}^N \tilde{K}_m \otimes \tilde{L}_m \right\rangle : Z \in \text{SepD}(\mathbb{C}^L : \mathbb{C}^M) \right\}, \quad (2.22)$$

where $\tilde{K}_m = G_{\mathcal{A}}^{-1/2} \Psi^* K_m \Psi G_{\mathcal{A}}^{-1/2}$ and $\tilde{L}_m = G_{\mathcal{B}}^{-1/2} \Phi^* L_m \Phi G_{\mathcal{B}}^{-1/2}$. Depending on the nature of each K_m and L_m and our choices of ansatz states, we could have $\tilde{K}_m$ and $\tilde{L}_m$ easily computable on a quantum co-processor. Moreover, the structure is the same as discussed in Section 2.1.2, meaning that we can dramatically speed up the see-saw algorithm when applied to this problem.

2.3 Applications to finding separable ground states of Hamiltonians

In this section, we use see-saw and our dimension-reduction idea to solve for the separable state of the greatest³ energy of a certain Hamiltonian. The discussion in this section is broken into several parts. The first part introduces a *measure of how much entanglement* is in the ground space of a Hamiltonian. The second part discusses how one can approach choosing the ansatz states to run the dimension-reducing heuristic. We also graph how well the heuristics perform with respect to the number of ansatz states chosen. The third part compares how well our heuristic performs compared to the (a) full see-saw algorithm, (b) full ground energy calculation, and (c) the heuristic ground energy calculation from [BHVK22]. We run all of our tests on the one-dimensional Ising Hamiltonian, noting that our methods apply to many other popular Hamiltonians as well.

2.3.1 A ground space entanglement measure

For a Hamiltonian H , let us denote the largest and smallest eigenvalues by $\lambda_{\max}$ and $\lambda_{\min}$, respectively.

Definition 2.1 (Separable ground energy). The separable ground energy of a Hamiltonian H is the optimal value of the following optimization problem

$$\alpha = \text{maximize } \left\{ \langle \tau, H \rangle : \tau \in \text{SepD}(\mathcal{A} : \mathcal{B}) \right\}. \quad (2.23)$$

³Note that since we are maximizing throughout the discussions in this work, we think of the largest eigenvalue as the ground energy. This convention is without loss of generality since one can replace the Hamiltonian H with $-H$ if one prefers to minimize.

Since we cannot always efficiently compute the exact value of α , due to the intrinsic difficulty of optimizing over separable states, we typically rely on approximations of α , denoted $\hat{\alpha}$.

Using these values, we define a measure of entanglement as the (normalized) difference between the ground energy (i.e., $\lambda_{\max}$) and α , and an approximate version using $\hat{\alpha}$. Roughly speaking, we define a measure of how close a separable state is to the ground space.

Definition 2.2 (Ground space entanglement measure, and an approximation). For a non-zero Hamiltonian H , we define a measure of ground space entanglement as

$$\delta = \frac{\lambda_{\max} - \alpha}{\lambda_{\max} - \lambda_{\min}} \in [0, 1]. \quad (2.24)$$

We note that $\delta = 0$ if and only if the ground space contains a separable state (even though it can also contain entangled states as well). When any of the above quantities are not known, we define an approximate ground space entanglement measure as

$$\hat{\delta} = \frac{\hat{\lambda}_{\max} - \hat{\alpha}}{\hat{\lambda}_{\max} - \hat{\lambda}_{\min}} \quad (2.25)$$

where the hats represent the best approximations to the actual values that we can obtain.⁴

We note that we cannot place bounds on $\hat{\delta}$ due to the inability to place general bounds on $\hat{\alpha}$ relative to α most of the time.

At first glance, it might be tempting to define the ground space entanglement as quantifying the amount of entanglement in a ground state. While this could have its applications, we have chosen a different definition, which we now motivate with an illustrative example.

⁴Sometimes we can calculate $\lambda_{\max}$ and $\lambda_{\min}$ easily, and for large instances one can approximate them using the heuristic eigenvalue solver in [BHVK22].

Consider the following two-qubit Hamiltonian

$$H = (1 - \varepsilon) |00\rangle\langle 00| + |\psi^-\rangle\langle\psi^-|, \quad (2.26)$$

where $|\psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ and $\varepsilon > 0$ is a small constant. For any ε , the (unique) ground state of H is $|\psi^-\rangle$ which is a maximally entangled state (to which most entanglement measures would assign a nontrivial value). However, we have $\lambda_{\max} = 1$, $\lambda_{\min} = 0$, $\alpha \geq 1 - \varepsilon$, and thus $\delta \leq \varepsilon$. Therefore, our measure is small, indicating that there is a separable state that has energy *close* to that in the ground space.

The Ising Hamiltonian

We consider the one-dimensional Ising model of N qubits with transverse field h , longitudinal field g , and coupling term J , defined as

$$H_{\text{Ising}} = H_z + H_x = - \sum_{n=1}^N [J\sigma_n^z \sigma_{n+1}^z + g\sigma_n^z + h\sigma_n^x] \quad (2.27)$$

where $H_z = - \sum_n [J\sigma_n^z \sigma_{n+1}^z + g\sigma_n^z]$ and $H_x = - \sum_n h\sigma_n^x$, and σ^z and σ^x denote the Pauli Z and X matrices, respectively. The shorthand notation σ_n^z indicates that in the sequence of N qubits, the Pauli Z matrix is affecting the n th qubit while the remaining qubits are left alone. For example, $\sigma_3^z := \mathbb{1}_1 \otimes \mathbb{1}_2 \otimes \sigma^z \otimes \mathbb{1}_4 \otimes \cdots \otimes \mathbb{1}_N$.

For this Hamiltonian with parameters J , g , and h , we first want to know at what values of these parameters will the ground space entanglement measure (i.e., $\hat{\delta}$) peak. To answer this, we set $J = 1$, $g = 0$, and vary the value of h in the range $[0, 5]$. Figure 2.1 illustrates our entanglement measure for 12 and 14 qubits.

We observe that for the 12-qubit case, $\hat{\delta}$ peaks around $h = 1.3$ and for the 14-qubit case

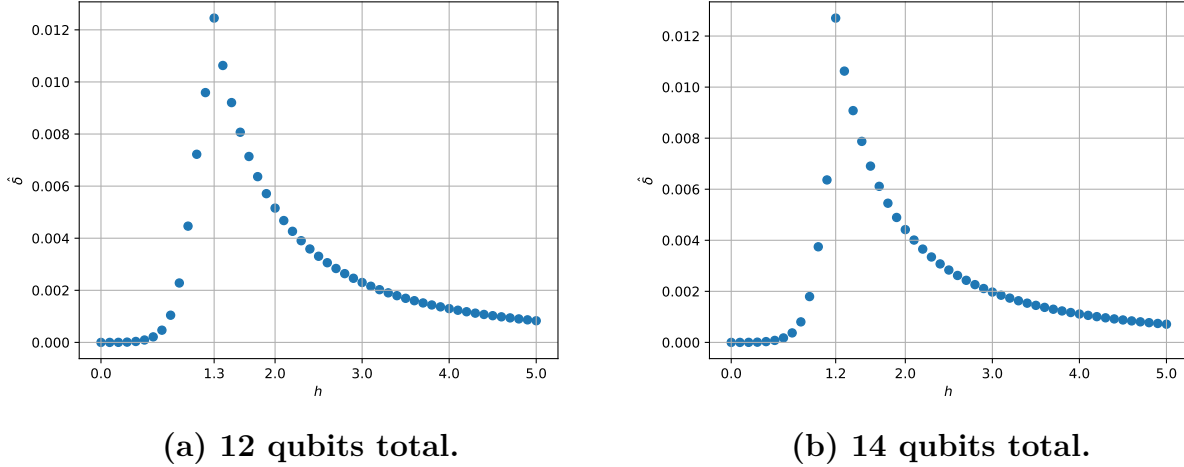


Figure 2.1: Approximate ground-space entanglement measure as a function of the transverse field h for H_{Ising} with 12 qubits in panel (a) and 14 qubits in panel (b), with $J = 1$ and $g = 0$. In each case, $\mathcal{A}$ consists of the first half of the qubits and $\mathcal{B}$ consists of the second half. Both $\lambda_{\max}$ and $\lambda_{\min}$ can be computed for these graphs.

the peak is around $h = 1.2$. These graphs suggest that there is a non-trivial amount of entanglement in these ground spaces. Note that while this model has a phase transition for the critical point $h = 1$, the entanglement is actually maximal at a value that is close, but not exactly at the critical point [OAFF02], which matches our results. Going forward, we set h to be in this range and test other figures of interest.

2.3.2 Evaluating the performance of our heuristic with varying numbers of ansatz states

We now study the performance of our heuristic for reducing the dimensionality of large separability problems discussed in Section 2.2 for the Hamiltonian H_{Ising} . To formulate our problem as an instance of (2.22), we require two things: (a) the ansatz states $\{|\psi_1\rangle, \dots, |\psi_L\rangle\} \subset \mathcal{A}$ and $\{|\phi_1\rangle, \dots, |\phi_M\rangle\} \subset \mathcal{B}$, and (b) the matrices K_m and L_m for all values of m .

Choosing the ansatz states

We use the *NISQ-friendly version of the Krylov subspace* approach proposed by [BHVK22] to generate the ansatz space $\mathcal{S}$. The original Krylov subspace approach approximates the ground state of the Hamiltonian H up to order K , using a reference state $|\eta\rangle$ as

$$|\xi(\chi)\rangle^{(K)} = \chi_0 |\eta\rangle + \chi_1 H |\eta\rangle + \cdots + \chi_K H^K |\eta\rangle \quad (2.28)$$

where χ is a $(K + 1)$ -dimensional vector. If a Hamiltonian can be expressed as a weighted sum of Pauli strings, i.e., $H = \sum_m c_m P_m$, then H^k can be expressed as

$$H^k = \left(\sum_m c_m P_m \right)^k = \sum_{m_1, \dots, m_k} c_{m_1, \dots, m_k} P_{m_1} \cdots P_{m_k}, \quad (2.29)$$

which can be further rewritten as the weighted sum of products of Pauli strings. Each product of Pauli strings applied to the reference state, $P_{m_1} \cdots P_{m_k} |\eta\rangle$, is a state that we consider to be added to our set of ansatz states. This process repeated up to power K results in an ansatz space that contains the original Krylov subspace. The first order of the ansatz states for H_{Ising} is as follows:

$$\{\sigma_1^z |\eta\rangle, \sigma_2^z |\eta\rangle, \dots, \sigma_N^z |\eta\rangle, \sigma_1^z \sigma_2^z |\eta\rangle, \sigma_2^z \sigma_3^z |\eta\rangle, \dots, \sigma_N^z \sigma_1^z |\eta\rangle, \sigma_1^x |\eta\rangle, \sigma_2^x |\eta\rangle, \dots, \sigma_N^x |\eta\rangle\}. \quad (2.30)$$

In our case, we generate the ansatz states $\{|\psi_1\rangle, \dots, |\psi_L\rangle\} \subset \mathcal{A}$ and $\{|\phi_1\rangle, \dots, |\phi_M\rangle\} \subset \mathcal{B}$, in the context of H_{Ising} , by picking the reference states $|\psi\rangle \in \mathcal{A}$ and $|\phi\rangle \in \mathcal{B}$, and using the NISQ-friendly version of the Krylov subspace approach. We construct $\mathcal{S}_{\mathcal{A}}$ by first generating the states in Eq. (2.30) with Pauli strings that act trivially on $\mathcal{B}$ with a reference state $|\psi\rangle \in \mathcal{A}$ (so that the states are understood to be in $\mathcal{A}$) and choosing L linearly independent

states from this set. We construct $\mathcal{S}_B$ analogously, by considering Pauli strings that act trivially on $\mathcal{A}$, then choosing M linearly independent states.

Representing the Ising Hamiltonian in the form $\sum_m K_m \otimes L_m$

To obtain the matrices K_m and L_m to represent the Ising Hamiltonian, suppose that of the N qubits, N_A of them correspond to the first set of ansatz states, while the remaining N_B correspond to the second set of ansatz states, i.e., $N_A + N_B = N$. Then

$$\begin{aligned}
-\sum_{n=1}^N J \sigma_n^z \sigma_{n+1}^z = \\
\sqrt{J} H_A^{zz} \otimes \sqrt{J} \mathbb{1}_B + \sqrt{J} \mathbb{1}_A \otimes \sqrt{J} H_B^{zz} + \sqrt{J} \sigma_{N_A}^z \otimes \sqrt{J} \sigma_{N_A+1}^z + \sqrt{J} \sigma_1^z \otimes \sqrt{J} \sigma_N^z,
\end{aligned} \tag{2.31}$$

where $H_A^{zz} = -\sum_{l=1}^{N_A-1} \sigma_l^z \sigma_{l+1}^z$ represents the pairwise terms contained only on the first N_A qubits, $H_B^{zz} = -\sum_{m=N_A+1}^{N-1} \sigma_m^z \sigma_{m+1}^z$ represents the pairwise terms contained only on the remaining N_B qubits, and the last two terms above represent the two terms which act non-trivially on qubits $N_A, N_A + 1$ and on qubits $1, N$. Also, we have

$$-\sum_{n=1}^N g \sigma_n^z = \sqrt{g} H_A^z \otimes \sqrt{g} \mathbb{1}_B + \sqrt{g} \mathbb{1}_A \otimes \sqrt{g} H_B^z, \tag{2.32}$$

where $H_A^z = -\sum_{l=1}^{N_A} \sigma_l^z$ and $H_B^z = -\sum_{m=N_A+1}^N \sigma_m^z$ represent the single-qubit σ^z terms. Lastly,

$$-\sum_{n=1}^N h \sigma_n^x = \sqrt{h} H_A^x \otimes \sqrt{h} \mathbb{1}_B + \sqrt{h} \mathbb{1}_A \otimes \sqrt{h} H_B^x, \tag{2.33}$$

where $H_A^x = -\sum_{l=1}^{N_A} \sigma_l^x$ and $H_B^x = -\sum_{m=N_A+1}^N \sigma_m^x$ represent the single-qubit σ^x terms.

Varying the number of ansatz states

We now address the following question:

How does the number of ansatz states L and M affect the performance of our heuristic?

To answer this, we consider the 12-qubit Ising Hamiltonian (with $J = 1, g = 0$ and $h = 1.3$) and use the see-saw algorithm for special-case instances discussed in Section 2.1.1 to compute the lower bound $\hat{\alpha}$. We now compare $\hat{\alpha}$, which is our approximation to the actual separable ground energy, to the value attained if we were to use our dimension-reducing ansatz. We note that while we do not need to use dimension reduction here, it serves as a test bed to see how well we could expect it to perform on larger examples where we might not know the answer.

Suppose that these 12 qubits are split evenly between Alice and Bob and we have up to $2^6 = 64$ ansatz states each. To begin, we randomly choose a reference state $|\eta\rangle$ then generate $L \in \{1, \dots, 64\}$ ansatz states for each of the varying values of L . More precisely, we pick the first L and numerically verify that they are linearly independent so that the inverses of their Gram matrices are well-defined. Since the reference state is chosen randomly, we repeat this 10 times for each L . We compute the lower bound, denoted $\hat{\alpha}_{L,i}$, for the i th choice of reference state for the choice of L . We denote the mean of these 10 values as $\tilde{\alpha}_L$ and its maximum by $\hat{\alpha}_L$. Figure 2.2 describes the difference between $\hat{\alpha}$ and these values, for each value of L .

This gives an illustration of how many ansatz states one could choose to use. We observe that the error decreases to near 0 as the number of ansatz states increases. However, the computation gets more demanding as this number increases, so there is a trade-off. From

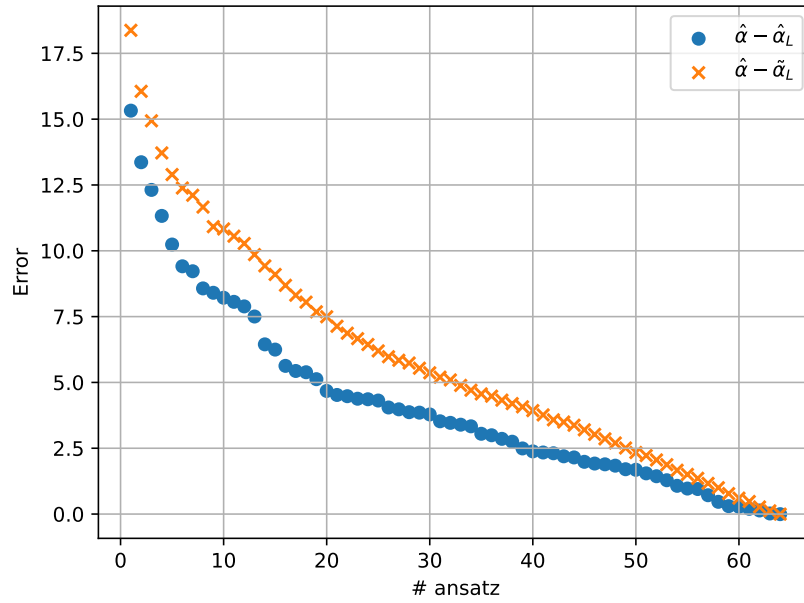


Figure 2.2: The error generated by our heuristic as the number of ansatz states is varied for the Ising Hamiltonian on 12 qubits (with $J = 1$, $g = 0$, and $h = 1.3$). Here, $\hat{\alpha}_L$ and $\tilde{\alpha}_L$ denote, respectively, the maximum and mean of the $\hat{\alpha}_{L,i}$ values computed using 10 randomly selected reference states.

this numerical experiment, it seems that the errors decrease at a slower rate starting around $L = 20$.

2.3.3 Calculating ground energies and separable ground energies on large Ising Hamiltonians

In this subsection, we consider the $2N$ -qubit Ising Hamiltonian with $J = 1$, $g = 0$, and $h = 1.4$ where $\mathcal{A}$ is the first $N_{\mathcal{A}}$ qubits and $\mathcal{B}$ is the remaining $N_{\mathcal{B}}$ qubits. Our computations here are split into two camps:

- First, we compute the separable ground energy approximation $\hat{\alpha}$, when possible, and also our heuristic approximation $\hat{\alpha}_L$ as discussed in the previous subsection, using the maximum number of ansatz states L that would fit into memory, computed over 10 randomly chosen reference states.
- Second, we compute the ground energy $\lambda_{\max}$ using an eigenvalue solver, when possible, and $\hat{\lambda}_{\max}$ using the approximate eigenvalue solver in [BHVK22].

The values are shown in Figure 2.3.

We note there are several points of interest in the above figure. First, at $N = 7$ we observe that there is a gap between $\hat{\lambda}_{\max}$ and $\lambda_{\max}$ noting that this is also a heuristic-based approach to computing eigenvalues. After this value of N , we stopped computing this heuristic as it became too expensive.

For $N = 9$, we can store an entire basis of ansatz states while for $N \geq 10$, we do not have sufficient memory to store enough ansatz states to choose a basis. Since we do not have a basis, we are solving a smaller dimensional optimization problem and therefore we see a small dip in performance like in Figure 2.2.

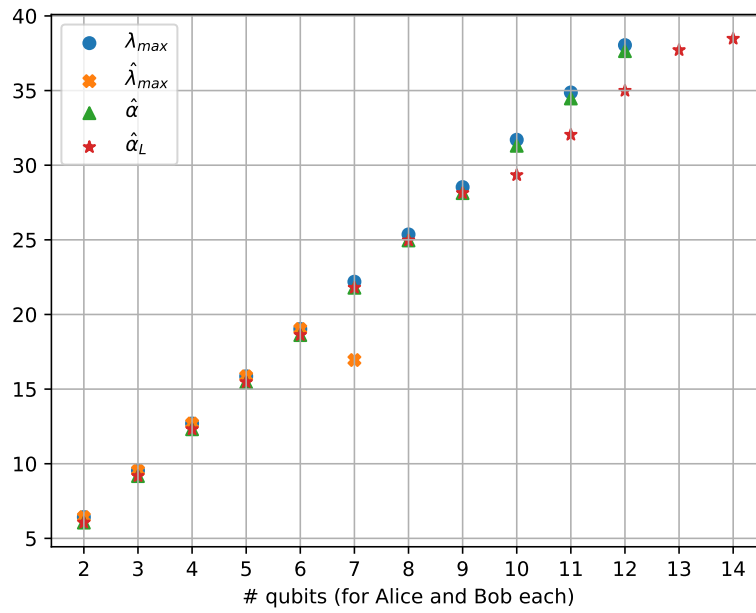


Figure 2.3: Comparison of calculated values (where available) and approximations of the ground energy and separable ground energy for the Ising Hamiltonian ($J = 1$, $g = 0$, and $h = 1.4$) for systems of up to 28 qubits.

When $N \geq 13$, the computation of our inner approximation $\hat{\alpha}$ (without ansatz) exceeds 24 hours and therefore had to be terminated. However, the approximation $\hat{\alpha}_L$ is still able to be calculated, and we do so for the largest possible L for each number of qubits. Also, $\lambda_{\max}$ can be computed up to $N = 12$, noting that this is a $2N$ -qubit Hamiltonian and thus this computation involves computing the eigenvalue of a $2^{24} \times 2^{24}$ matrix. Despite the sparsity of H_{Ising} , for values of $N \geq 13$, we are unable to compute the value of $\lambda_{\max}$ because the size of the Hamiltonian exceeds the available memory (32 GB, as described below).

At $N = 15$, the size of each of the K_m and L_m matrices exceeds the available memory.

Computational platform

All computations were performed on a 32 GB 10th Generation Intel Core i9-10885H CPU (16 MB cache, 2.40 GHz, 8 cores).

2.4 Related work for optimizing over separable states

An algorithm was proposed in [BCY11] that solves the optimization problem (2.4) in time

$$\exp(\mathcal{O}(\delta^{-2} \log(\dim \mathcal{A}) \log(\dim \mathcal{B}) \|C\|_F^2)), \quad (2.34)$$

for additive error δ which is based on *symmetric extensions* introduced in [DPS02]. Another approach that combines KKT conditions with symmetric extensions was studied in [HNW17]. In [SW15], the authors propose a quasi-polynomial time algorithm which takes as input the Hermitian matrix C and its decomposition $C = \sum_{i=1}^M C_i^1 \otimes C_i^2$ for small M . Then optimizing over product states $\langle \rho \otimes \sigma, C \rangle$ becomes $\sum_{i=1}^M \langle \rho, C_i^1 \rangle \langle \sigma, C_i^2 \rangle$. By modifying the

construction of ε -nets [SW15, Definition 1] using the multiplicative matrix weights (MMW) update method [AHK12], we can enumerate all possible values of $\langle C_i^1, \rho \rangle$ for all values of i . Thus the problem can be solved efficiently when M is small. Moreover, when C is positive semidefinite, they propose another algorithm which combines this ε -net approach with Schmidt decompositions to approximate the optimal value of problem (2.4) in time

$$\exp(\mathcal{O}(\log(d) + \delta^{-2} \|C\|_F^2 \ln(\|C\|_F/\delta))) \quad (2.35)$$

where $d = \dim(\mathcal{A}) = \dim(\mathcal{B})$. There are also several variational approaches to solving this problem [CAW22, PRRW24].

Comparisons to our algorithm. One issue that still arises in working with the hierarchy of symmetric extensions, or other popular outer approximations to Sep (such as PPT (Positive-Partial Transpose) operators), is the curse of dimensionality. The sizes of the variables in the semidefinite programs still scale exponentially, and this problem is only exacerbated at higher levels. While these algorithms are great in general, their size is still an issue, especially with current numerical solvers. We do note that these algorithms, as well as others, can be combined with our dimension-reducing quantum heuristics though.

Our approach has some similarities to [SW15]. We look at general C matrices as well as those that can be decomposed as a sum of tensor products. While their approach uses MMW, we use dimension-reduction techniques to get the size of the problem smaller, then generalized eigensolvers, see Section 2.1. Thus, the approaches are complimentary in this setting. We do note that since our approach is heuristics-based, it can be used in general settings, sometimes even when the size of C is exponentially large, as discussed in Section 2.2.

Comparing to the previously-mentioned variational algorithms, our algorithm promises to

give feasible solutions whereas this is not always the case in variational algorithms. This could be an important feature, depending on the application. Unlike variational algorithms, our approach does not involve the optimization of parameters in any parametric quantum circuit, thereby circumventing the barren plateau problem [MBS⁺18] by design. Furthermore, the absence of a classical-quantum feedback loop, characteristic of Variational Quantum Algorithms, significantly reduces the overhead associated with quantum measurements.

2.4.1 Related work for optimizing over separable ground states of Hamiltonians

The mean-field method is a powerful class of approximations for complex interacting Hamiltonians, reducing them to effective one-body problems that are easier to analyze. A prominent example is the Hartree-Fock approach for fermionic Hamiltonians, which simplifies the interaction terms to one-body fermionic operators [Q⁺20]. The core idea is to replace the original Hamiltonian H with a simpler, mean-field Hamiltonian H_{MF} . By construction, the ground state of H_{MF} is a separable state (i.e., a product state of single-particle wave functions). A folklore result is that the computation of the ground-state energy of a mean-field Hamiltonian H is equivalent to solving a problem of the form (2.4). However, the mean-field ansatz is highly constrained. This assumed separability (often defined in a specific, potentially non-local basis corresponding to the particle wave functions) means that the ground state of the mean-field Hamiltonian frequently lacks important correlations present in the true ground state, potentially rendering it an inadequate approximation.

Comparisons to our algorithm. Our algorithm is distinct from the mean-field approach in a crucial way. The mean-field method first approximates the Hamiltonian to get H_{MF} and then finds its exact (separable) ground state. In contrast, our approach directly finds

an approximate separable ground state for the original Hamiltonian H . The separable states obtained by these two different methods are generally not the same. Our method's flexibility stems from an ansatz that can systematically incorporate higher-order terms, which are explicitly neglected in standard mean-field and Hartree-Fock approximations. In fact, our approach can be built directly on top of these methods. For instance, the Hartree-Fock ground state can serve as the initial reference state for our heuristic, which can then be improved by including higher-order Pauli terms that go beyond the mean-field limitations.

Chapter 3

A hybrid quantum-classical algorithm for Bayes-optimal quantum state discrimination using the source code

The contents of this chapter are based on work with Jamie Sikora and Sarvagya Upadhyay in [\[MSU26\]](#).

This chapter develops a unified framework for quantum state discrimination based on Helstrom’s general Bayes approach. Through a reward matrix that assigns a payoff to each possible measurement outcome conditioned on every possible state, this framework captures both standard discrimination problems and ordered or structured tasks such as changepoint identification.

The main technical contribution of the chapter is an exact semidefinite programming reduction for this general Bayes problem. The original SDP optimizes over L measurement operators acting on the full Hilbert space, so its variables scale with the state dimension

d , which becomes prohibitive for many-qubit or long-sequence instances. We show that the same optimal reward can instead be computed using an SDP whose variables are L positive semidefinite matrices of size $N \times N$, constrained to sum to the Gram matrix G of the candidate states. Thus, the optimization depends only on the pairwise inner products of the states, not on their ambient Hilbert-space representation. This converts the problem from one scaling with the physical dimension of the states to one scaling with the number of candidate hypotheses.

This Gram-matrix viewpoint also leads to a hybrid quantum–classical algorithm. When the candidate states are efficiently preparable, a quantum device can estimate the required inner products using overlap-estimation primitives such as the Hadamard test, or swap test. These estimates define the Gram matrix, which is then passed to a classical SDP solver to compute the optimal reward and an associated reduced measurement description. In this way, the quantum component supplies state-overlap information that may be inaccessible classically, while the classical component performs the optimization over the reduced SDP. The reduction is most useful for pure or low-rank states; for high-rank mixed states, the effective number of pure components can grow and reduce the computational advantage.

The chapter illustrates the framework through two main applications. The first is quantum changepoint identification, where each hypothesis corresponds to a different location, or set of locations, at which a quantum source changes from one state to another. The reward-matrix formulation allows one to move beyond exact changepoint recovery and study objectives such as the horseshoe reward, where guesses within a specified distance are fully rewarded, and the closer-the-better reward, where the payoff decays with distance from the true changepoint. For single-changepoint examples, the reduced SDP enables computations for sequences far beyond what the original Hilbert-space SDP could handle, including instances with tens to hundreds of qubits. The chapter also introduces a Toeplitz-structured dual heuristic for

large changepoint instances, which empirically gives close approximations while significantly reducing runtime in the tested regimes.

The second application is quantum error type classification. Here an n -qubit state is sent through a single-qubit Pauli error channel, and the goal is not to identify which qubit was affected, but only which type of error occurred: I , X , Z , or ZX . This is naturally expressed as a classification problem within the reward framework. For a family of states, the chapter derives an analytic solution for $n = 2$ and then uses the reduced SDP numerically for larger n . These computations would be infeasible in the original formulation, since the full SDP variables would scale as $2^n \times 2^n$, but become accessible through the Gram-matrix reduction when the overlaps have exploitable structure.

This chapter is organized as follows:

- Section 3.1: We introduce the quantum state discrimination problem and some of its variants.
- Section 3.2: We discuss the general Bayes approach for the state discrimination problem and show how these variants are subsumed by the framework. This is also shown to extend to the case of mixed states by considering a classification problem.
- Section 3.3: We describe our semidefinite programming reduction and the hybrid algorithm for computing the optimal reward.
- Section 3.4: We consider two problems that can be formulated as instances of the general Bayes approach and apply our reduced SDPs. We first look at the effect of reward functions for the quantum changepoint identification problem, introduce a heuristic and evaluate its performance as the number of states is increased. Next, we view a quantum error classification task and illustrate the performance of our reduced SDP for this problem.

- Appendix ??: We introduce a general version of the quantum changepoint identification problem that considers at most p changepoints and show how this can be expressed using the framework for the state discrimination problem and discuss heuristics for speedups. We consider the case where two or more changepoints can occur and illustrate through numerical experiments the performance of our heuristic approach for some candidate set of states.

3.1 Quantum state discrimination

Formally, suppose Alice selects the state ρ_i from the set of states $\{\rho_1, \dots, \rho_N\}$ with corresponding a priori probabilities $\{p_1, \dots, p_N\}$. For Bob to make a guess, he must construct measurement operators $\{M_1, \dots, M_L\}$, such that a detection event on the i -th operator, M_i , corresponds to Bob guessing i . Note that the number of these operators, L , depends on the figure of merit adopted. By Born's rule, the probability of a detection event on the i -th operator upon receiving the state ρ_j is given by

$$\Pr[i|j] = \langle M_i, \rho_j \rangle. \quad (3.1)$$

We now discuss several variants, each of which correspond to how we want i and j to be related.

Minimum error discrimination

In this variant, Bob wishes to guess which state was sent. That is, upon observing a detection outcome on the i -th operator, Bob concludes that he received the state ρ_i . Bob's goal is to maximize the correlation between i and j in this case. The optimal guessing probability can

be computed as the optimal value of the optimization problem below

$$\mathcal{P}_{\text{Best}} = \text{maximize: } \left\{ \sum_{i=1}^N p_i \langle M_i, \rho_i \rangle \right\}. \quad (3.2)$$

Distinguishing between two states is well-studied and the solution is given by the *Helstrom measurement* [Hel69]. When dealing with more than two states, a solution is known for geometrically uniform states [EMV04], for mirror-symmetric states [ABGH02], and for symmetric states generated by a specific class of unitary matrices [Bar01, CDPS04, CDPS06]. Also, [Bae13] proposes a general approach by exploiting the geometry of the states. However, for most cases, a closed-form solution is unknown and one must resort to numerical solvers [CVX12] to find the optimal measurements.

Unambiguous discrimination

This variant is similar to the minimum error case, except we further restrict Bob to *never make a mistake*. In other words, Bob only makes a guess when he is certain about his received state, and in all other cases he returns an inconclusive outcome corresponding to “I do not know!”. This can be achieved through constraining the measurement operators $\{M_1, \dots, M_N, M_{N+1}\}$ to satisfy $\Pr[i|j] = \langle M_i, \rho_j \rangle = 0$, for all $i \neq j$. Here, we have $L = N + 1$ with M_{N+1} corresponding to the inconclusive outcome. The guessing probability in this setting is equal to the optimal value of the following optimization problem

$$\text{maximize: } \left\{ \sum_{i=1}^N q_i \langle M_i, \rho_i \rangle : \langle M_i, \rho_j \rangle = 0, \text{ for all } i \neq j \right\}. \quad (3.3)$$

The unambiguous case is often restricted to the cases where the states are either linearly independent for pure states [Che98], or when their supports do not completely overlap for

mixed states [RST03]. Similar to the minimum error case, discriminating between two states unambiguously has a closed form solution given by Jaeger and Shimony [JS95].

Generalizations of these strategies

From the discussion so far, we can express the probability of correctly identifying the states, the error probability, and the probability of obtaining an inconclusive outcome below

$$P_D := \sum_{i=1}^N p_i \langle M_i, \rho_i \rangle, \quad P_E := \sum_{i,j:i \neq j}^N p_j \langle M_i, \rho_j \rangle, \quad \text{and} \quad P_I := \sum_{j=1}^N p_j \langle M_{N+1}, \rho_j \rangle, \quad (3.4)$$

respectively. In unambiguous discrimination, the operators must minimize P_I , equivalently maximize P_D , subject to $P_E = 0$. In these cases, it may be prudent to relax this constraint by setting $P_E \leq \epsilon$ for some $\epsilon \in (0, 1)$. Introduction of such error is useful in quantum key distribution, for example, where the amount of information that can be learned by an eavesdropper can be decreased through this error [TKI03]. Some constraints were established in [TAS07] on the optimal solution for the unambiguous case with fixed error probability. A similar idea for minimum-error discrimination was investigated in [Eld03a].

Quantum state exclusion

Consider the same game where Alice picks a state and sends it to Bob, but instead of Bob identifying what state he received, here the task is to *exclude* a state he did *not* receive. For instance, if Bob receives the state ρ_1 and he replies “I did not receive ρ_2 ”, this corresponds to a winning scenario. Here, Bob wishes to construct measurements $\{M_1, \dots, M_N\}$ that minimizes the detection outcome M_j upon receiving the state ρ_j . The error probability for minimum-error state exclusion can be computed as the optimal value of the following

optimization problem

$$\mathcal{P}_{\text{Worst}} = \text{minimize: } \left\{ \sum_{i=1}^N p_i \langle M_i, \rho_i \rangle \right\}. \quad (3.5)$$

This problem has been studied in detail in [BJOP14, UBK⁺20, DS20]. A set of quantum states is said to be *antidistinguishable* if there exists a POVM that can perfectly exclude the state Bob did not receive, i.e., the error above is 0. Necessary and sufficient conditions when a set is antidistinguishable are derived in [RS23], tight bounds are given in [JRS25], and optimal error exponents are discussed in [MNW24].

For the unambiguous variant of quantum state exclusion, Bob now wishes to construct measurements $\{M_1, \dots, M_N, M_{N+1}\}$ such that detection outcome M_j excludes with certainty the state ρ_j and the detection on M_{N+1} leads to a situation where Bob does not exclude any state. The goal is to minimize the probability of obtaining the inconclusive outcome, while ensuring that a detection on M_j does not exclude the state ρ_j . The optimal error probability of this scenario is given by the optimal value of the following problem

$$\text{minimize: } \left\{ \sum_{j=1}^N p_j \langle M_{N+1}, \rho_j \rangle : \langle M_j, \rho_j \rangle = 0 \text{ for all } j \right\}. \quad (3.6)$$

We now discuss a general framework to study quantum state discrimination problems.

3.2 The general Bayes approach

Recall the setting that Bob expects one of the states from the set $\{\rho_1, \dots, \rho_N\}$ with corresponding a priori probabilities $\{p_1, \dots, p_N\}$. Helstrom's Bayes approach [Hel69] considers measurements $\{M_1, \dots, M_L\}$ along with the parameters R_{ij} , where R_{ij} is Bob's re-

ward¹ if he outputs i , given by M_i , upon receiving the state ρ_j . The idea behind the reward is that it generalizes the aforementioned discrimination tasks while opening up the possibilities for new ones. The parameter R_{ij} is also referred to as the cost matrix [Hel69, Hol73, Hol78, BM96, Hel03, YKL03, NU12], loss function [WP18] or the payoff function [Wat18].

In general, we can assume that Bob wishes to maximize his reward function given as

$$\text{maximize: } \left\{ \sum_{i=1}^L \sum_{j=1}^N R_{ij} q_j \langle M_i, \rho_j \rangle \right\}. \quad (3.7)$$

We describe below how some discrimination tasks fit into this framework.

- **Minimum-error state discrimination:** This is the case when $L = N$, and

$$R_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}. \quad (3.8)$$

Notice here that Bob is given a unit reward only when he is able to correctly identify the state he received, and gets no reward when he makes an error. Applying this reward scheme to (3.7), we can recover (3.2).

- **Minimum-error state exclusion:** This is the case when $L = N$, and

$$R_{ij} = \begin{cases} 0 & \text{if } i = j \\ 1 & \text{otherwise} \end{cases}. \quad (3.9)$$

Substituting this reward scheme in (3.7) recovers (3.5).

¹The reward can also be negative which can be interpreted as a penalty.

- **Unambiguous state discrimination:** This is the case when $L = N + 1$, with $i = N + 1$ indicating an inconclusive measurement, and

$$R_{ij} = \begin{cases} 1 & \text{if } i = j \\ -\infty & \text{if } i \neq j \\ 0 & \text{if } i = N + 1 \end{cases} . \quad (3.10)$$

Bob is awarded a unit reward when he correctly identifies his received state and is granted a penalty of $-\infty$ for all erroneous situations to ensure that such errors are not committed. Also, he receives no reward (or penalty) for the inconclusive outcome $N + 1$. Using this reward scheme in (3.7) gives us (3.3).

A variant of this scheme is

$$R_{ij} = \begin{cases} 1 & \text{if } i = j \\ -\beta & \text{if } i \neq j \\ 0 & \text{if } i = N + 1 \end{cases} \quad (3.11)$$

for some $\beta > 0$. This allows a penalty for an incorrect guess, but is not completely forbidden.

- **Unambiguous state exclusion:** This is the case when

$$R_{ij} = \begin{cases} -\infty & \text{if } i = j \\ 0 & \text{if } i \neq j \\ -1 & \text{if } i = N + 1 \end{cases} . \quad (3.12)$$

Here Bob is not allowed to make any errors, therefore he receives a penalty of $-\infty$

whenever he excludes his received state.

We also consider other reward functions relevant to a wide range of quantum problems. In many settings, an incorrect guess may still be preferred over other incorrect guesses. An important example of this perspective arises in anomaly detection [LSMT24] and changepoint detection [SBC⁺16, SCMT17].

- **The horseshoe reward:** This is the same as the minimum error case, but one does not need to be exact to be correct, *just close enough*. By setting μ to be the *closeness parameter*, a non-negative integer, and $L = N$, define

$$R_{ij} = \begin{cases} 1 & \text{if } |i - j| \leq \mu \\ 0 & \text{otherwise} \end{cases}. \quad (3.13)$$

This gives a unit reward if one is close enough, and nothing otherwise. By setting $\mu = 0$, we recover minimum error discrimination, (3.2).

- **The closer-the-better reward:** This is when one does not need to be exact to be correct, *but closer is better*. By setting $\gamma \in [0, 1]$ to be some fixed reward, and $L = N$, define

$$R_{ij} = \gamma^{|i-j|}. \quad (3.14)$$

This gives a greater reward the closer one is to correctly identifying the state. By setting $\gamma = 0$, we recover minimum error discrimination, (3.2). By setting $\gamma = 1$, we get the every-guess-is-a-winner reward.

- **The high school exam reward:** This is when one is rewarded partial marks for

saying “I do not know”. Here $L = N + 1$, and

$$R_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \\ 0.25 & \text{if } i = N + 1 \end{cases} . \quad (3.15)$$

This deters from random guessing and awards partial credit to an inconclusive guess. This variant is studied in [CF15] which they call the $0 - 1 - \lambda$ reward where $1 - \lambda$ is the reward that Bob receives for saying “I do not know”. They examine how the measurement changes as the value of λ is varied.

3.2.1 Classification and mixed states

This approach also allows one to capture a large class of discrimination problems known as *classification*. In this task, each state belongs to a subset of the classes $\{C_1, \dots, C_L\}$, and Bob does not need to identify exactly which state was given, he just needs to identify a class that it belongs to. For example, if Bob is given a picture of an animal, a sufficient guess could be “cat” as opposed to identifying exactly which cat.

For this task, we can define

$$R_{ij} = \begin{cases} 1 & \text{if } \rho_j \in C_i \\ 0 & \text{otherwise} \end{cases} . \quad (3.16)$$

(Here, i runs over the number of classes.) This means that Bob gets a unit reward if he correctly guesses a class that ρ_j belongs to, i.e., he correctly classifies the given state.

Mixed states

While many state discrimination results apply to the pure state case, we now show how this framework can be applied to mixed states as well. Suppose that the spectral decomposition of the state ρ_j is given by

$$\rho_j = \sum_{k_j=1}^{r_j} \lambda_{jk_j} |\psi_{jk_j}\rangle\langle\psi_{jk_j}|, \quad (3.17)$$

where r_j is the rank of the state ρ_j . Then the reward function in (3.7) can be expressed as

$$\begin{aligned} \sum_{i=1}^L \sum_{j=1}^N R_{ij} p_j \langle M_i, \rho_j \rangle &= \sum_{i=1}^L \sum_{j=1}^N R_{ij} p_j \left\langle M_i, \sum_{k_j=1}^{r_j} \lambda_{jk_j} |\psi_{jk_j}\rangle\langle\psi_{jk_j}| \right\rangle \\ &= \sum_{i=1}^L \sum_{j=1}^N \sum_{k_j=1}^{r_j} R_{ij} p_j \lambda_{jk_j} \langle M_i, |\psi_{jk_j}\rangle\langle\psi_{jk_j}| \rangle \\ &= \sum_{i=1}^L \sum_{j=1}^N \sum_{k_j=1}^{r_j} R_{ij} \tilde{p}_{jk_j} \langle M_i, |\psi_{jk_j}\rangle\langle\psi_{jk_j}| \rangle \end{aligned} \quad (3.18)$$

where $\tilde{p}_{jk_j} = p_j \lambda_{jk_j}$ denotes the apriori probability of Bob receiving the state $|\psi_{jk_j}\rangle$.

Note that the number of pure states is no longer N , but rather r where $r = \sum_{j=1}^N r_j$ is the sum of the ranks. Therefore, if all of the states have low rank then we still have a small (pure state) instance of the problem. Thus, by defining the rewards $R_{ijk_j} = R_{ij}$, we can model the general setting for mixed states using the pure state version.

As a concrete example, the case of minimum error discrimination of mixed states can be defined using the rewards

$$R_{ijk_j} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}. \quad (3.19)$$

Remark 3.1. Although our results are presented for pure states, the preceding discussion illustrates how this framework can handle the mixed state case as well.

Remark 3.2. We emphasize that the advantage of our reduction is most pronounced when the states are pure or low-rank. When the ranks are large, the effective number of pure states can become large, and the computational savings diminish.

3.3 Technical results

In this section, we describe our semidefinite programming reduction and the hybrid algorithm for computing the optimal reward. In Section 3.4 we show applications and numerical simulations.

3.3.1 Semidefinite programming formulations and reductions

Given a set of rewards R_{ij} , we can model the optimization of the optimal reward as the optimal objective function value of the following semidefinite program.

$$\alpha = \text{maximize: } \left\{ \sum_{i=1}^L \sum_{j=1}^N R_{ij} p_j \langle M_i, |\psi_j\rangle\langle\psi_j| \rangle \right\} \quad (3.20)$$

We now investigate how hard it is to compute α and to find optimal measurements.

We remark that the quantity given in (3.20) is the optimization of a linear function over affine constraints $\sum_{i=1}^L M_i = \mathbb{1}$ and each variable is positive semidefinite. Thus, it can be written as an SDP. However, this computation involves finding L operators each of size $d \times d$ which is the size of each of the states. Thus, if the states involve many qubits, this is an intractable problem.

To reduce the size of the SDP problem, we investigate the following equivalent SDP, below

$$\alpha' = \text{maximize: } \left\{ \sum_{i=1}^L \sum_{j=1}^N R_{ij} p_j \langle j | W_i | j \rangle : \sum_{i=1}^L W_i = G, W_i \succcurlyeq 0 \right\} \quad (3.21)$$

where G is the Gram matrix of the states $\{|\psi_i\rangle\}$, i.e.,

$$G = \sum_{i,j=1}^N \langle \psi_i | \psi_j \rangle |i\rangle\langle j|. \quad (3.22)$$

Theorem 3.3 shows this equivalence. This SDP has the advantage of (typically) being much smaller in size compared to the one given by (3.20) and in general much easier to solve. Indeed, they involve L SDP variables each of size N now. Their utility is given by the following theorem.

Theorem 3.3. *Given N fixed pure states $\{|\psi_1\rangle, \dots, |\psi_N\rangle\}$ and a priori probabilities $\{p_1, \dots, p_N\}$, the general Bayes approach given in (3.20) can be calculated by (3.21). More precisely, $\alpha = \alpha'$.*

Proof. The optimal reward value of the general Bayes approach, given in (3.20) can be formally described as the primal-dual pair, below

$$\begin{aligned} \alpha = \max \quad & \sum_{i=1}^L \sum_{j=1}^N R_{ij} p_j \langle M_i, \rho_j \rangle \\ \text{s.t.} \quad & \sum_{i=1}^L M_i = \mathbb{1} \\ & M_i \succcurlyeq 0, i \in [L], \end{aligned} \quad \begin{aligned} \beta = \min \quad & \text{Tr}(Y) \\ \text{s.t.} \quad & Y \succcurlyeq \sum_{j=1}^N R_{ij} q_j \rho_j, i \in [L]. \end{aligned} \quad (3.23)$$

Note that $\alpha = \beta$ from strong duality when R_{ij} is finite.

We consider the case when the states are pure, i.e., $\{|\psi_1\rangle, \dots, |\psi_N\rangle\}$, with corresponding a

priori probabilities $\{p_1, \dots, p_N\}$. This gives us the SDPs

$$\begin{aligned}
\alpha = \max \quad & \sum_{i=1}^L \sum_{j=1}^N R_{ij} q_j \langle M_i, |\psi_j\rangle \langle \psi_j| \rangle \beta = \min \quad \text{Tr}(Y) \\
\text{s.t.} \quad & \sum_{i=1}^L M_i = \mathbb{1} \quad \text{s.t. } Y \succcurlyeq \sum_{j=1}^N R_{ij} q_j |\psi_j\rangle \langle \psi_j|, \quad i \in [L]. \\
& M_i \succcurlyeq 0, \quad i \in [L],
\end{aligned} \tag{3.24}$$

We now suppose a particular structure on the variables in the primal above. We show that there is no loss of generality in considering this structure. To this end, define the following matrix

$$\Psi := \sum_{k=1}^N |\psi_k\rangle \langle k| \tag{3.25}$$

and note that $\Psi^* \Psi = G$, the Gram matrix of the set of states $\{|\psi_1\rangle, \dots, |\psi_N\rangle\}$. For a primal feasible solution $\{M_1, \dots, M_L\}$, define $W_i := \Psi^* M_i \Psi$ for $i \in \{1, \dots, L\}$, since $M_i \succcurlyeq 0$ for all $i \in [L]$ and $\sum_{i=1}^L M_i = \mathbb{1}$, we have that

$$W_i \succcurlyeq 0, \text{ for all } i \in [L], \text{ and } \sum_{i=1}^L W_i = G. \tag{3.26}$$

Also, we have $|\psi_j\rangle = \Psi |j\rangle$, so we have

$$\langle M_i, |\psi_j\rangle \langle \psi_j| \rangle = \langle \psi_j | M_i | \psi_j \rangle = \langle j | \Psi^* M_i \Psi | j \rangle = \langle j | W_i | j \rangle. \tag{3.27}$$

Thus, under this assumption, we have the following primal-dual pair of SDPs

$$\begin{aligned}
 \alpha' &= \max \sum_{i=1}^L \sum_{j=1}^N R_{ij} q_j \langle j | W_i | j \rangle & \beta' &= \min \langle X, G \rangle \\
 \text{s.t. } \sum_{i=1}^L W_i &= G & \text{s.t. } X &\succcurlyeq \sum_{j=1}^N R_{ij} q_j |j\rangle\langle j|, \quad i \in [L]. \\
 W_i &\succcurlyeq 0, \quad i \in [L].
 \end{aligned} \tag{3.28}$$

Since we can take a feasible $\{M_1, \dots, M_L\}$ and construct feasible $\{W_1, \dots, W_L\}$ with the same objective function value, we have that $\alpha \leq \alpha'$. By strong duality, one can check that $\alpha' = \beta'$. All that remains to be proved is that $\alpha \geq \alpha'$.

Suppose $\{W_1, \dots, W_L\}$ is a feasible solution of the primal SDP in (3.27). If $\Psi = \sum_{i=1}^r \sigma_i |\mu_i\rangle\langle \nu_i|$ is the singular value decomposition of Ψ where r denotes its rank, then $\Psi^+ = \sum_{i=1}^r \sigma_i^{-1} |\nu_i\rangle\langle \mu_i|$ is the Moore-Penrose generalized inverse of Ψ . Define $M_i = (\Psi^+)^* W_i (\Psi^+) + \frac{1}{L} (\mathbb{1} - \Psi\Psi^+)$ for each $i \in [L]$. Since $W_i \succcurlyeq 0$ and $\Psi\Psi^+ = \sum_{i=1}^r |\mu_i\rangle\langle \mu_i|$ is a projection, we have that $M_i \succcurlyeq 0$ for each $i \in [L]$. If we sum over all the elements $\{M_1, \dots, M_L\}$, we get

$$\begin{aligned}
 \sum_{i=1}^L M_i &= \sum_{i=1}^L \left((\Psi^+)^* W_i (\Psi^+) + \frac{1}{L} (\mathbb{1} - \Psi\Psi^+) \right) \\
 &= (\Psi^+)^* \left(\sum_{i=1}^L W_i \right) \Psi^+ + \left(\sum_{i=1}^L \frac{1}{L} \right) (\mathbb{1} - \Psi\Psi^+) = \mathbb{1}.
 \end{aligned} \tag{3.29}$$

The last equality is obtained from noting that $\sum_{i=1}^L W_i = G = \Psi^* \Psi$. For each value of i , we

have

$$\langle \psi_j | M_i | \psi_j \rangle = \langle j | \Psi^* M_i \Psi | j \rangle = \langle j | \Psi^* \left((\Psi^+)^* W_i (\Psi^+) + \frac{1}{L} (\mathbb{1} - \Psi \Psi^+) \right) \Psi | j \rangle = \langle j | W_i | j \rangle. \quad (3.30)$$

The last equality holds because $\sum_{i=1}^L W_i = G = \Psi^* \Psi$, implying that

$$\text{Range}(W_i) \subseteq \text{Range}(G) = \text{Range}(\Psi^*) \quad (3.31)$$

and $\Psi^+ \Psi$ is the projection onto the range of Ψ^* (and thus leaves W_i unchanged). Therefore, we get $\alpha \geq \alpha'$.

□

Note that this proof is similar in structure to a reduction in [JRS25]. In that work, the focus was on quantum state exclusion. The proof above is the generalization to the general Bayes approach.

While restricting to the support subspace is well known², this reduction is the explicit reformulation as an SDP over $N \times N$ matrices parameterized solely by the Gram matrix G . This reformulation: (i) decouples the SDP from any Hilbert-space representation, (ii) enables direct construction from experimentally accessible inner products via the hybrid algorithm of Strategy 1.3, and (iii) yields a concrete computational pipeline that can scale with the number of states in the problem, which is illustrated in Section 3.4.

Related works that focus on Gram matrices. Several works exploit the fact that the discrimination properties for a set of states are captured by the Gram matrix *for fixed applica-*

²The minimal reducing subspace technique [Hol82, Proposition 1] is an example of such an approach.

tions. For instance, [SBC⁺16, SCMT17, SMVMT18] use this to prove bounds on the success probability for the changepoint problem, [LSMT24] draws on this for the multi-anomaly detection problem, and [SLH⁺24] utilizes this for error position identification. In [DPP15], the authors characterize the optimal success probability of discriminating a set of linearly independent pure states in the minimum error regime as a function of the Gram matrix. For discriminating between linearly independent pure states with an intrinsic ordering, [VMT19] obtains a bound for the success probability when using a combination of the horseshoe reward (Eq. (3.15)) and unambiguous discrimination. In [JRS25, JLRS25], analyzing the task of *k*-learnability makes significant use of a reduction of the problem to properties of the Gram matrix. These works demonstrate that not only is the reduction to a Gram matrix attractive from a computational perspective, but also from an analytical perspective. Our reduction in this work takes the above approaches and unifies them for general, application-agnostic state discrimination tasks. Perhaps surprisingly, this has not been yet been considered and we suspect this will be helpful for many other future applications.

3.3.2 A hybrid algorithm for calculating the optimal reward

As mentioned above, if we had access to the inner products, we would be able to solve for the optimal reward function values (for any reward of our choosing). However, computing inner products of exponentially large vectors is expensive. Therefore, we now discuss means of doing this on a quantum computer. After all, the quantities involve a property of physical quantum states.

Since the states $\{|\psi_i\rangle\}$ are prepared and sent to Bob, we make the assumption that they are efficiently preparable. Given this, Bob can perform some pre-computations on them before the discrimination game starts. For instance, Bob can perform the Hadamard test to learn

$\text{Re}(\langle\psi_i|\psi_j\rangle)$ and/or $\text{Im}(\langle\psi_i|\psi_j\rangle)$ for all values of i and j (see Section 1.1.4), assuming he has access to a unitary U_{ij} that can map the state $|\psi_i\rangle$ to $|\psi_j\rangle$ for all values of i and j .

We also note that if Bob is given access to unitaries that prepare each of the states $|\psi_i\rangle$, then a block-encoding of the Gram matrix can be efficiently implemented (see Lemma 47 of [GSLW19]).

For completeness, we mention that in the case where it is hard to implement the controlled unitary required for the Hadamard test, if the unitary U can be decomposed into either the (1) sum of Pauli products with a number of terms that is polynomial in the number of qubits n , or (2) tensor product of unitaries U_q where each U_q acts on at most $\mathcal{O}(\text{poly}(\log n))$ qubits, then the term $\langle\psi_i|\psi_j\rangle$ can be estimated using a direct measurement method [MF19]. Moreover, if each of the states $\{|\psi_i\rangle\}$ can be prepared as $P_i|\psi\rangle$ using an initial state $|\psi\rangle$ and Pauli string $P_i \in \{\mathbb{1}, X, Y, Z\}^{\otimes t}$ for some t , then note that $\langle\psi_i|\psi_j\rangle = \langle\psi|P_iP_j|\psi\rangle = a\langle\psi|P'|\psi\rangle$ where P' is some Pauli string and $a \in \{+1, -1, +i, -i\}$. Then calculating the inner product amounts to computing the expectation value of the Pauli string P' [BH21].

In the special case where the inner products are all non-negative, we can use the standard swap test to compute these values. Given two states $|\psi\rangle$ and $|\phi\rangle$, the swap test has two outcomes, one occurring with probability $\frac{1}{2} + \frac{1}{2}|\langle\psi|\phi\rangle|^2$ and the other occurring with probability $\frac{1}{2} - \frac{1}{2}|\langle\psi|\phi\rangle|^2$ (we discuss this more in Section 1.1.4).

Figure 3.1 depicts this hybrid algorithm.

Strategy 3.4. *Suppose we are given N pure states $\{|\psi_1\rangle, \dots, |\psi_N\rangle\}$ and a priori probabilities $\{p_1, \dots, p_N\}$, and suppose further that we are given efficient preparation unitaries that prepare each of the states. Then we can approximate the optimal reward function using a hybrid quantum-classical algorithm via the Hadamard test. If the quantum states are not known but the inner products are promised to be non-negative, the same holds only given access*

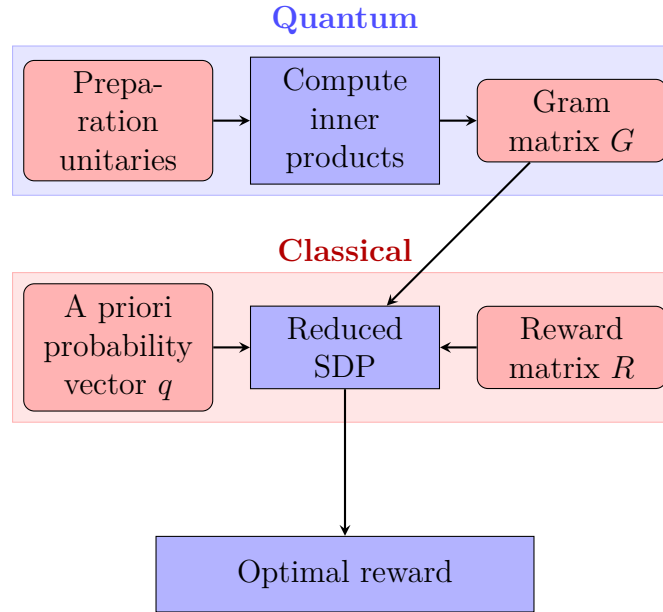


Figure 3.1: Flowchart of the hybrid quantum-classical algorithm. The quantum stage computes the Gram matrix from state-preparation unitaries. The classical stage supplies the Gram matrix, reward matrix, and prior-probability vector to a reduced semidefinite program, whose solution gives the optimal reward.

to preparation devices via the swap test. In each case, the classical part of the algorithm involves solving SDPs with L variables each of size $N \times N$.

We remark that in this work we do not examine estimation errors (say, in computing the inner products) and how sensitive the SDPs are to these errors. As far as we are aware, general sensitivity analyses are not known for SDP algorithms. As such, we leave this as an interesting open problem and rely simply on using machine-accurate estimations of such parameters.

3.4 Applications

We now discuss two problems that can be formulated as instances of the general Bayes approach and then apply our reduced SDPs followed by a discussion of numerical performance.

3.4.1 The quantum changepoint identification problem

Suppose that Alice sends to Bob a sequence of states $|\psi\rangle^{\otimes c_1} \otimes |\phi_1\rangle^{\otimes (c_2 - c_1)} \otimes |\phi_2\rangle^{\otimes (N - c_2)}$, one state at a time. We can think of $|\psi\rangle$ as the original state and $|\phi_1\rangle$ and $|\phi_2\rangle$ as mutated states. Perhaps Alice bought a cheap QKD device which started breaking down at time steps c_1 and $(c_2 - c_1)$. In this problem, Bob wishes to guess the index of the time steps, see Figure 3.2 for an illustration. This is a quantum version of a *changepoint problem* which is a widely studied field in statistical analysis with diverse applications, see the survey [AC17].

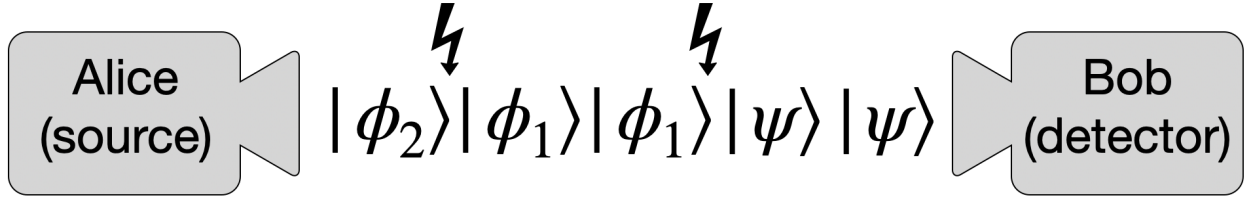


Figure 3.2: A quantum source sends a stream of states $|\psi\rangle$, with two mutations occurring in this example. The task is to design an optimal detector that determines when the mutations occurred.

Here we assume that both the promised state as well as the mutated ones are known to Bob. (We also discuss the case when they are unknown but we have a source of them to perform some pre-computations.) Observe that this problem can be formulated as an instance of the state discrimination problem. Consider the set of states consisting of every possible sequence. Then Bob's task is to construct measurements that can identify the changepoints, where each possible guess corresponds to a particular state in our set.

The changepoint problem *with one mutation* has been introduced and studied in [SBC⁺16] for the minimum error case and in [SCMT17] in the unambiguous case. In both works, they present the optimal success probabilities and analyze restricted measurements given by online algorithms. Achieving the bound in [SBC⁺16] requires a measurement performed jointly on all states in the sequence. Here we make the same assumption, that we have the availability of quantum memory capable of storing each of these particles sent by Alice, thereby enabling such joint measurements.

We refer the reader to Appendix ?? for the formulation of the general case.

3.4.2 At most one changepoint

Using the formulation in ??, when there is at most one changepoint Bob discriminates between the following states

$$\begin{aligned}
 |\tau_1\rangle &:= |\tau_{1NN\dots NN}\rangle = |\psi\rangle \otimes |\phi_1\rangle^{\otimes N-1} \\
 |\tau_2\rangle &:= |\tau_{2NN\dots NN}\rangle = |\psi\rangle^{\otimes 2} \otimes |\phi_1\rangle^{\otimes N-2} \\
 &\vdots \\
 |\tau_{N-1}\rangle &:= |\tau_{N-1,NN\dots NN}\rangle = |\psi\rangle^{\otimes N-1} \otimes |\phi_1\rangle \\
 |\tau_N\rangle &:= |\tau_{NNN\dots NN}\rangle = |\psi\rangle^{\otimes N}.
 \end{aligned} \tag{3.32}$$

Notice that the last state $|\tau_N\rangle$ indicates the sequence where no changepoint has occurred, i.e, Alice generated all the states as promised. All the remaining states correspond to the sequences where exactly one changepoint occurred. Therefore, Bob wishes to not only determine whether a changepoint occurred, but also the exact point of said occurrence.

Here we assume that the changepoint follows a uniform prior, i.e, each of the states

$|\tau_k\rangle, k \in \{1, \dots, N\}$, are equally likely.

Lemma 3.5. *Gram matrix G of the possible sequences generated by Alice when there is at most one changepoint is a Hermitian Toeplitz matrix.*

Proof. By direct computation, we have

$$G_{ij} = \begin{cases} \langle \phi_1 | \psi \rangle^{(j-i)} & i < j \\ \langle \psi | \phi_1 \rangle^{(i-j)} & \text{otherwise} \end{cases}$$

G is clearly a Hermitian Toeplitz matrix from the above. $\square$

Lemma 3.6. *When there is at most one changepoint, the Gram matrix G has the same reward as the Gram matrix $T^{(1)}$ where $(T^{(1)})_{ij} = |G_{ij}|$. Moreover, $T^{(1)}$ is a symmetric Toeplitz matrix.*

Proof. Define $|\phi'_1\rangle = e^{i\theta} |\phi_1\rangle$, where we choose $\theta \in \mathbb{R}$ such that $\langle \phi'_1 | \psi \rangle = |\langle \phi_1 | \psi \rangle|$. From (3.20), the value of the SDP clearly remains same if we replace any state with one that is equivalent up to a global phase. Define $|\tau'_i\rangle = |\psi\rangle^{\otimes i} \otimes |\phi'_1\rangle^{\otimes (N-i)}$, for all values of i . Then $|\tau_i\rangle$ and $|\tau'_i\rangle$ differ by a global phase, thus the SDPs have the same value. Let $T^{(1)}$ be the Gram matrix of $|\tau'_1\rangle, \dots, |\tau'_N\rangle$. Notice that $(T^{(1)})_{ij} = |\langle \psi | \phi_1 \rangle|^{|i-j|} = |G_{ij}|$. Finally, $T^{(1)}$ is symmetric Toeplitz, as desired. $\square$

The dual SDP in (??) can be expressed as

$$\begin{aligned} \beta'_{T1} = \min \quad & \langle X, T^{(1)} \rangle \\ \text{s.t.} \quad & X \succcurlyeq \frac{1}{N} \sum_{j=1}^N R_{ij} |j\rangle \langle j|, i \in \{1, \dots, N\} \end{aligned} \quad (3.33)$$

Consider the following reward scheme

$$(R_{1CP})_{ij} = \begin{cases} r_{|i-j|}, & i \in \{1, \dots, N\} \\ c, & i = N + 1 \end{cases} \quad (3.34)$$

for $j \in \{1, \dots, N\}$, where $r_0, \dots, r_{N-1}$ and c are real scalars. The motivation behind this strategy is as follows. When Bob is able to correctly identify the changepoint, he is awarded a reward of r_0 , regardless of when the changepoint occurred. Similarly, if the changepoint occurred at the j -th state, and Bob guessed the i -th state, irrespective of whether $i < j$ or $i > j$, he is given a reward of $r_{|i-j|}$. Notwithstanding the state Bob received, if he gives an inconclusive outcome, he is awarded a constant reward of c .

Accounting for all this, we can rewrite the dual SDP in (??) as follows

$$\begin{aligned} \beta'_{1CP} = \min \quad & \langle X_{1CP}, T_{1CP} \rangle \\ \text{s.t.} \quad & X_{1CP} \succcurlyeq \frac{1}{N} \sum_{j=1}^N r_{|i-j|} |j\rangle \langle j|, i \in \{1, \dots, N\} \\ & X_{1CP} \succcurlyeq \frac{c}{N} \mathbb{1}_N. \end{aligned} \quad (3.35)$$

A heuristic

Observe that $(R_{1CP})_{ij} = r_{|i-j|}$, $i, j \in \{1, \dots, N\}$ corresponds to a symmetric Toeplitz matrix. Since T_{1CP} is a symmetric Toeplitz matrix and R_{1CP} has a Toeplitz-like structure, we investigate a heuristic solution X'_{1CP} which is restricted to be symmetric Toeplitz. The incentive behind such a restriction is that a symmetric Toeplitz matrix can be fully constructed from its first row, and thus has fewer parameters. Then both T_{1CP} and X'_{1CP} can be expressed as

$$T_{1CP} = \sum_{k=-(N-1)}^{(N-1)} \gamma^{|k|} \Theta_k \quad X'_{1CP} = \sum_{k=-(N-1)}^{(N-1)} x_{|k|} \Theta_k, \quad (3.36)$$

where $\gamma = |\langle \psi | \phi_1 \rangle|$, and Θ_k is a $N \times N$ Toeplitz matrix with ones on the k -th diagonal and zeros elsewhere. Here, $k = 0$ indicates the principal diagonal, $k > 0$ denotes the sub-diagonals, and $k < 0$ corresponds to the super-diagonals.

We can express (3.35) with this heuristic as

$$\begin{aligned} \beta''_{1CP} = \min \quad & \langle x, p \rangle \\ \text{s.t.} \quad & \sum_{k=0}^{N-1} x_k \Theta_k \succcurlyeq \frac{1}{N} \sum_{j=1}^N r_{|i-j|} |j\rangle\langle j|, \quad i \in \{1, \dots, N\} \\ & \sum_{k=0}^{N-1} x_k \Theta_k \succcurlyeq \frac{c}{N} \mathbb{1}_N, \end{aligned} \quad (3.37)$$

where $p = (N, 2(N-1)\gamma, 2(N-2)\gamma^2, \dots, 2\gamma^{N-1})^\top$ and $x = (x_0, x_1, x_2, \dots, x_{N-1})^\top$.

Numerical experiment

We look at the specific case where the promised state is $|\psi\rangle = |0\rangle$ and the mutated state is $|\phi_1\rangle = |+\rangle$. Here we consider the following reward function

$$(R_{1CP})_{ij} = \begin{cases} \left(\frac{1}{\sqrt{2}}\right)^{|i-j|}, & i \in \{1, \dots, N\} \\ 0, & i = N + 1 \end{cases} \quad (3.38)$$

Figure 3.3 considers the solution of the reduced SDP (3.35) and the heuristic (3.37) for this case. We note the following two observations: (1) Our reduction (3.35) is the probability of both detecting as well as localizing a single changepoint for sequences of length 220. Without this reduction, solving (3.20) involves finding operators of size $2^{220} \times 2^{220}$. (2) Observe that solving our heuristic (3.37) is roughly *seven* times faster, with a gap of only about 10^{-3} .

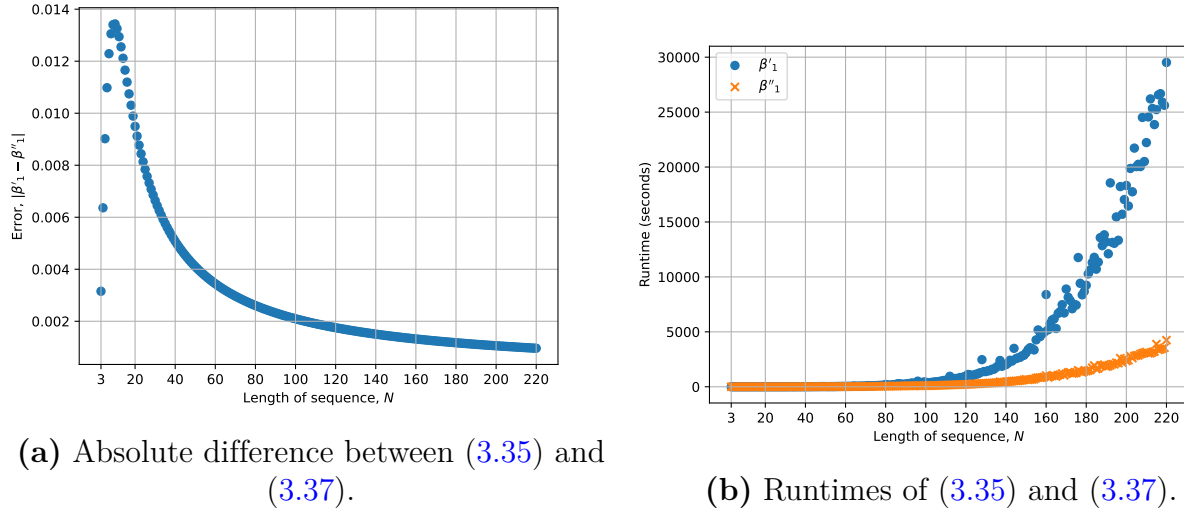


Figure 3.3: Here we consider sequences in which the state $|0\rangle$ mutates to the state $|+\rangle$. Using the closer-the-better reward scheme with $\gamma = 1/\sqrt{2}$, panel (a) shows the error between the heuristic (3.37) and the reduced SDP (3.35), while panel (b) shows the runtimes of both methods. The difference between (3.35) and (3.37) decreases rapidly, approaching 10^{-3} at $N = 220$. Although there is little timing advantage for sequences up to length 80, computing (3.37) is roughly *seven* times faster than computing (3.35) for larger values of N .

3.4.3 A single changepoint with varying reward functions

One changepoint, with horseshoe reward. In this example, we consider the horseshoe reward with varying closeness parameters μ . Recall the reward function is given as

$$R_{ij} = \begin{cases} 1 & \text{if } |i - j| \leq \mu \\ 0 & \text{otherwise} \end{cases}. \quad (3.39)$$

Figure 3.4 plots the optimal reward value for a single changepoint example.

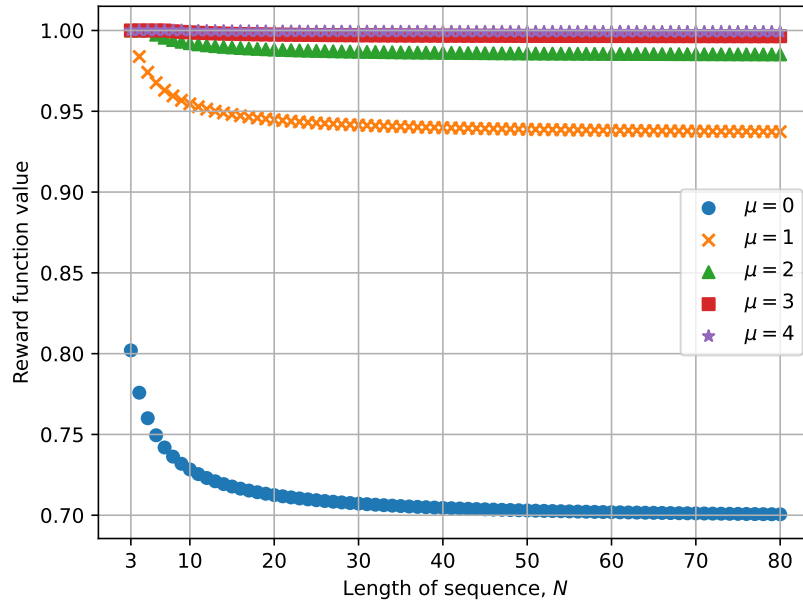


Figure 3.4: We consider a sequence of N states that begins in $|0\rangle$ and may switch to $|+\rangle$ at an unknown position. The optimal reward, corresponding to the optimal probability of identifying the location of the change point, is plotted as a function of N for different values of the closeness parameter μ . The reward stabilizes for large sequence lengths, and larger values of μ substantially improve the reward, as expected.

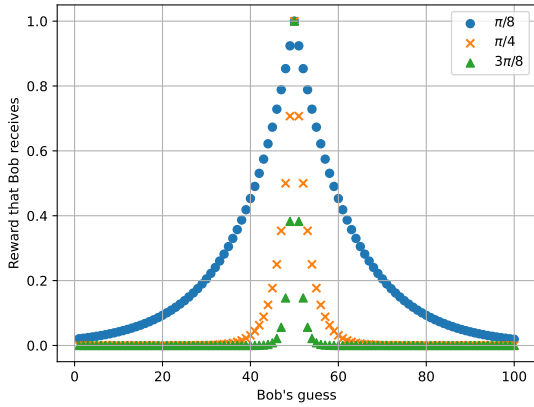
We note that in this figure, we were easily able to compute the reduced SDPs for this problem up to 80 qubits. Using the original SDP, this would be completely intractable as the variables

would each be of size $2^{80} \times 2^{80}$.

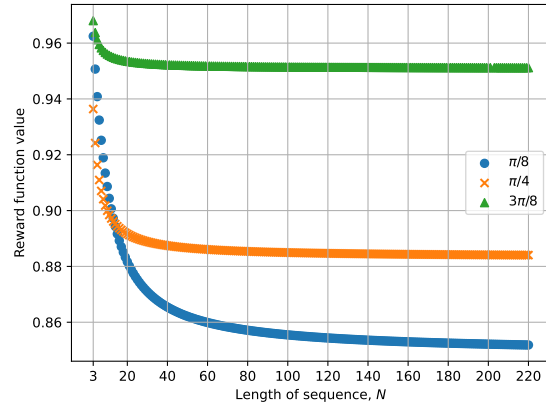
One changepoint, with the closer-the-better reward. In this example, we consider the closer-the-better reward with varying parameters γ . Recall the reward function is given as

$$R_{ij} = \gamma^{|i-j|}.$$

An example reward function is illustrated in Figure 3.5 (a) while Figure 3.5 (b) plots the optimal reward value for three particular examples.



(a) Reward that Bob receives for the sequence $|\psi\rangle^{\otimes 50} \otimes |\phi\rangle^{\otimes 50}$ under the closer-the-better reward scheme.



(b) Optimal reward function value as a function of N for varying values of θ .

Figure 3.5: We consider the closer-the-better reward scheme with $\gamma = |\langle \psi | \phi \rangle| = \cos(\theta)$ for $\theta \in \{\pi/8, \pi/4, 3\pi/8\}$, where $|\psi\rangle = |0\rangle$ and $|\phi\rangle = \cos(\theta)|0\rangle + \sin(\theta)|1\rangle$. Panel (a) shows the reward Bob receives for the sequence $|\psi\rangle^{\otimes 50} \otimes |\phi\rangle^{\otimes 50}$ under this reward scheme. Panel (b) shows the optimal reward function value as a function of N , the number of states sent. As the states become closer to one another, locating the change point becomes more difficult. However, the rewards are scaled so that more difficult settings correspond to larger rewards.

In this setting, we are able to easily compute the reward function for states up to 220 qubits.

3.4.4 Related work

For the minimum-error case, [SBC⁺16] showed that in the limit of the sequence length N , the probability of success is bounded by

$$P_D \leq \frac{4\sqrt{1-\gamma^2}}{\pi} K^2(\gamma^2) + 4 \left(\frac{1+\gamma}{1-\gamma} \right)^{3/2} \frac{1}{N^{1-\varepsilon}} \quad (3.40)$$

where $K(x)$ is the complete elliptic function of the first kind [Dar00], and $\varepsilon > 0$ is an arbitrary constant. For a sequence of length $N = 50$, Figure 3.6 compares the difference between the heuristic (3.37) and the optimal SDP (3.35) where $r_{|i-j|} = \delta_{ij}$ for all i, j , and $c = 0$), with the difference between Eq. (3.40) and the aforementioned optimal SDP, for varying values of γ^2 . Observe that while our heuristic outperforms the bound in Eq. (3.40), the heuristic requires us to solve an SDP while Eq. (3.40) provides a closed-form expression.

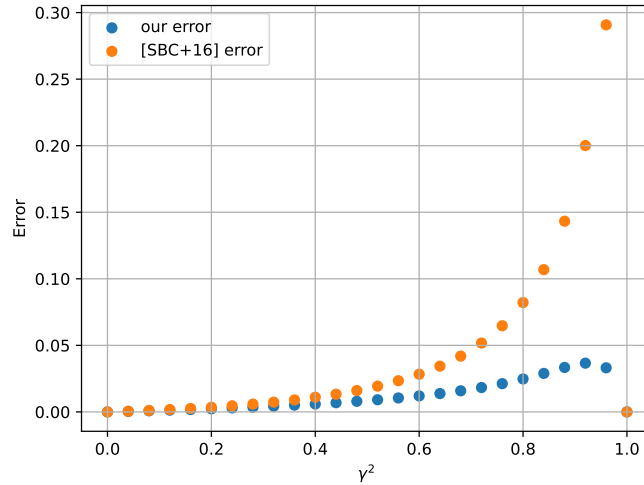


Figure 3.6: For a sequence of length $N = 50$, we compare the difference between the heuristic (3.37) and the optimal SDP with the difference between Eq. (3.40) and the optimal SDP, as the value of $\gamma^2 = |\langle \psi | \phi_1 \rangle|^2$ is varied. Although our heuristic provides a tighter bound than Eq. (3.40), it requires solving an SDP, whereas Eq. (3.40) is a closed-form expression.

Similarly for the unambiguous case, [SCMT17] found that the success probability P_u can be

approximated by the expression

$$P_u \approx \begin{cases} \frac{1-\gamma}{1+\gamma} + \frac{1}{N} \frac{2\gamma}{(1+\gamma)^2}, & 0 \leq \gamma \leq \gamma^* = (\sqrt{5} - 1)/2 \\ \frac{1-\gamma}{1+\gamma} + \frac{1}{N} \frac{2\gamma}{(1+\gamma)^2} - \frac{2}{N} \left(\frac{1-\gamma-\gamma^2}{1+\gamma} \right)^2, & \gamma^* < \gamma \leq 1. \end{cases} \quad (3.41)$$

For a sequence of length $N = 15$, Figure 3.7 compares the difference between the heuristic (3.37) and the optimal SDP (3.35) where $r_{|i-j|} = 1$ if $i = j$, $r_{|i-j|} = -M$ if $i \neq j$, and $c = 0$, for $M = 10^4$, with the difference between Eq. (3.41) and this optimal SDP, for varying values of γ . For $\gamma \leq \gamma^* = (\sqrt{5} - 1)/2$, the error between the heuristic and this optimal SDP is of the order 10^{-5} . When $\gamma > \gamma^*$, observe that Eq. (3.41) provides a much better bound compared to the heuristic.

Remark 3.7. We remark that our original SDP value is an approximation on unambiguous discrimination since we cannot numerically set $M = \infty$. Thus, Figure 3.7 compares the two heuristics compared to this SDP approximation. We see that in this case, both heuristics perform well.

We defer the discussion of two or more changepoints to Appendix ??.

3.4.5 Quantum error type classification

Suppose now that Alice sends Bob an n -qubit state $|\phi\rangle$ via a single-qubit Pauli error channel. We assume that any of the n qubits can be affected by one of these errors with equal probability. Bob's task is to simply find out which of these errors occurred, i.e., I, X, Z , or ZX (which equals Y up to global phase).

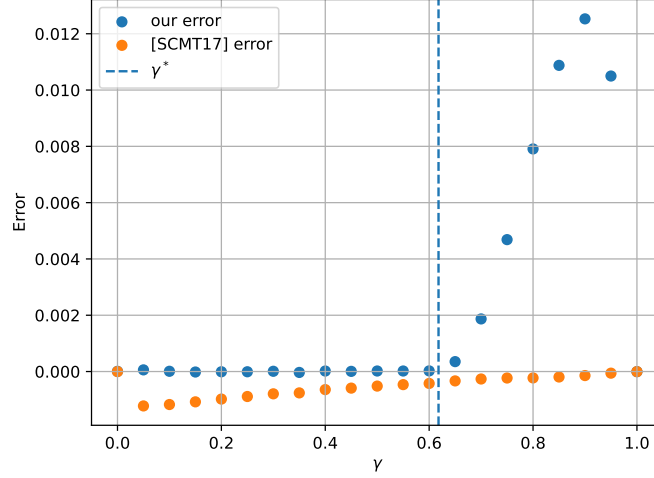


Figure 3.7: For a sequence of length $N = 15$, we compare the difference between the heuristic (3.37) and the optimal SDP with the difference between Eq. (3.40) and the optimal SDP for varying values of γ . For $\gamma \leq \gamma^* = (\sqrt{5} - 1)/2$, the error between the heuristic and the optimal SDP is of order 10^{-5} . When $\gamma > \gamma^*$, Eq. (3.40) provides a substantially tighter bound than the heuristic.

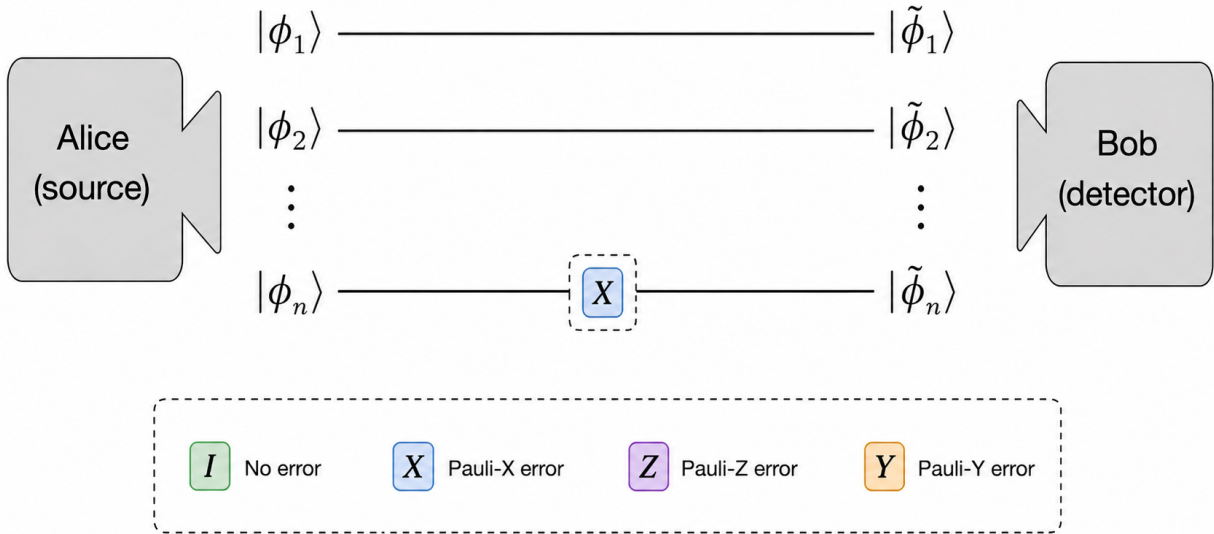


Figure 3.8: A quantum source sends an n -qubit state $|\phi\rangle$ through a single-qubit Pauli error channel. The task is to classify the type of error, I , X , Z , or Y , that affected the state without determining *which qubit* the error acted on.

For the n -qubit state, all of the $N = 3n + 1$ possibilities are

$$\mathcal{S} := \{|\phi\rangle, X_1|\phi\rangle, \dots, X_n|\phi\rangle, Z_1|\phi\rangle, \dots, Z_n|\phi\rangle, Z_1X_1|\phi\rangle, \dots, Z_nX_n|\phi\rangle\}. \quad (3.42)$$

Bob can formulate this as a problem of the form described in Eq. (3.18) with the reward given in Eq. (3.16), as follows. Define the set $\mathcal{I} = \{I, X, Z, ZX\}$ and let C_i for $i \in \mathcal{I}$ denote that Pauli- i error occurred. Let $R_{ij} = 1$ if $|\psi_j\rangle \in \mathcal{S}$ is correctly classified into C_i , and 0 otherwise. The a priori probability of the states are $q_1 = 1/4$ and $1/4n$ for each of the other states in $\mathcal{S}$. Thus, the optimal classification probability can be formulated as the following SDP

$$\max \left\{ \sum_{i \in \mathcal{I}} \sum_{j=1}^N R_{ij} q_j \langle M_i, |\psi_j\rangle\langle\psi_j| \rangle : \sum_{i \in \mathcal{I}} M_i = I, M_i \succcurlyeq 0 \right\}. \quad (3.43)$$

For our numerical experiments, we consider the state

$$|\psi_n(\theta)\rangle = \cos(\theta) |\text{GHZ}_n\rangle + \sin(\theta) |\text{W}_n\rangle, \quad (3.44)$$

where

$$|\text{GHZ}_n\rangle = \frac{|0\rangle^{\otimes n} + |1\rangle^{\otimes n}}{\sqrt{2}} \text{ and } |\text{W}_n\rangle = \frac{1}{\sqrt{n}}(|100 \dots 0\rangle + |010 \dots 0\rangle + |001 \dots 0\rangle + \dots + |000 \dots 1\rangle)$$

for $n \geq 2$.

For $n = 2$, the Bell states are

$$|\phi^\pm\rangle = (|00\rangle \pm |11\rangle)/\sqrt{2} \quad \text{and} \quad |\psi^\pm\rangle = (|01\rangle \pm |10\rangle)/\sqrt{2}.$$

We get $|\text{GHZ}_2\rangle = |\phi^+\rangle$ and $|\text{W}_2\rangle = |\psi^+\rangle$. Let us define the state

$$|\chi(\theta)^\pm\rangle = \cos(\theta) |\phi^\pm\rangle + \sin(\theta) |\psi^\pm\rangle,$$

$\Phi^\pm = |\phi^\pm\rangle\langle\phi^\pm|$, $\Psi^\pm = |\psi^\pm\rangle\langle\psi^\pm|$, and let $\bar{\theta} = \pi/2 - \theta$.

We get the following states

$$\begin{aligned} |\psi_2(\theta)\rangle &= |\chi(\theta)^+\rangle, X_1 |\psi_2(\theta)\rangle = X_2 |\psi_2(\theta)\rangle = |\chi(\bar{\theta})^+\rangle, \\ Z_1 |\psi_2(\theta)\rangle &= |\chi(\theta)^-\rangle, Z_2 |\psi_2(\theta)\rangle = |\chi(-\theta)^-\rangle, \\ Z_1 X_1 |\psi_2(\theta)\rangle &= |\chi(\bar{\theta})^-\rangle, Z_2 X_2 |\psi_2(\theta)\rangle = |\chi(-\bar{\theta})^-\rangle. \end{aligned}$$

Define

$$\begin{aligned} \rho_I &:= |\chi(\theta)^+\rangle\langle\chi(\theta)^+| = \cos^2(\theta) \Phi^+ + \sin^2(\theta) \Psi^+ + \frac{\sin(\theta)\cos(\theta)}{2} (\mathbb{1} \otimes X + X \otimes \mathbb{1}), \\ \rho_X &:= |\chi(\bar{\theta})^+\rangle\langle\chi(\bar{\theta})^+| = \sin^2(\theta) \Phi^+ + \cos^2(\theta) \Psi^+ + \frac{\sin(\theta)\cos(\theta)}{2} (\mathbb{1} \otimes X + X \otimes \mathbb{1}), \\ \rho_Z &:= \frac{1}{2} |\chi(\theta)^-\rangle\langle\chi(\theta)^-| + \frac{1}{2} |\chi(-\theta)^-\rangle\langle\chi(-\theta)^-| = \cos^2(\theta) \Phi^- + \sin^2(\theta) \Psi^-, \\ \rho_{ZX} &:= \frac{1}{2} |\chi(\bar{\theta})^-\rangle\langle\chi(\bar{\theta})^-| + \frac{1}{2} |\chi(-\bar{\theta})^-\rangle\langle\chi(-\bar{\theta})^-| = \sin^2(\theta) \Phi^- + \cos^2(\theta) \Psi^-. \end{aligned}$$

The problem can be expressed as

$$P_{\text{succ}} = \max \left\{ \frac{1}{4} \langle M_I, \rho_I \rangle + \frac{1}{4} \langle M_X, \rho_X \rangle + \frac{1}{4} \langle M_Z, \rho_Z \rangle + \frac{1}{4} \langle M_{ZX}, \rho_{ZX} \rangle \right\}. \quad (3.45)$$

Note that $\rho_I, \rho_X \in \text{span}(\Phi^+, \Psi^+)$ and $\rho_Z, \rho_{ZX} \in \text{span}(\Phi^-, \Psi^-)$. Thus, we can write Eq. (3.45)

as $P_{\text{succ}} = P_{\text{succ}}^+ + P_{\text{succ}}^-$, where

$$\begin{aligned} P_{\text{succ}}^+ &= \max \left\{ \frac{1}{4} \langle M_I, \rho_I \rangle + \frac{1}{4} \langle M_X, \rho_X \rangle : M_I + M_X = \Pi_+, M_I, M_X \succcurlyeq 0 \right\} \\ P_{\text{succ}}^- &= \max \left\{ \frac{1}{4} \langle M_Z, \rho_Z \rangle + \frac{1}{4} \langle M_{ZX}, \rho_{ZX} \rangle : M_Z + M_{ZX} = \Pi_-, M_Z, M_{ZX} \succcurlyeq 0 \right\} \end{aligned}$$

with $\Pi_{\pm} = \Phi^{\pm} + \Psi^{\pm}$. It is straightforward to see that $\Pi_+ + \Pi_- = \mathbb{1}$.

Consider P_{succ}^+ . Let us set $M_X = \Pi_+ - M_I$. Then

$$P_{\text{succ}}^+ = \frac{1}{4} \langle M_I, \rho_I - \rho_X \rangle + \frac{1}{4} \langle \Pi_+, \rho_X \rangle = \frac{1}{4} + \frac{\cos(2\theta)}{4} \langle M_I, \Phi^+ - \Psi^+ \rangle.$$

Similarly, by setting $M_{ZX} = \Pi_- - M_Z$, we get

$$P_{\text{succ}}^- = \frac{1}{4} + \frac{\cos(2\theta)}{4} \langle M_Z, \Phi^- - \Psi^- \rangle.$$

Thus, Eq. (3.45) becomes

$$P_{\text{succ}} = \frac{1}{2} + \frac{\cos(2\theta)}{4} (\langle M_I, \Phi^+ - \Psi^+ \rangle + \langle M_Z, \Phi^- - \Psi^- \rangle)$$

For $\theta < \pi/4$, we have $\cos(2\theta) > 0$ and so by choosing $M_I = \Phi^+$ and $M_Z = \Phi^-$ we get $P_{\text{succ}} = (1 + \cos(2\theta))/2 = \cos^2(\theta)$. Similarly, for $\theta > \pi/4$, we have $\cos(2\theta) < 0$ and so by choosing $M_I = \Psi^+$ and $M_Z = \Psi^-$ we get $P_{\text{succ}} = (1 - \cos(2\theta))/2 = \sin^2(\theta)$. At $\theta = \pi/4$, we have $\cos(2\theta) = 0$ which gives us $P_{\text{succ}} = 1/2$. Combining these, we get $P_{\text{succ}} = \max\{\cos^2(\theta), \sin^2(\theta)\}$.

While finding an analytical expression for the SDP value for $n = 2$ is not too hard, this is much harder for larger values of n and thus we turn to computing these numerically. Figure 3.9 illustrates the optimal error type classification probability as a function of the

angle θ of the state in Eq. (3.44).

To further examine the classification probabilities at $\theta = 49.8^\circ$, where there is seemingly an inflection point, we calculated the success probability for larger values of n at this value of θ . We observe that the success probability keeps shifting very slightly as the value of n increases, with a probability of 0.904764 at $n = 400$ which took 801398 seconds (roughly 9.28 days) of computation time (refer to the computational platform section for hardware details). Using the original SDP, this would correspond to dealing with variables of size $2^{400} \times 2^{400}$ which is computationally infeasible.

Remark 3.8. Note that this classical simulation was feasible since the state overlaps had an exploitable structure. However, for generic states and error unitaries, one would likely need to take advantage of a quantum computer to compute the Gram matrix.

Computational platform

For the quantum changepoint problems, the SDPs were solved using CVX [CVX12] on a 32 GB 10th generation Intel Core i9-10885H CPU (16 MB cache, 2.40 GHz, 8 cores).

For the quantum error type identification problem, the SDPs were solved using CVXPY [DB16, AVDB18] on a 32 GB AMD Ryzen 9 3950X (32) @ 3.50 GHz CPU with NVIDIA GeForce RTX 2080 Ti Rev. A GPU.

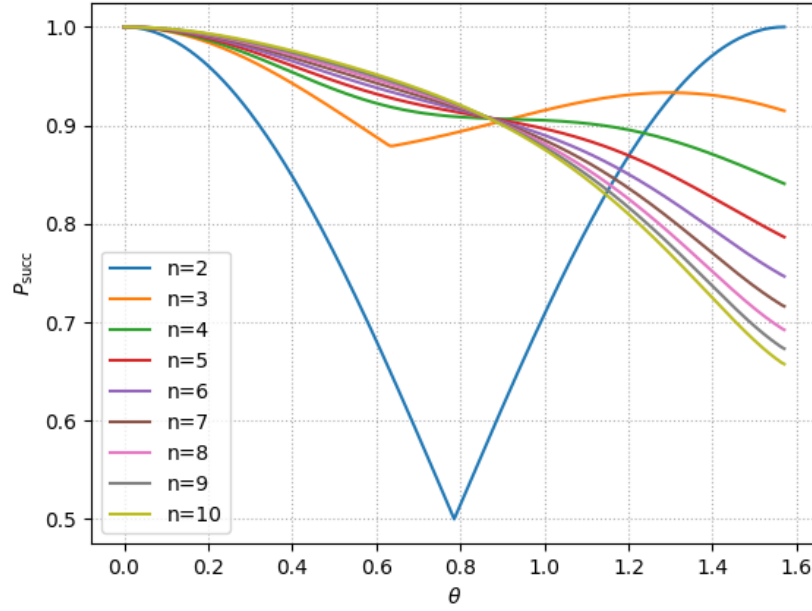


Figure 3.9: The optimal error-type classification probability for $|\psi_n(\theta)\rangle = \cos(\theta)|\text{GHZ}_n\rangle + \sin(\theta)|\text{W}_n\rangle$ as a function of $\theta \in [0, \pi/2]$. For $n = 2$, the success probability is $\max\{\cos^2(\theta), \sin^2(\theta)\}$. For $n \geq 3$, the curves appear to have an inflection point near $\theta = 49.8^\circ$, corresponding to a success probability of approximately 0.907.

Chapter 4

The pretty bad measurement

The contents of this chapter is based on work with Caleb McIrvine and Jamie Sikora in [\[MMS24\]](#).

The pretty good measurement (PGM) is a way to approximate the best possible measurement for the minimum error state discrimination task. This chapter introduces the pretty bad measurement (PBM) as an analogue for the minimum error state exclusion problem. We compare the performance of these measurements with the blind guessing probability. We observe that while PGM can perform no worse than blind guessing, PBM can perform no better than blind guessing. Lastly, we look at the application of the PBM for the quantum anomaly detection/avoidance problem.

We develop the pretty bad measurement through the following steps.

- Section [4.1](#): We discuss the pretty good measurement (PGM).
- Section [4.2](#): We describe how we can offset a measurement using an offset parameter.
- Section [4.3](#): We define the pretty bad measurement (PBM) as the counterpart of PGM

for the quantum state exclusion problem.

- Section 4.4: We discuss an interpretation of PBM as PGM in a different setting.
- Section 4.5: We illustrate an example of a state for which the PGM achieves the optimal probability for state discrimination while the PBM is a decent approximation to the optimal probability for state exclusion.
- Section 4.6: We describe how PGM performs no worse than blind guessing and PBM performs no better than blind guessing on randomly generated examples.
- Section 4.7: We discuss how PBM can be applied to the quantum anomaly detection problem.

4.1 The pretty good measurement (PGM)

For an ensemble $\{(p_i, \rho_i) : i \in [N]\}$, the pretty good measurement (PGM), also known as the square root measurement, introduced by Hausladen and Wootters [HW94] and further analyzed by Barnum and Knill [BK02], are the measurements $\{G_1, \dots, G_N\}$, where each operator is defined as

$$G_i = P^{-1/2} \left(p_i \rho_i \right) P^{-1/2} \text{ where } P = \sum_{i=1}^N p_i \rho_i. \quad (4.1)$$

If Bob uses the PGM for quantum state discrimination, his success probability is

$$\mathcal{P}_{\text{PGM}} = \sum_{i=1}^N p_i \langle G_i, \rho_i \rangle = \sum_{i=1}^N p_i^2 \text{tr}(P^{-1/2} \rho_i P^{-1/2} \rho_i). \quad (4.2)$$

Barnum and Knill [BK02] gave the following bound on the performance of the PGM.

Theorem 4.1 (Barnum-Knill Inequality [BK02]). *Let $\mathcal{P}_{\text{Best}}$ denote the optimal success probability (Eq. (1.1)), then*

$$\mathcal{P}_{\text{Best}}^2 \leq \mathcal{P}_{\text{PGM}} \leq \mathcal{P}_{\text{Best}}. \quad (4.3)$$

This bound ensures that the PGM always achieves a success probability at least as large as the square of the optimal probability, guaranteeing near-optimal performance.

We now present a theorem which compares $\mathcal{P}_{\text{PGM}}$ to the blind guessing probability, i.e., $1/N$.

Theorem 1. [Ren17] For any set of states $\{\rho_1, \dots, \rho_N\}$ and probability distribution $\{p_1, \dots, p_N\}$, we have

$$\mathcal{P}_{\text{PGM}} \geq \frac{1}{N} + \frac{(1 - N\mathcal{P}_{\text{Best}})^2}{N(N-1)} \quad (4.4)$$

where $1/N$ can be interpreted as the blind guessing probability. Moreover, if $\mathcal{P}_{\text{PGM}} = 1/N$, then we have $\rho_i = \rho_j$ and $p_i = p_j = 1/N$ for all $i \neq j$.

We now discuss our first step towards defining a pretty bad measurement.

4.2 Purposely messing up measurements with offsets

Suppose that one is given a decent measurement for state discrimination and one wishes to make it perform worse. Then one way to do this is to take the outcome of this measurement and *change the answer*. To this end, suppose we start with measurements $\{M_1, \dots, M_N\}$ and an *offset parameter* s . We define the offset measurement $\{M_1^s, \dots, M_N^s\}$ as

$$M_i^s = M_{i \oplus s} \quad (4.5)$$

where we use the notation $\oplus$ to denote cyclic addition, e.g., $5 \oplus 1 = 1$ when $N = 5$. Conceptually, an offset should mess up a good measurement and, conversely, could have the ability to improve a bad measurement. We have the need to quantify how well these offset measurements perform, thus we define

$$\alpha_s = \sum_{i=1}^N p_i \langle M_i^s, \rho_i \rangle = \sum_{i=1}^N p_i \langle M_{i \oplus s}, \rho_i \rangle. \quad (4.6)$$

It turns out that all of these quantities are related to each other via the equality

$$\sum_{s=0}^{N-1} \alpha_s = 1 \quad (4.7)$$

which is easily seen by noting that, for any fixed i , $\{M_i, M_{i \oplus 1}, \dots, M_{i \oplus (N-1)}\}$ yields a valid POVM, and thus $\sum_{s=0}^{N-1} M_{i \oplus s} = \mathbb{1}$.

We now prove a slightly weaker version of Theorem 1.

4.2.1 Proof of a slightly weaker version of Theorem 1

Given the quantum states $\{\rho_1, \dots, \rho_N\}$ and the probability distribution $\{p_1, \dots, p_N\}$, define $P = \sum_{i=1}^N p_i \rho_i$ and the measurements $\{G_1, \dots, G_N\}$ where $G_i = P^{-1/2} (p_i \rho_i) P^{-1/2}$ as in Eq. (4.1). Moreover, define the N -dimensional vector r where

$$r_i = \|P^{-1/4} (p_i \rho_i) P^{-1/4}\|_F. \quad (4.8)$$

The proof begins as follows. First, we establish that $\mathcal{P}_{\text{PGM}} = \|r\|_2^2$. Second, we bound $\|r\|_1 \geq 1$. Then showing $\mathcal{P}_{\text{PGM}} \geq 1/N$ follows by Cauchy-Schwarz.

To see that $\|r\|_2^2 = \mathcal{P}_{\text{PGM}}$, we have

$$\begin{aligned} \|r\|_2^2 &= \sum_{i=1}^N r_i^2 = \sum_{i=1}^N \|P^{-1/4} (p_i \rho_i) P^{-1/4}\|_F^2 = \sum_{i=1}^N \text{tr} (P^{-1/4} (p_i \rho_i) P^{-1/2} (p_i \rho_i) P^{-1/4}) \\ &= \sum_{i=1}^N p_i \text{tr} (\rho_i P^{-1/2} (p_i \rho_i) P^{-1/2}) = \sum_{i=1}^N p_i \text{tr} (\rho_i G_i) = \mathcal{P}_{\text{PGM}}. \end{aligned} \quad (4.9)$$

We now bound the following quantity:

$$\begin{aligned} p_i \langle G_{i \oplus s}, \rho_i \rangle &= p_i \text{tr} (P^{-1/2} (p_{i \oplus s} \rho_{i \oplus s}) P^{-1/2} \rho_i) \\ &= \text{tr} (P^{-1/4} (p_{i \oplus s} \rho_{i \oplus s}) P^{-1/4} P^{-1/4} (p_i \rho_i) P^{-1/4}) \\ &\leq \|P^{-1/4} (p_{i \oplus s} \rho_{i \oplus s}) P^{-1/4}\|_F \|P^{-1/4} (p_i \rho_i) P^{-1/4}\|_F \\ &= r_{i \oplus s} \cdot r_i. \end{aligned} \quad (4.10)$$

This bound is useful in showing $1 \leq \|r\|_1$. Using Eq. (4.7), we have

$$1 = \sum_{s=0}^{N-1} \alpha_s = \sum_{s=0}^{N-1} \sum_{i=1}^N p_i \langle G_{i \oplus s}, \rho_i \rangle \leq \sum_{s=0}^{N-1} \sum_{i=1}^N r_{i \oplus s} \cdot r_i = \left(\sum_{i=1}^N r_i \right)^2 = \|r\|_1^2 \quad (4.11)$$

noting the last equality holds since $r_i \geq 0$ for all i .

Let e be the N -dimensional all-ones vector and note that $\|r\|_1 = \langle r, e \rangle$. Then by the Cauchy-Schwarz inequality (for vectors), we have

$$1 \leq \|r\|_1 = \langle r, e \rangle \leq \|r\|_2 \|e\|_2 = \sqrt{\mathcal{P}_{\text{PGM}}} \cdot \sqrt{N} \quad (4.12)$$

proving that $\mathcal{P}_{\text{PGM}} \geq 1/N$.

Now, if we have that $\mathcal{P}_{\text{PGM}} = 1/N$, we must have equality throughout (4.12). This means that the vector r must be proportional to e and $\|r\|_1 = 1$. These together imply that

$r_i = 1/N$ for all i which further implies that $P^{-1/4}(p_i \rho_i)P^{-1/4} \neq 0$ for all i , as well as $p_i \neq 0$ for all i .

We further require inequality Eq. (4.10) to hold with equality, i.e., we must saturate Cauchy-Schwarz *at every i and $i \oplus s$* . This can only be true if there exists $\lambda_{i,j} \neq 0$ such that

$$P^{-1/4}(p_i \rho_i)P^{-1/4} = \lambda_{i,j} P^{-1/4}(p_j \rho_j)P^{-1/4} \quad (4.13)$$

for all i and j . Multiplying both sides by $P^{1/4}$, we have

$$p_i \rho_i = \lambda_{i,j} p_j \rho_j. \quad (4.14)$$

Taking the trace of both sides, we get

$$\lambda_{i,j} = \frac{p_i}{p_j}. \quad (4.15)$$

From Eq. (4.14), we have $\rho_i = \rho_j$ for all i and j .

All that remains to prove is that $p_i = 1/N$ for all i . Recall that $r_i = 1/N$ for all i . Thus, we have

$$1 = \frac{r_i}{r_j} = \frac{\|P^{-1/4}(p_i \rho_i)P^{-1/4}\|_F}{\|P^{-1/4}(p_j \rho_j)P^{-1/4}\|_F} = \frac{p_i}{p_j} \frac{\|P^{-1/4} \rho_i P^{-1/4}\|_F}{\|P^{-1/4} \rho_j P^{-1/4}\|_F} = \frac{p_i}{p_j} \quad (4.16)$$

where the last equality holds since $\rho_i = \rho_j$ for all i and j . Thus, $p_i = 1/N$ for all i , as required.

4.3 The pretty bad measurement (PBM)

We now define the counterpart to the PGM which now approximates $\mathcal{P}_{\text{Worst}}$ instead. Our approach is quite simple and borrows much of the elegance from the PGM. We define it below and then give some interpretations of it.

Definition 1. Let $\{G_1, \dots, G_N\}$ be the pretty good measurement. The pretty bad measurement (PBM) is defined as the POVM $\{B_1, \dots, B_N\}$ where

$$B_i = \frac{1}{N-1} \left(\mathbb{1} - G_i \right). \quad (4.17)$$

One way to interpret the PBM is by viewing it as the PGM with a uniformly random offset. In other words,

$$B_i = \frac{1}{N-1} \sum_{s=1}^{N-1} G_i^s \quad (4.18)$$

noting that we omit the $s = 0$ index since $G_i^0 = G_i$. With this, we can calculate the success probability of the PBM as

$$\mathcal{P}_{\text{PBM}} = \sum_{i=1}^N p_i \operatorname{tr}(\rho_i B_i) = \frac{1}{N-1} \sum_{s=1}^{N-1} \sum_{i=1}^N p_i \operatorname{tr}(\rho_i G_i^s) = \frac{1}{N-1} \sum_{s=1}^{N-1} \alpha_s. \quad (4.19)$$

Since $\alpha_0 = \mathcal{P}_{\text{PGM}}$, as this is the no-offset success probability, we have the following equation which follows directly from Eq. (4.7)

$$\mathcal{P}_{\text{PGM}} + (N-1)\mathcal{P}_{\text{PBM}} = 1. \quad (4.20)$$

By combining with Theorem 1, we have our main result, below, which is the counterpart to

the PGM bound, Theorem 1.

Theorem 2. For any set of states $\{\rho_1, \dots, \rho_N\}$ and probability distribution $\{p_1, \dots, p_N\}$, we have

$$\mathcal{P}_{\text{PBM}} \leq \frac{1}{N} - \frac{(1 - N\mathcal{P}_{\text{Best}})^2}{N(N-1)^2}. \quad (4.21)$$

Moreover, if $\mathcal{P}_{\text{PBM}} = 1/N$, then we have $\rho_i = \rho_j$ and $p_i = p_j = 1/N$ for all $i \neq j$.

As a consequence of our discussion so far, we have

$$\mathcal{P}_{\text{Best}} \geq \mathcal{P}_{\text{PGM}} \geq \frac{1}{N} \geq \mathcal{P}_{\text{PBM}} \geq \mathcal{P}_{\text{Worst}}. \quad (4.22)$$

4.4 Another interpretation

There is another way to interpret the PBM which is as the PGM *in a different setting*. For the set of states $\{\rho_1, \dots, \rho_N\}$ with probabilities $\{p_1, \dots, p_N\}$, define a new problem instance $\{\tilde{\rho}_1, \dots, \tilde{\rho}_N\}$ with probabilities $\{\tilde{p}_1, \dots, \tilde{p}_N\}$ where

$$\tilde{\rho}_i = \frac{1}{1 - p_i} \sum_{j \neq i} p_j \rho_j \quad \text{and} \quad \tilde{p}_i = \frac{1 - p_i}{N - 1} \quad (4.23)$$

assuming $p_i \neq 1$ for all i . We can think of $\tilde{\rho}_i$ as averaging over all of the ρ_j s with different indices. Thus, by guessing i accurately in the tilde setting one is actually guessing poorly in the original setting. Before analyzing the PGM in the tilde setting, notice that

$$\tilde{p}_i \tilde{\rho}_i = \frac{1}{N - 1} \sum_{j \neq i} p_j \rho_j = \frac{1}{N - 1} (P - p_i \rho_i) \quad (4.24)$$

recalling the definition of P from Eq. (4.1). We also have that $\tilde{P} = \sum_{i=1}^N \tilde{p}_i \tilde{\rho}_i = P$. Denoting the PGM in the tilde setting as $\{\tilde{G}_1, \dots, \tilde{G}_N\}$, we can calculate that

$$\tilde{G}_i = \tilde{P}^{-1/2} \left(\tilde{p}_i \tilde{\rho}_i \right) \tilde{P}^{-1/2} = B_i.$$

Thus, one can view the PBM as the PGM if the problem instance is changed accordingly.

Remark. We note that the mixing of POVM elements considered above is studied more generally in the context of *reverse channels* and *reverse observables* which have been used in the study of compatibility [FHL17].

4.5 Illustrative examples

Consider the three trine states, defined as

$$\begin{aligned} |\psi_1\rangle &= |0\rangle, & |\psi_2\rangle &= \cos(2\pi/3) |0\rangle + \sin(2\pi/3) |1\rangle, \\ |\psi_3\rangle &= \cos(4\pi/3) |0\rangle + \sin(4\pi/3) |1\rangle. \end{aligned} \tag{4.25}$$

These are shown in Fig. 4.1, along with the five quantities of interest.

We remark that this example sees the PGM being a perfect approximation of $\mathcal{P}_{\text{Best}}$ while the PBM is a decent, but not perfect, approximation of $\mathcal{P}_{\text{Worst}}$. It is an interesting question to see in which cases the PBM is the worst measurement. The above example suggests that this is indeed tricky to answer since it does not coincide with the settings where the PGM is the best measurement.

If the set of states is antidistinguishable, that is, if $\mathcal{P}_{\text{Worst}} = 0$, then if $\mathcal{P}_{\text{PBM}} = \mathcal{P}_{\text{Worst}}$, then by Eq. (4.20), we must have $\mathcal{P}_{\text{PGM}} = 1$. In this case, the states must have orthogonal

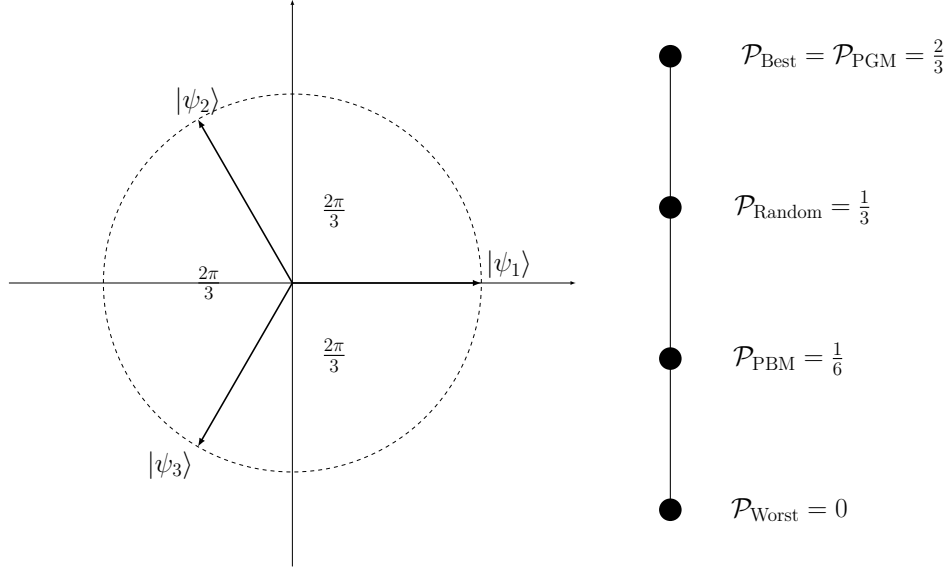


Figure 4.1: If given a trine state, each with equal probability, we have $\mathcal{P}_{\text{Best}} = \mathcal{P}_{\text{PGM}} = 2/3$, with $\mathcal{P}_{\text{Random}} = 1/3$ being the blind-guessing probability, and $\mathcal{P}_{\text{PBM}} = 1/6$ approximating $\mathcal{P}_{\text{Worst}} = 0$. The pretty good measurement is given by the POVM $G_i = \frac{2}{3}|\psi_i\rangle\langle\psi_i|$; the pretty bad measurement is given by the POVM $B_i = \frac{1}{3}\sum_{j \neq i} |\psi_j\rangle\langle\psi_j|$; and the worst measurement is given by $W_i = \frac{2}{3}|\psi_i^\perp\rangle\langle\psi_i^\perp|$, where $|\psi_i^\perp\rangle\langle\psi_i^\perp|$ is a $\pi/2$ rotation of $|\psi_i\rangle$.

supports and we have

$$\mathcal{P}_{\text{Best}} = \mathcal{P}_{\text{PGM}} = 1 \quad \text{and} \quad \mathcal{P}_{\text{PBM}} = \mathcal{P}_{\text{Worst}} = 0. \quad (4.26)$$

On the other hand, if the set of states is maximally indistinguishable, i.e., $\rho_i = \rho_j$ and $p_i = p_j = 1/N$, for all $i \neq j$, then we have

$$\mathcal{P}_{\text{Best}} = \mathcal{P}_{\text{PGM}} = \frac{1}{N} = \mathcal{P}_{\text{PBM}} = \mathcal{P}_{\text{Worst}}. \quad (4.27)$$

What is interesting in the above two examples is that PBM being the worst measurement implies that PGM is the best measurement. This is actually true in general, which we state below, however the trine state example shows that the converse is not true.

Lemma 1. If $\mathcal{P}_{\text{PBM}} = \mathcal{P}_{\text{Worst}}$, then $\mathcal{P}_{\text{PGM}} = \mathcal{P}_{\text{Best}}$.

Proof. Suppose $\mathcal{P}_{\text{PBM}} = \mathcal{P}_{\text{Worst}}$. Consider the *best* measurement $\{M_1, \dots, M_N\}$ and its random offset measurement probabilities α_s . With this, we have $\alpha_0 = \mathcal{P}_{\text{Best}}$ and note that $\alpha_s \geq \mathcal{P}_{\text{Worst}} = \mathcal{P}_{\text{PBM}}$ for all s . By Eq. (4.7) we get

$$1 = \sum_{s=0}^{N-1} \alpha_s = \mathcal{P}_{\text{Best}} + \sum_{s=1}^{N-1} \alpha_s \geq \mathcal{P}_{\text{Best}} + \sum_{s=1}^{N-1} \mathcal{P}_{\text{PBM}} \geq \mathcal{P}_{\text{PGM}} + (N-1)\mathcal{P}_{\text{PBM}} = 1 \quad (4.28)$$

where the last equality holds by Eq. (4.20). Thus, we have equality throughout, proving that $\mathcal{P}_{\text{PGM}} = \mathcal{P}_{\text{Best}}$. $\square$

4.6 Numerical simulations

We have discussed how the PGM always performs no worse than blind guessing and consequently proved that the PBM always performs no better than blind guessing. To get a better idea of how these two probabilities compare, we calculate them on many randomly generated examples. To generate random states, we can use the Bloch sphere to define a parameterized qubit $\rho(\theta, \phi)$ as

$$\rho(\theta, \phi) = \frac{1}{2}\mathbb{1} + \frac{\sin(\theta)\cos(\phi)}{2}X + \frac{\sin(\theta)\sin(\phi)}{2}Y + \frac{\cos(\theta)}{2}Z$$

where X , Y , and Z are the Pauli operators and $\theta, \phi \in (0, 2\pi)$ are real parameters which we randomly generate. We generate 1000 random qubit instances $\{\rho_1, \dots, \rho_N\}$ for $N = 2$ in Fig. 4.2 and for $N = 6$ in Fig. 4.3, each with randomly generated probabilities $(p_1, \dots, p_N)$.

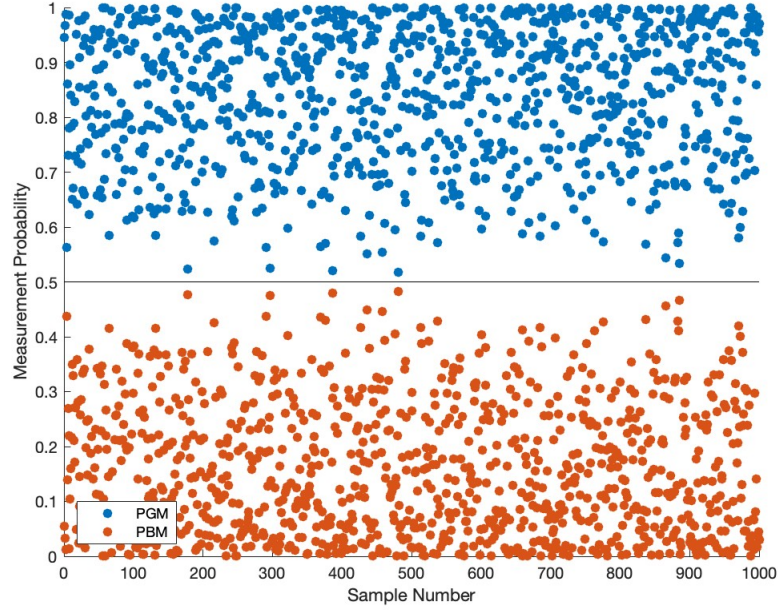


Figure 4.2: Success probabilities of the pretty good measurement and pretty bad measurement for 1000 randomly generated pairs of qubit states with randomly chosen prior probabilities. The pretty good measurement always achieves a success probability greater than $1/2$, whereas the pretty bad measurement always achieves a success probability less than $1/2$.

4.7 Application: Quantum anomaly detection/avoidance

Suppose Alice promises to send Bob n copies of a state ρ . However, she has a faulty device which is known to produce one copy of an anomaly (read: a *pretty bad state*) σ at some point in this sequence. Upon receiving the n states, Bob wants to identify a state *which is not faulty*, i.e., not σ .

Our approach is to cast this as a discrimination problem with the states

$$\rho_i = \rho^{\otimes(i-1)} \otimes \sigma \otimes \rho^{\otimes(N-i)},$$

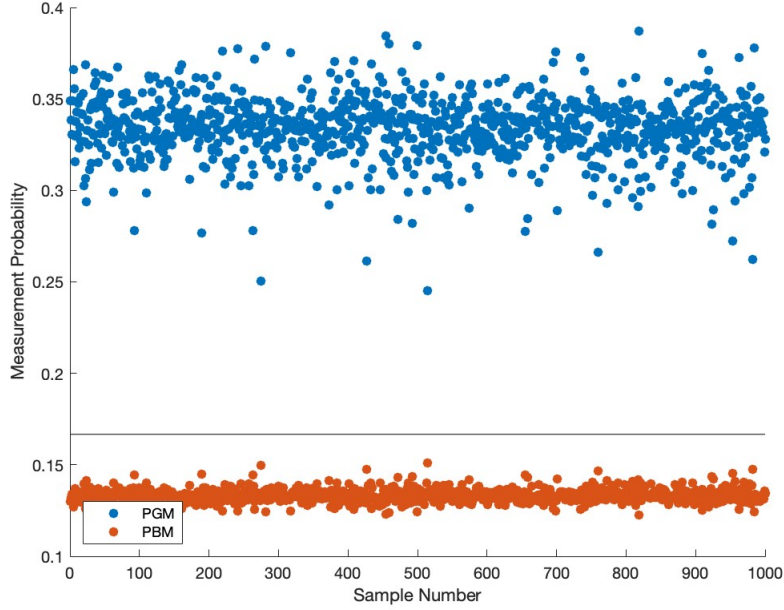


Figure 4.3: Success probabilities of the pretty good measurement and pretty bad measurement for 1000 randomly generated ensembles of six qubit states with randomly selected prior probabilities. For small values of n , $\mathcal{P}_{\text{PGM}}$ appears to cluster around $\frac{2}{n}$, while $\mathcal{P}_{\text{PBM}}$ appears to cluster around $\frac{n-2}{n(n-1)}$. The case $n = 6$ is shown here.

for $i \in [N]$, and to use the pretty bad measurement. Note that ρ_i is simply N copies of ρ , but with σ in the i -th spot. In the traditional discrimination task, Bob is trying to guess where the anomaly has occurred. However, this is overkill if we just want to find a place where it *did not occur*. Thus, taking the perspective of state exclusion makes more sense here. In other words, we do want to determine where the anomaly occurred as much as where it *did not occur*. We focus on these two concepts in the following example and graphs.

As an example, we consider the case where Alice promises to send to Bob 7 copies of the state $\rho = |0\rangle\langle 0|$ but an anomaly occurs creating the state $\sigma = |\psi\rangle\langle\psi|$ where $|\psi\rangle = \gamma|0\rangle + \sqrt{1-\gamma^2}|1\rangle$ for some fixed parameter $\gamma \in [0, 1]$ known to both Alice and Bob. Fig. 4.4 illustrates the probabilities $\mathcal{P}_{\text{Best}}$, $\mathcal{P}_{\text{PGM}}$, $\mathcal{P}_{\text{PBM}}$, $\mathcal{P}_{\text{Worst}}$ and the blind guessing probability $1/N$, as a function of γ , below. For anomaly detection, [LSMT24, SLH⁺24] showed that PGM is indeed

the optimal measurement. Moreover they provide a closed-form expression for the optimal success probability. Therefore, using Eq. (4.20) we also have an expression for $\mathcal{P}_{\text{PBM}}$.

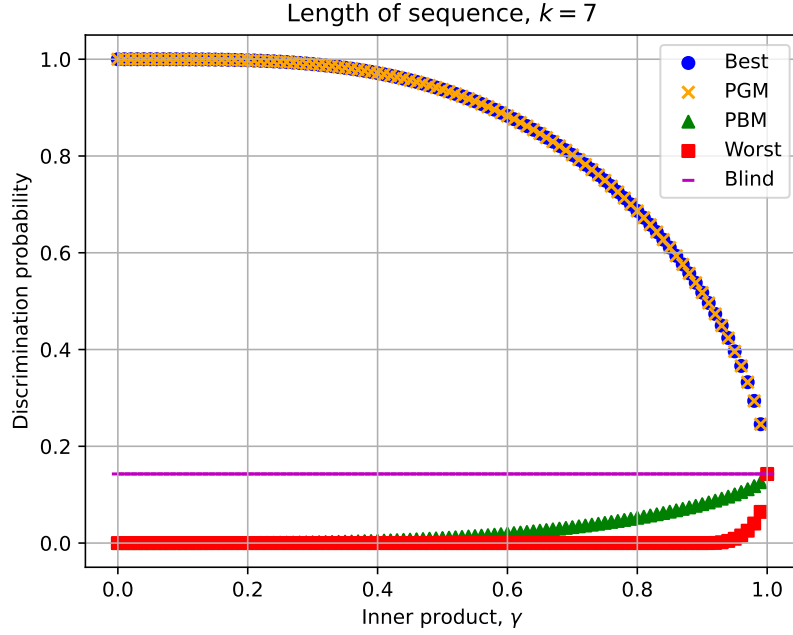


Figure 4.4: Probabilities for the quantum state discrimination task defined by $|\psi_i\rangle = |0\rangle^{\otimes(i-1)} \otimes |\psi\rangle \otimes |0\rangle^{\otimes(7-i)}$, where $|\psi\rangle = \gamma|0\rangle + \sqrt{1-\gamma^2}|1\rangle$ and $i \in \{1, \dots, 7\}$. While $\mathcal{P}_{\text{PGM}} = \mathcal{P}_{\text{Best}}$ for all values of γ , $\mathcal{P}_{\text{PBM}}$ begins to depart from the worst success probability near $\gamma = 0.4$. All four probabilities coincide at $\gamma = 1$, since all the states are identical in this case; see Eq. (4.27).

Recall that in the anomaly avoidance problem, we want to identify a state that did not occur. Since PBM is trying to minimize $\sum_{i=1}^N \frac{1}{N} \langle M_i, \rho_i \rangle$, what PBM is approximating in the context of anomaly avoidance is the minimum average error. In this case, the probability of *success* using PBM is $1 - \mathcal{P}_{\text{PBM}}$. In Fig. 4.5, we compare this probability to that given by PGM which approximates the best average identification probability.

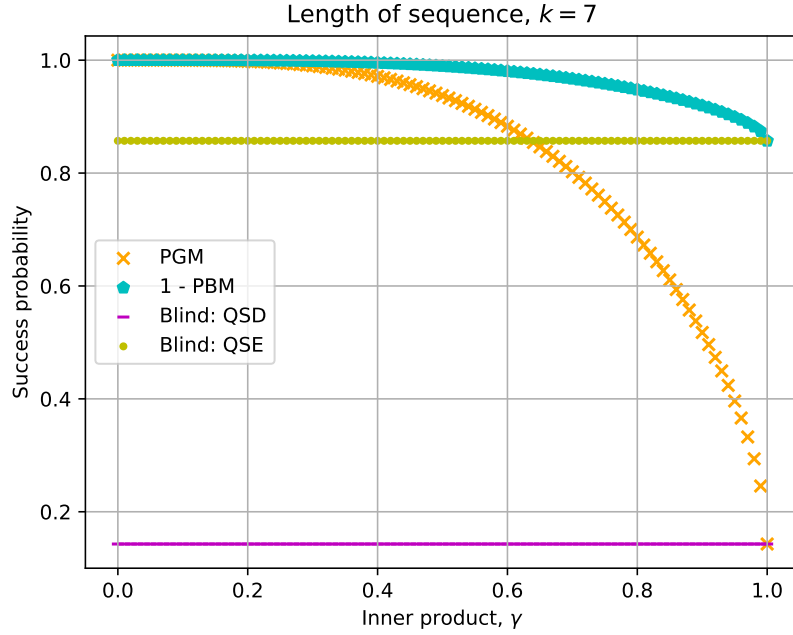


Figure 4.5: Given one of the equally likely states $|\psi_i\rangle = |0\rangle^{\otimes(i-1)} \otimes |\psi\rangle \otimes |0\rangle^{\otimes(7-i)}$, where $|\psi\rangle = \gamma|0\rangle + \sqrt{1-\gamma^2}|1\rangle$ and $i \in \{1, \dots, 7\}$, we plot the probability of successfully excluding the anomaly $|\psi\rangle$ using the PBM and the probability of successfully identifying $|\psi\rangle$ using the PGM. The probability $\mathcal{P}_{\text{PGM}}$ decreases rapidly beginning near $\gamma = 0.2$ and approaches the blind identification probability of $1/7$ at $\gamma = 1$. In contrast, the exclusion success probability $1 - \mathcal{P}_{\text{PBM}}$ decreases more slowly and approaches the blind exclusion probability of $6/7$ at $\gamma = 1$. As expected, the exclusion success probability is substantially greater than the identification success probability.

Chapter 5

Local strategies are pretty good at computing Boolean properties of quantum sequences

The contents of this chapter is based on work with Tathagata Gupta, Shayeeef Murshid, Vincent Russo, Jamie Sikora, and Alice Zheng in [\[GMM⁺26\]](#).

In the previous two chapters we have focused on optimizing and approximating global measurements for various aspects of the state discrimination problem. We did not take into account the fact that quantum memory is a scarce and costly resource. In this chapter, we study the problem of computing global properties of quantum sequences when quantum systems must be measured individually, without storing or jointly processing them. In our setting, a bit string $x \in \{0, 1\}^n$ is encoded into an n -qubit product state $|\psi_{x_1}\rangle \otimes \cdots \otimes |\psi_{x_n}\rangle$, and the goal is to infer $f(x) \in \{0, 1\}$ from measurements of this quantum encoding. We consider a simple local strategy, which we call the *greedy strategy*, that applies the same optimal single-system measurement independently to each subsystem and then infers $f(x)$

from the outcomes. Our main result gives a complete characterization of when the greedy strategy is optimal: it achieves the same maximum success probability as an unrestricted global measurement if and only if the target Boolean function is affine (in all but finitely many cases). We establish a universal performance guarantee for general Boolean functions, showing that the success probability of the greedy strategy is always at least the square of the optimal global success probability, in direct analogy with the Barnum-Knill bound for the pretty good measurement. These results demonstrate that even under extreme memory constraints, simple local measurement strategies can remain provably competitive for learning global properties of quantum sequences.

This chapter is organized in the following manner.

- Section 5.1: We formulate the problem as an instance of quantum state discrimination and present the main results.
- Section 5.2: We describe the greedy strategy and prove that for all Boolean functions, the performance of this greedy strategy is bounded below by the square of the performance of the optimal global strategy. For affine Boolean functions, we show that the greedy strategy is optimal.
- Section 5.3: We prove that for Boolean functions that are not affine, the greedy strategy is not globally optimal.
- Section 5.5: We consider two non-affine functions, the AND and majority functions. For the AND function, we show that both the greedy as well as the optimal global strategy converge to perfect success exponentially quickly. For the majority function, we observe that the greedy strategy converges to a value strictly bounded away from 1, while the global strategy achieves a slightly better performance.

5.1 Problem formulation and main results

In this section we formulate our problem as a task of quantum state discrimination and present the main results.

5.1.1 Problem formulation

In classical computing, data are modeled by bit strings. In our setting, we consider encoding the classical symbols 0 and 1 into two arbitrary, but fixed and known quantum states $|\psi_0\rangle$ and $|\psi_1\rangle$. This induces a map from an n -bit string $x \in \{0, 1\}^n$ to a product quantum state of length n ,

$$x = x_1x_2 \dots x_n \in \{0, 1\}^n \longmapsto |\psi_x\rangle := |\psi_{x_1}\rangle \otimes |\psi_{x_2}\rangle \otimes \dots \otimes |\psi_{x_n}\rangle. \quad (5.1)$$

Any Boolean property of the resulting quantum sequence can be described by a Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$. Learning this property, equivalently, determining the value of $f(x)$ does not require full identification of the underlying string x . Rather, it suffices to discriminate between the two sets of quantum sequences corresponding to inputs for which $f(x) = 0$ and for which $f(x) = 1$. Thus, given an unknown quantum sequence $|\psi_x\rangle$, the task is to decide whether the classical label x belongs to S_0 or S_1 , where $S_0 = f^{-1}(0)$ and $S_1 = f^{-1}(1)$ are the pre-images of 0 and 1 respectively. Without loss of generality we can consider $|\psi_0\rangle$ and $|\psi_1\rangle$ to be qubit states with a nonnegative inner product [BC09, Section 3.1]. Furthermore, since orthogonal and identical states represent trivial discrimination cases, we restrict the inner product to the open interval $\langle \psi_0 | \psi_1 \rangle = s \in (0, 1)$.

To learn the property¹ f , the learner must discriminate between two mixed quantum states

¹ Henceforth, we use the terms “property” and “function” interchangeably.

σ_0 and σ_1 where σ_i is the density matrix obtained by mixing the states $\{|\psi_x\rangle\langle\psi_x| : f(x) = i\}$. Assuming a uniform prior over the input strings in $\{0, 1\}^n$, these mixed states can be written as

$$\sigma_0 = \frac{1}{|S_0|} \sum_{x \in S_0} |\psi_x\rangle\langle\psi_x| \quad \text{and} \quad \sigma_1 = \frac{1}{|S_1|} \sum_{x \in S_1} |\psi_x\rangle\langle\psi_x|, \quad (5.2)$$

which occur with prior probabilities $p_0 = |S_0|/2^n$ and $p_1 = |S_1|/2^n$. We refer to the resulting ensemble

$$\mathcal{E} = \{(p_0, \sigma_0), (p_1, \sigma_1)\} \quad (5.3)$$

as the Boolean ensemble. In this work, we consider the *minimum-error* evaluation of the function f , where the learner measures the unknown quantum state $|\psi_x\rangle$ to infer the value of $f(x)$, seeking to minimize the probability of error. Operationally, this corresponds to discriminating between σ_0 and σ_1 using a two-outcome POVM $\mathcal{M} = \{M_0, M_1\}$, where the outcome i is interpreted as the guess $f(x) = i$. Conditioned on the state being σ_i , the probability of a correct guess is $\text{Tr}(\sigma_i M_i)$. The optimal success probability is obtained by maximizing the average probability of a correct decision over all such POVMs:

$$\max \{p_0 \langle M_0, \sigma_0 \rangle + p_1 \langle M_1, \sigma_1 \rangle : M_0 + M_1 = \mathbb{1}, M_0, M_1 \succcurlyeq 0\}. \quad (5.4)$$

This is a semidefinite program and generalizes to discrimination among any finite ensemble of quantum states. Since we are only dealing with two states here, the optimal minimum-error success probability is given by the Helstrom bound [Hel69, Hol76]

$$\frac{1}{2} + \frac{1}{2} \|p_0 \sigma_0 - p_1 \sigma_1\|_1, \quad (5.5)$$

where $\|X\|_1 = \text{Tr}(\sqrt{X^* X})$ is the trace norm.

An optimal POVM achieving the maximum success probability may require a joint measure-

ment acting on the entire n -qubit state $|\psi_x\rangle$, which is technologically challenging [LST09, MMZ⁺21, CVM⁺23]. A more feasible alternative is a local measurement strategy, in which each state is measured individually. Such a strategy effectively extracts classical information from each subsystem and uses it to infer the value of $f(x)$. In this chapter, we focus on a simple procedure: a fixed local strategy that applies a predetermined measurement to each subsystem, independent of all of the previous outcomes, thus requiring neither classical nor quantum memory. A natural local measurement to consider is the one that aims to optimally distinguish between $|\psi_0\rangle$ and $|\psi_1\rangle$ at each position. Given any Boolean function f , this strategy learns each bit optimally, constructs a guess for the encoded string y , and evaluates $f(y)$. This can also be seen as a *non-adaptive greedy* strategy². One might be tempted to think that this naïve and simple strategy would perform rather poorly when compared with a global measurement. Our first result shows that, remarkably, this strategy performs rather well—but first we establish some notation.

Definition 5.1. Given a positive integer n , a function $f : \{0, 1\}^n \rightarrow \{0, 1\}$, and two encoding qubit states $|\psi_0\rangle$ and $|\psi_1\rangle$ with inner product $s = \langle\psi_0|\psi_1\rangle$, the success probability of learning f using a strategy “strat” when each $x \in \{0, 1\}^n$ is chosen uniformly at random is denoted by $P(n, f, s, \text{strat})$.

Throughout the paper, we denote the fixed local (greedy) strategy and a globally optimal strategy by “greedy” and “global”, respectively.

² Note, however, that this notion of greediness may not be optimal in the class of local memoryless strategies for computing the overall function value.

5.1.2 Main results

We now state our first main result which establishes a universal performance guarantee for the greedy strategy.

Theorem 5.2. *For all Boolean functions f , positive integers n , and $s \in (0, 1)$, the performance of the greedy strategy can be bounded below by the following relation*

$$P(n, f, s, \text{global})^2 \leq P(n, f, s, \text{greedy}). \quad (5.6)$$

The proof consists in showing that the greedy strategy is implemented by the pretty good measurement (PGM) [HW94] for discriminating the Boolean ensemble $\mathcal{E} = \{(p_0, \sigma_0), (p_1, \sigma_1)\}$. The bound then follows from Barnum and Knill's theorem (Theorem 4.1). This is presented in Section 5.2.2.

This raises a natural question: for which Boolean functions does the greedy strategy actually achieve the global optimum? The next result identifies a class of functions that satisfy this.

Theorem 5.3 (Sufficiency). *Let $f : \{0, 1\}^n \rightarrow \{0, 1\}$ be of the form*

$$f(x_1, \dots, x_n) = b_0 \oplus b_1 x_1 \oplus b_2 x_2 \oplus \dots \oplus b_n x_n, \quad (5.7)$$

for some $b_0, b_1, \dots, b_n \in \{0, 1\}$. Then, the greedy strategy is globally optimal, that is

$$P(n, f, s, \text{greedy}) = P(n, f, s, \text{global}) \quad (5.8)$$

for all $s \in (0, 1)$.

The functions of the form in Eq. (5.7) are called *affine Boolean functions*. For affine Boolean

functions, both the globally optimal success probability (via the Helstrom bound) and the greedy success probability can be computed in closed form. A direct calculation using properties of the trace norm shows that they are equal for all $s \in (0, 1)$. See Section 5.2.3 for details.

Our final result shows that this class is essentially the only one for which the greedy strategy is optimal, establishing a converse to Theorem 5.3.

Theorem 5.4 (Necessity). *Suppose a Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ is not affine, meaning it is not of the form in Eq. (5.7). Then, the greedy strategy is not globally optimal, that is*

$$P(n, f, s, \text{greedy}) < P(n, f, s, \text{global}) \quad (5.9)$$

for all but possibly finitely many values of $s \in (0, 1)$.

Together, Theorems 5.3 and 5.4 provide a complete characterization: affine Boolean functions are precisely those for which memoryless measurement strategies suffice for optimal learning. Note that Theorem 5.4 holds for all $s \in (0, 1)$ except possibly for a finite set, hence for all but a set of measure zero. The proof is presented in Section 5.3.

Section 5.4 contains supplementary results for the proof of Theorem 5.4. Section 5.5 briefly discusses the performance of the greedy strategy for some non-affine functions. We consider the AND and MAJ (majority) functions as representative non-affine examples. The former is highly unbalanced ($|S_0| = 2^n - 1, |S_1| = 1$), while the latter is balanced ($|S_0| = |S_1|$). In both cases, globally optimal strategies outperform the greedy strategy, as illustrated by numerical results for small sequence lengths ($3 \leq n \leq 9$). Their asymptotic behavior, however, differs: for AND, both the greedy and global strategies achieve perfect success as $n \rightarrow \infty$, whereas for MAJ, the greedy strategy remains bounded away from unity in this limit.

5.2 Proofs of Theorems 5.2 and 5.3: Performance of the greedy strategy

Recall from Chapter 4 that the pretty good measurement (PGM) is defined for an ensemble $\{(p_i, \rho_i)\}_{i=1}^n$ as the POVM $\{G_1, \dots, G_n\}$ such that

$$G_i = P^{-1/2} (p_i \rho_i) P^{-1/2} \quad \text{with} \quad P = \sum_{i=1}^n p_i \rho_i.$$

5.2.1 The greedy strategy

Let $\mathcal{M} = \{M_0, M_1\}$ be an optimal POVM for discriminating the ensemble $\{(1/2, |\psi_0\rangle), (1/2, |\psi_1\rangle)\}$ on a single qubit. Since $|\psi_0\rangle$ and $|\psi_1\rangle$ appear with equal probabilities, the PGM is optimal for this single-qubit discrimination problem [HW94], with

$$M_i = \frac{1}{2} \rho^{-1/2} |\psi_i\rangle\langle\psi_i| \rho^{-1/2}, \quad (5.10)$$

for $i \in \{0, 1\}$ where $\rho = \frac{1}{2} |\psi_0\rangle\langle\psi_0| + \frac{1}{2} |\psi_1\rangle\langle\psi_1|$ is the average single-qubit state.

The greedy strategy is as follows: measure each qubit of the sequence $|\psi_x\rangle$ independently via the POVM $\mathcal{M}$, obtaining outcome $y_i \in \{0, 1\}$ for the i -th qubit, yielding a classical string $y = y_1 \cdots y_n$. Then, evaluate $f(y)$: if $y \in S_0$, guess that the input was from S_0 ; if $y \in S_1$, guess S_1 .

The success probability of this strategy can be computed as follows. Given that the true

input is x , the probability of obtaining outcome string y through local measurements is

$$P(y \mid x) = \langle M_y, |\psi_x\rangle\langle\psi_x| \rangle = \prod_{i=1}^n \langle M_{y_i}, |\psi_{x_i}\rangle\langle\psi_{x_i}| \rangle, \quad (5.11)$$

where $M_y = \bigotimes_{i=1}^n M_{y_i}$. We succeed if x and y belong to the same set (S_0 or S_1). Averaging over the uniform distribution on inputs, the greedy success probability is

$$P(n, f, s, \text{greedy}) = \sum_{x \in \{0,1\}^n} \frac{1}{2^n} \sum_{\substack{y \in \{0,1\}^n \\ f(y)=f(x)}} P(y \mid x) = \frac{|S_0|}{2^n} \sum_{y \in S_0} \langle M_y, \sigma_0 \rangle + \frac{|S_1|}{2^n} \sum_{y \in S_1} \langle M_y, \sigma_1 \rangle. \quad (5.12)$$

5.2.2 Lower-bounding the greedy strategy via the PGM

Now consider distinguishing the Boolean ensemble in Eq. (5.3) via the PGM. The average state is

$$\rho_{\text{avg}} = p_0 \sigma_0 + p_1 \sigma_1 = \frac{|S_0|}{2^n} \sigma_0 + \frac{|S_1|}{2^n} \sigma_1 = \frac{1}{2^n} \sum_{x \in \{0,1\}^n} |\psi_x\rangle\langle\psi_x| = \left(\frac{1}{2} |\psi_0\rangle\langle\psi_0| + \frac{1}{2} |\psi_1\rangle\langle\psi_1| \right)^{\otimes n} = \rho^{\otimes n}. \quad (5.13)$$

The PGM operators for distinguishing σ_0 and σ_1 are

$$M_{\sigma_0}^{\text{PGM}} = \frac{|S_0|}{2^n} \rho_{\text{avg}}^{-1/2} \sigma_0 \rho_{\text{avg}}^{-1/2} \quad \text{and} \quad M_{\sigma_1}^{\text{PGM}} = \frac{|S_1|}{2^n} \rho_{\text{avg}}^{-1/2} \sigma_1 \rho_{\text{avg}}^{-1/2}. \quad (5.14)$$

We can now establish the key connection:

Theorem 5.5. *For any Boolean function $f : \{0,1\}^n \rightarrow \{0,1\}$ and any inner product $s \in (0,1)$, the pretty good measurement achieves the same success probability as the greedy strategy for evaluating f .*

Proof. We show that the PGM operators can be decomposed as a sum of product measurements that precisely match the greedy strategy. For $i \in \{0, 1\}$, starting with $M_{\sigma_i}^{\text{PGM}}$, we have

$$\begin{aligned}
 M_{\sigma_i}^{\text{PGM}} &= \frac{|S_i|}{2^n} (\rho^{\otimes n})^{-1/2} \sigma_i (\rho^{\otimes n})^{-1/2} = \frac{|S_i|}{2^n} (\rho^{-1/2})^{\otimes n} \left(\frac{1}{|S_i|} \sum_{x \in S_i} |\psi_x\rangle\langle\psi_x| \right) (\rho^{-1/2})^{\otimes n} \\
 &= \frac{1}{2^n} \sum_{x_1 \dots x_n \in S_i} (\rho^{-1/2} |\psi_{x_1}\rangle\langle\psi_{x_1}| \rho^{-1/2}) \otimes \dots \otimes (\rho^{-1/2} |\psi_{x_n}\rangle\langle\psi_{x_n}| \rho^{-1/2}) \\
 &= \sum_{y_1 \dots y_n \in S_i} M_{y_1} \otimes \dots \otimes M_{y_n} = \sum_{y \in S_i} M_y.
 \end{aligned} \tag{5.15}$$

The success probability of the PGM is therefore

$$\begin{aligned}
 P(n, f, s, \text{PGM}) &= p_0 \langle M_{\sigma_0}^{\text{PGM}}, \sigma_0 \rangle + p_1 \langle M_{\sigma_1}^{\text{PGM}}, \sigma_1 \rangle \\
 &= \frac{|S_0|}{2^n} \sum_{y \in S_0} \langle M_y, \sigma_0 \rangle + \frac{|S_1|}{2^n} \sum_{y \in S_1} \langle M_y, \sigma_1 \rangle \\
 &= P(n, f, s, \text{greedy}),
 \end{aligned} \tag{5.16}$$

where the last equality is by Eq. (5.12). $\square$

The proof of Theorem 5.2 follows immediately.

Proof of Theorem 5.2. From Barnum and Knill's theorem (Theorem 4.1) and Theorem 5.5 we have

$$P(n, f, s, \text{global})^2 \leq P(n, f, s, \text{PGM}) = P(n, f, s, \text{greedy}), \tag{5.17}$$

completing the proof. $\square$

Theorem 5.5 reveals a remarkable structural property: although the PGM operators $M_{\sigma_i}^{\text{PGM}}$ are formulated as general operators on the full n -qubit Hilbert space, for the ensemble arising from Boolean function evaluation they decompose exactly into the local measurement

operators of the greedy strategy:

$$M_{\sigma_i}^{\text{PGM}} = \sum_{y_1 \dots y_n \in S_i} M_{y_1} \otimes \dots \otimes M_{y_n}. \quad (5.18)$$

This decomposition shows that what appears to be a global measurement is, in fact, equivalent to independently measuring each qubit optimally. Therefore,

$P(n, f, s, \text{greedy}) = P(n, f, s, \text{PGM})$, and Theorem 5.2 follows immediately by applying the Barnum and Knill's theorem (Theorem 4.1).

5.2.3 Optimality for affine functions

Having provided a universal guarantee for the greedy strategy for arbitrary Boolean functions, we find a class of Boolean functions for which it is optimal. This is the content of Theorem 5.3, which we restate for convenience and prove below.

Theorem 5.3 (Sufficiency). *Let $f : \{0, 1\}^n \rightarrow \{0, 1\}$ be of the form*

$$f(x_1, \dots, x_n) = b_0 \oplus b_1 x_1 \oplus b_2 x_2 \oplus \dots \oplus b_n x_n,$$

for some $b_0, b_1, \dots, b_n \in \{0, 1\}$. Then the greedy strategy is globally optimal; that is,

$$P(n, f, s, \text{greedy}) = P(n, f, s, \text{global})$$

for all $s \in (0, 1)$.

Proof. Any affine Boolean function is either constant or balanced. We consider the cases separately.

Case 1: Constant functions. If $b_i = 0$ for all $i \in \{1, \dots, n\}$, then $f(x) = b_0$ is constant. In this case, S_0 and S_1 are trivially distinguishable (one is empty and the other contains all strings), and both global and greedy strategies achieve success probability 1. Thus, the greedy strategy is optimal.

Case 2: Balanced functions. Without loss of generality³, we assume $b_0 = 0$ and at least one $b_i \neq 0$ for $i \in \{1, \dots, n\}$. Let $I = \{i \in [n] : b_i = 1\}$ denote the set of indices where the coefficient is 1, and let $m = |I| \geq 1$. Then,

$$f(x_1, \dots, x_n) = \bigoplus_{i \in I} x_i. \quad (5.19)$$

The set S_0 consists of all strings where $\bigoplus_{i \in I} x_i = 0$, and S_1 of all strings where $\bigoplus_{i \in I} x_i = 1$. Thus, $|S_0| = |S_1| = 2^{n-1}$. The optimal global success probability is given by the Helstrom measurement:

$$P(n, f, s, \text{global}) = \frac{1}{2} + \frac{1}{4} \|\sigma_0 - \sigma_1\|_1. \quad (5.20)$$

We compute

$$\begin{aligned} \sigma_0 - \sigma_1 &= \frac{1}{2^{n-1}} \sum_{x \in S_0} |\psi_x\rangle\langle\psi_x| - \frac{1}{2^{n-1}} \sum_{x \in S_1} |\psi_x\rangle\langle\psi_x| \\ &= \frac{1}{2^{n-1}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} |\psi_x\rangle\langle\psi_x| \\ &= \frac{1}{2^{n-1}} \sum_{x \in \{0,1\}^n} (-1)^{\bigoplus_{i \in I} x_i} |\psi_x\rangle\langle\psi_x|. \end{aligned} \quad (5.21)$$

For the variables indexed by $i \in I$, we can factor out their contribution. For variables with $i \notin I$, summing over their values contributes a factor of $(|\psi_0\rangle\langle\psi_0| + |\psi_1\rangle\langle\psi_1|)$ for each such

³ If $b_0 = 1$, then we are effectively just flipping the value of f and also our guess.

variable. Thus,

$$\begin{aligned}
 \sigma_0 - \sigma_1 &= \frac{1}{2^{n-1}} \sum_{x \in \{0,1\}^n} (-1)^{\bigoplus_{i \in I} x_i} |\psi_x\rangle\langle\psi_x| \\
 &= \frac{1}{2^{n-1}} \sum_{x_I \in \{0,1\}^m} \sum_{x_{\bar{I}} \in \{0,1\}^{n-m}} \left[\bigotimes_{i \in I} (-1)^{x_i} |\psi_{x_i}\rangle\langle\psi_{x_i}| \right] \otimes \left[\bigotimes_{j \notin I} |\psi_{x_j}\rangle\langle\psi_{x_j}| \right] \\
 &= \frac{1}{2^{n-1}} \left[\bigotimes_{i \in I} (|\psi_0\rangle\langle\psi_0| - |\psi_1\rangle\langle\psi_1|) \right] \otimes \left[\bigotimes_{j \notin I} (|\psi_0\rangle\langle\psi_0| + |\psi_1\rangle\langle\psi_1|) \right].
 \end{aligned} \tag{5.22}$$

The second equality uses the property $(-1)^{\bigoplus_{i \in I} x_i} = \bigotimes_{i \in I} (-1)^{x_i}$. Summation over x_I and $x_{\bar{I}}$ indicates choosing x_i where $i \in I$ and $j \notin I$, respectively. Using multiplicativity of the trace norm $\|A \otimes B\|_1 = \|A\|_1 \cdot \|B\|_1$ and noting that $\left\| |\psi_0\rangle\langle\psi_0| + |\psi_1\rangle\langle\psi_1| \right\|_1 = 2$ (since both are density operators),

$$\begin{aligned}
 \|\sigma_0 - \sigma_1\|_1 &= \frac{1}{2^{n-1}} \left\| |\psi_0\rangle\langle\psi_0| - |\psi_1\rangle\langle\psi_1| \right\|_1^m \cdot \left\| |\psi_0\rangle\langle\psi_0| + |\psi_1\rangle\langle\psi_1| \right\|_1^{n-m} \\
 &= \frac{1}{2^{n-1}} \left\| |\psi_0\rangle\langle\psi_0| - |\psi_1\rangle\langle\psi_1| \right\|_1^m \cdot 2^{n-m} \\
 &= \frac{1}{2^{m-1}} \left\| |\psi_0\rangle\langle\psi_0| - |\psi_1\rangle\langle\psi_1| \right\|_1^m.
 \end{aligned} \tag{5.23}$$

Therefore,

$$P(n, f, s, \text{global}) = \frac{1}{2} + \frac{1}{2^{m+1}} \left\| |\psi_0\rangle\langle\psi_0| - |\psi_1\rangle\langle\psi_1| \right\|_1^m. \tag{5.24}$$

Now, consider the greedy strategy. Let p be the optimal success probability for discriminating $\{\frac{1}{2}|\psi_0\rangle, \frac{1}{2}|\psi_1\rangle\}$ on a single qubit. The greedy strategy measures each qubit independently and computes f on the resulting classical string y . Success occurs when $f(y) = f(x)$, which happens when $\bigoplus_{i \in I} y_i = \bigoplus_{i \in I} x_i$. For the m qubits indexed by I , the parity of measurement outcomes must match the parity of inputs. For the remaining $n - m$ qubits, their outcomes

are irrelevant. The probability that the parity is correct among the m relevant qubits is

$$\begin{aligned} \sum_{\substack{j=0 \\ j \text{ even}}}^m \binom{m}{j} p^{m-j} (1-p)^j &= \frac{1}{2} \left(\sum_{j=0}^m \binom{m}{j} p^{m-j} (1-p)^j + \sum_{j=0}^m (-1)^j \binom{m}{j} p^{m-j} (1-p)^j \right) \\ &= \frac{1}{2} ((p + (1-p))^m + (p - (1-p))^m) = \frac{1}{2} + \frac{1}{2} (2p - 1)^m. \end{aligned} \quad (5.25)$$

For the optimal single-qubit measurement, $p = \frac{1}{2} + \frac{1}{4} \left\| |\psi_0\rangle\langle\psi_0| - |\psi_1\rangle\langle\psi_1| \right\|_1$, resulting in $2p - 1 = \frac{1}{2} \left\| |\psi_0\rangle\langle\psi_0| - |\psi_1\rangle\langle\psi_1| \right\|_1$. Therefore,

$$\begin{aligned} P(n, f, s, \text{greedy}) &= \frac{1}{2} + \frac{1}{2^{m+1}} \left\| |\psi_0\rangle\langle\psi_0| - |\psi_1\rangle\langle\psi_1| \right\|_1^m \\ &= P(n, f, s, \text{global}), \end{aligned} \quad (5.26)$$

for all affine Boolean functions f . □

5.3 Proof of Theorem 5.4: Only affine functions achieve optimality

We now investigate the Boolean functions for which the PGM (and hence the greedy strategy) is globally optimal.

5.3.1 Optimality criterion for the PGM

The following lemma, due to Iten, Renes and Sutter [IRS16], provides a necessary and sufficient condition for the optimality of the PGM:

Lemma 5.6 (PGM optimality [IRS16]). *Let $\{(\eta_x, \rho_x)\}$ be an ensemble of quantum states, and let $|\xi_x\rangle_{BB'}$ denote a purification of $(\rho_x)_B$ for each x . Then the pretty good measurement*

is optimal for distinguishing $\{(\eta_x, \rho_x)\}$ if and only if

$$[G_{X'B'}, \tau_{X'B'}] = 0, \quad (5.27)$$

where $G_{X'B'}$ is the generalized Gram matrix defined as

$$G_{X'B'} := \sum_{x, x'} \sqrt{\eta_x \eta_{x'}} |x\rangle\langle x'|_{X'} \otimes \text{Tr}_B(|\xi_x\rangle\langle \xi_{x'}|_{BB'}) \quad (5.28)$$

and $\tau_{X'B'}$ is the block-diagonal operator

$$\tau_{X'B'} := \sum_{x'} |x'\rangle\langle x'|_{X'} \otimes \langle x'| \sqrt{G_{X'B'}} |x'\rangle_{X'}, \quad (5.29)$$

formed from the diagonal blocks of $\sqrt{G_{X'B'}}$.

This criterion is powerful because it reduces the question of PGM optimality to checking whether two specific operators commute. For our problem, the generalized Gram matrix of the Boolean ensemble is given by the following lemma.

Lemma 5.7. *booleangrams* The generalized Gram matrix G for the Boolean ensemble has the block form

$$G = \frac{1}{2^n} \begin{pmatrix} \Gamma_0 & \Gamma' \\ \Gamma'^* & \Gamma_1 \end{pmatrix} \quad (5.30)$$

where:

- Γ_i is the Gram matrix of the states in S_i , with entries $(\Gamma_i)_{xy} = \langle \psi_x | \psi_y \rangle$ for $x, y \in S_i$,
- Γ' is the cross-Gram matrix with entries $(\Gamma')_{xy} = \langle \psi_x | \psi_y \rangle$ for $x \in S_0$ and $y \in S_1$.

Proof. To apply the PGM optimality criterion from Lemma 5.6, we must compute the generalized Gram matrix G for the Boolean ensemble (Eq. (5.3)). We begin by constructing explicit purifications of the mixed states σ_0 and σ_1 .

For each $i \in \{0, 1\}$, a purification of σ_i is given by

$$|\xi_i\rangle_{BB'} = \frac{1}{\sqrt{|S_i|}} \sum_{x \in S_i} |\psi_x\rangle_B |x\rangle_{B'}, \quad (5.31)$$

where the auxiliary system B' has dimension $|S_i|$, and the states $\{|y\rangle_{B'}\}_{y \in S_i}$ form an orthonormal basis of B' . Here, B denotes the physical n -qubit system containing the state $|\psi_x\rangle$, while B' is the purifying system that tracks which element of S_i was selected.

Recall from Eq. (5.28) that the generalized Gram matrix is defined as

$$G_{X'B'} := \sum_{i,j \in \{0,1\}} \sqrt{p_i p_j} |i\rangle\langle j| \otimes \text{Tr}_B (|\xi_i\rangle\langle \xi_j|_{BB'}), \quad (5.32)$$

where $i, j \in \{0, 1\}$ index the two hypotheses (whether the input belongs to S_0 or S_1), and p_x are the prior probabilities: $p_0 = |S_0|/2^n$ and $p_1 = |S_1|/2^n$.

We compute the partial trace:

$$\begin{aligned} \text{Tr}_B (|\xi_i\rangle\langle \xi_j|_{BB'}) &= \text{Tr}_B \left(\frac{1}{\sqrt{|S_i||S_j|}} \sum_{\substack{x \in S_i \\ y \in S_j}} |\psi_x\rangle\langle \psi_y|_B \otimes |x\rangle\langle y|_{B'} \right) \\ &= \frac{1}{\sqrt{|S_i||S_j|}} \sum_{\substack{x \in S_i \\ y \in S_j}} \langle \psi_y | \psi_x \rangle |x\rangle\langle y|_{B'}. \end{aligned} \quad (5.33)$$

Substituting into the definition of G and using $\sqrt{p_i p_j} = \sqrt{|S_i| |S_j|} / 2^n$, we obtain

$$\begin{aligned}
 G_{X'B'} &= \sum_{i,j \in \{0,1\}} \sqrt{p_i p_j} |i\rangle\langle j| \otimes \text{Tr}_B (|\xi_i\rangle\langle \xi_j|_{BB'}) \\
 &= \sum_{i,j \in \{0,1\}} \frac{\sqrt{|S_i| |S_j|}}{2^n} |i\rangle\langle j| \otimes \frac{1}{\sqrt{|S_i| |S_j|}} \sum_{\substack{x \in S_i \\ y \in S_j}} \langle \psi_y | \psi_x \rangle |x\rangle\langle y|_{B'} \\
 &= \frac{1}{2^n} \sum_{i,j \in \{0,1\}} |i\rangle\langle j| \otimes \sum_{\substack{x \in S_i \\ y \in S_j}} \langle \psi_y | \psi_x \rangle |x\rangle\langle y|_{B'}.
 \end{aligned} \tag{5.34}$$

The block structure follows directly from this calculation. The (i, j) -block of G corresponds to the operator acting on $|i\rangle\langle j|$, which contains the matrix of inner products $\langle \psi_y | \psi_x \rangle$ for $x \in S_i$ and $y \in S_j$. Since the states have real inner products, we use the identity $\langle \psi_y | \psi_x \rangle = \langle \psi_x | \psi_y \rangle$ to match the order in the standard definition of Gram matrices. For $i = j$, this gives the Gram matrices Γ_0 and Γ_1 . For $i \neq j$, this gives the cross-Gram matrices Γ' and Γ'^* . The global factor $1/2^n$ appears uniformly across all blocks.

□

5.3.2 Structural constraints from PGM optimality

We now translate the PGM optimality condition into an explicit algebraic relation between the Gram matrices Γ_0 , Γ_1 , and Γ' .

Theorem 5.8. *If the PGM is optimal for evaluating a Boolean function f , then the Gram matrices satisfy*

$$\Gamma_0 \Gamma' = \Gamma' \Gamma_1. \tag{5.35}$$

Proof. By Lemma 5.6, the PGM is optimal if and only if $[G, \tau] = 0$, where τ is the block-diagonal operator formed from the diagonal blocks of $\sqrt{G}$. Ignoring the global factor $1/2^n$

(which does not affect the Gram matrix relation),

$$G = \begin{pmatrix} \Gamma_0 & \Gamma' \\ \Gamma'^* & \Gamma_1 \end{pmatrix}, \quad \sqrt{G} = \begin{pmatrix} X & Y \\ Y^* & Z \end{pmatrix}, \quad \tau = \begin{pmatrix} X & 0 \\ 0 & Z \end{pmatrix} \quad (5.36)$$

where X and Z are positive semidefinite matrices, and Y satisfies the constraint imposed by $(\sqrt{G})^2 = G$. Expanding this constraint block-wise, we have

$$X^2 + YY^* = \Gamma_0, \quad XY + YZ = \Gamma', \quad Z^2 + Y^*Y = \Gamma_1. \quad (5.37)$$

Computing both sides of $G\tau = \tau G$, we have

$$G\tau = \begin{pmatrix} \Gamma_0 X & \Gamma' Z \\ \Gamma'^* X & \Gamma_1 Z \end{pmatrix}, \quad \tau G = \begin{pmatrix} X\Gamma_0 & X\Gamma' \\ Z\Gamma'^* & Z\Gamma_1 \end{pmatrix}. \quad (5.38)$$

Equating the $(1, 2)$ -blocks yields

$$\Gamma' Z = X\Gamma'. \quad (5.39)$$

We substitute the second equation of Eq. (5.37), $\Gamma' = XY + YZ$, into Eq. (5.39) and simplify to get

$$YZ^2 = X^2 Y. \quad (5.40)$$

Applying the other two equations of Eq. (5.37), we have

$$Y(\Gamma_1 - Y^*Y) = (\Gamma_0 - YY^*)Y, \quad (5.41)$$

which we expand and cancel the common term YY^*Y to obtain

$$Y\Gamma_1 = \Gamma_0 Y. \quad (5.42)$$

Finally, we compute $\Gamma_0\Gamma'$ using $\Gamma' = XY + YZ$ and the relations we have established. Note that $[\Gamma_0, X] = 0$ follows from comparing the $(1, 1)$ -blocks of $G\tau = \tau G$, and $[Z, \Gamma_1] = 0$ follows similarly from the $(2, 2)$ -blocks. Therefore,

$$\Gamma_0\Gamma' = \Gamma_0XY + \Gamma_0YZ = X\Gamma_0Y + Y\Gamma_1Z = XY\Gamma_1 + YZ\Gamma_1 = \Gamma'\Gamma_1, \quad (5.43)$$

completing the proof. $\square$

5.3.3 Consequences of the matrix relation

Having established that PGM optimality implies $\Gamma_0\Gamma' = \Gamma'\Gamma_1$, we now extract the structural consequences of this relation. A key observation is that the inner product between any two product states can be expressed in terms of the inner product $s = \langle\psi_0|\psi_1\rangle$ and the Hamming distance:

$$\langle\psi_x|\psi_y\rangle = \prod_{i=1}^n \langle\psi_{x_i}|\psi_{y_i}\rangle = s^{d(x,y)}, \quad (5.44)$$

where $d(x, y)$ is the Hamming distance between two n -bit strings x and y . This means that every entry of the matrices Γ_0 , Γ_1 , and Γ' is a monomial in s , allowing us to view these matrices as matrix-valued polynomials in s . The relation $\Gamma_0\Gamma' = \Gamma'\Gamma_1$ thus becomes a polynomial identity that must hold for all $s \in (0, 1)$.

The polynomial structure of the matrix relation $\Gamma_0\Gamma' = \Gamma'\Gamma_1$ yields two immediate and powerful consequences. First, by comparing the number of monic terms in corresponding matrix entries, we show that the function f must be constant ($|S_0||S_1| = 0$) or balanced ($|S_0| = |S_1|$).

Theorem 5.9. *If the Gram matrix relation in Eq. (5.35) holds, then the function f must be either constant ($|S_0||S_1| = 0$) or balanced ($|S_0| = |S_1|$), except possibly for a finite set of*

values of $s \in (0, 1)$.

Proof. By Eq. (5.44), every entry of the matrices Γ_0 , Γ_1 , and Γ' is a monic monomial in $s = \langle \psi_0 | \psi_1 \rangle$. Thus, every term in $\Gamma_0 \Gamma'$ is a sum of $|S_0|$ monic monomials, and every term in $\Gamma' \Gamma_1$ is a sum of $|S_1|$ monic monomials. We aim to show that $\Gamma_0 \Gamma' = \Gamma' \Gamma_1$ implies the function to be constant or balanced, which we do via its contrapositive. If the function is not constant, i.e., $|S_0| |S_1| \neq 0$, then $\Gamma_0 \Gamma'$ and $\Gamma' \Gamma_1$ are non-empty. If the function is additionally not balanced, i.e., $|S_0| \neq |S_1|$, then the elements of $\Gamma_0 \Gamma'$ and $\Gamma' \Gamma_1$ cannot coincide, except possibly for finitely many values of s . $\square$

Two comments are in order. First, the implication of this theorem does not hold for some s due to the Gram matrix relation being true regardless of the function structure. Additionally, the finite set of s exceptions depends on f . In this and the following theorems, this clause means: Let f be a Boolean function. For all except possibly finitely many $s \in (0, 1)$, the implication holds. Second, this constant or balanced consequence is used in Theorem 5.10 below, and later in Lemma 5.11. As their statements are trivial for constant functions, we assume balanced functions when the Gram matrix relation holds. We do this for ease of comprehension, noting that the same reasoning may hold for arbitrary functions when accounting for vacuously true statements.

The second consequence could be understood in terms of the *Boolean hypercube* graph (see Section 5.4 for details), and we use it shortly after to derive details about function structure. In this graph, nodes are n -bit strings and edges connect two nodes if and only if they differ by exactly one coordinate. For any pair of vertices $x \in S_0$ and $y \in S_1$ in the Boolean hypercube, consider all intermediate vertices w such that a path from x to y via w has total length L (where the path need not be shortest overall, just the sum $d(x, w) + d(w, y) = L$). By equating coefficients of powers of s in the polynomial identity, we show that, for balanced

functions, the number of such intermediate vertices in S_0 equals the number in S_1 , for every choice of x , y and L .

Theorem 5.10. *If the Gram matrix relation in Eq. (5.35) holds and the function f is balanced, then for all integers $L \in \{0, 1, \dots, 2n\}$ and for all $x \in S_0$, $y \in S_1$,*

$$|\{u \in S_0 : d(x, u) + d(u, y) = L\}| = |\{v \in S_1 : d(x, v) + d(v, y) = L\}| \quad (5.45)$$

for all but possibly finitely many values of $s \in (0, 1)$.

Proof. Fix any arbitrary $x \in S_0$ and $y \in S_1$. The (x, y) -entries of the matrix products in Eq. (5.35) are

$$(\Gamma_0 \Gamma')_{xy} = \sum_{u \in S_0} s^{d(x, u) + d(u, y)} \quad (\Gamma' \Gamma_1)_{xy} = \sum_{v \in S_1} s^{d(x, v) + d(v, y)}. \quad (5.46)$$

Equality of these expressions as polynomials in s means

$$\sum_{u \in S_0} s^{d(x, u) + d(u, y)} = \sum_{v \in S_1} s^{d(x, v) + d(v, y)}. \quad (5.47)$$

We can rewrite each sum by grouping terms according to the total distance

$$L = d(x, w) + d(w, y),$$

$$\sum_{L=0}^{2n} s^L |\{u \in S_0 : d(x, u) + d(u, y) = L\}| = \sum_{L=0}^{2n} s^L |\{v \in S_1 : d(x, v) + d(v, y) = L\}|. \quad (5.48)$$

This is a polynomial identity in s . For this to hold, either the coefficients of each power s^L on both sides must agree (giving Eq. (5.45)), or s must be one of (at most $2n$) roots of the above polynomial. □

5.3.4 Necessity of the affine structure

We now combine the results from the previous sections to obtain a complete characterization of when the greedy strategy is optimal. A crucial element in our proof is the following lemma about the Boolean hypercube, the proof of which can be found in Section 5.4.

Lemma 5.11 (Hypercube flip characterization). *hypercubeflip Let Q_n be the n -dimensional hypercube graph, and let $A, B \subseteq V(Q_n)$ be a partition satisfying*

- (i) $A \cup B = V(Q_n)$ and $A \cap B = \emptyset$;
- (ii) $|A| = |B| = 2^{n-1}$ (balanced partition);
- (iii) for all $L \in \mathbb{N}$ and for all $x \in A, y \in B$,

$$|\{u \in A : d(x, u) + d(u, y) = L\}| = |\{v \in B : d(x, v) + d(v, y) = L\}|. \quad (5.49)$$

Then there exists a unique index $i \in [n]$ such that for every $x = (x_1, \dots, x_n) \in \{0, 1\}^n$,

$$x \in A \iff x^{(i)} \in B, \quad (5.50)$$

where $x^{(i)} = (x_1, \dots, x_{i-1}, \overline{x_i}, x_{i+1}, \dots, x_n)$ denotes the bit-flip of x in the i -th coordinate.

We start with the following theorem, which states that the functions for which the greedy strategy is optimal must necessarily follow a recursive structure.

Theorem 5.12. *If the Gram matrix relation in Eq. (5.35) holds, then the function f is either constant or there exists an index $i \in [n]$ and a Boolean function $g : \{0, 1\}^{n-1} \rightarrow \{0, 1\}$ such that*

$$f(x_1, x_2, \dots, x_n) = g(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \oplus x_i \quad (5.51)$$

for all but possibly finitely many values of $s \in (0, 1)$.

Proof. Starting from the Gram matrix relation, Theorem 5.9 gives that the function is either constant (in which case the proof ends) or balanced. From Theorem 5.10, the Gram matrix relation additionally means that for all $x \in S_0$, $y \in S_1$ and integers L ,

$$|\{u \in S_0 : d(x, u) + d(u, y) = L\}| = |\{v \in S_1 : d(x, v) + d(v, y) = L\}| \quad (5.52)$$

for all but possibly finitely many values of the inner product s . By the hypercube structure of $\{0, 1\}^n$ and Lemma 5.11, the above two consequences imply that there exists an index $i \in [n]$ such that flipping the i -th bit maps S_0 bijectively onto S_1 . Equivalently, for every $x = (x_1, \dots, x_n) \in \{0, 1\}^n$,

$$x \in S_0 \iff x^{(i)} \in S_1, \quad (5.53)$$

where $x^{(i)}$ denotes the bit-flip of x in the i -th coordinate. Define

$$g(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) := f(x_1, \dots, x_{i-1}, 0, x_{i+1}, \dots, x_n). \quad (5.54)$$

Then, by construction,

$$f(x_1, \dots, x_n) = g(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \oplus x_i, \quad (5.55)$$

which proves Eq. (5.51), as needed. $\square$

The recursive structure revealed in Theorem 5.12 allows us to apply induction to completely characterize optimal functions by proving Theorem 5.4.

We briefly outline the essential steps of the proof for the reader's convenience. We have previously seen an equivalent condition for PGM optimality in Lemma 5.6, from which we

derived a necessary condition in Theorem 5.8: the Gram matrix relation in Eq. (5.35). The main consequence of this is Theorem 5.12, which constrains the function to a recursive form given by Eq. (5.51), which when shown via induction for each n implies the function is affine. In particular, our inductive hypothesis is: if g is an $(n - 1)$ -bit Boolean function for which Eq. (5.35) holds, then g must be affine. The remaining part of the induction step is to show the antecedent of the above, which we do in Lemma 5.13. Specifically, if Eq. (5.35) and Eq. (5.51) hold for an n -bit Boolean function f , then the $(n - 1)$ -bit Boolean function g that appears in the recursive form of f must satisfy the Gram matrix relation in Eq. (5.35). Therefore any Boolean function satisfying Eq. (5.35) must be affine, implying that f itself must be affine.

Theorem 5.4 (Necessity). *Suppose a Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ is not affine, meaning that it is not of the form in Eq. (5.7). Then the greedy strategy is not globally optimal; that is,*

$$P(n, f, s, \text{greedy}) < P(n, f, s, \text{global})$$

for all but possibly finitely many values of $s \in (0, 1)$.

Proof. We prove this statement via its contrapositive. Since the greedy strategy is globally optimal for evaluating a Boolean function f , Theorem 5.8 implies that the associated Gram matrices satisfy $\Gamma_0 \Gamma' = \Gamma' \Gamma_1$. Therefore, it suffices to show that this in turn means the function f must be an affine Boolean function. We prove this statement by induction on the number of variables n .

Base case: The case when $n = 1$ is trivial. For illustrative purposes, we additionally show the $n = 2$ case. By Theorem 5.9, f is either constant or balanced. If f is constant, the claim is again trivial. If f is balanced then $|f^{-1}(0)| = 2$. Without loss of generality, assume that $f(0, 0) = 0$; otherwise, we may replace f by its complement $\bar{f}$, which is equivalent to flipping

the value of b_0 . The possible choices for $f^{-1}(0) \subseteq \{00, 01, 10, 11\}$ are: $\{00, 01\}$, $\{00, 10\}$, and $\{00, 11\}$. Respectively, these correspond to the functions

$$f(x_1, x_2) = x_1, \quad f(x_1, x_2) = x_2, \quad f(x_1, x_2) = x_1 \oplus x_2. \quad (5.56)$$

All three are affine, confirming the claim for $n = 2$.

Induction step: Let $n \geq 2$ be an integer and assume the inductive hypothesis for $n - 1$, which is that if the Gram matrices of an $(n - 1)$ -bit Boolean function satisfy the Gram matrix relation in Eq. (5.35), then that function is affine.

Consider an arbitrary n -bit Boolean function f that (by the antecedent of the statement to be proven) satisfies the Gram matrix relation in Eq. (5.35). By Theorem 5.12, this function f is either constant (and thus affine with $b_1 = \dots = b_n = 0$ and b_0 equal to the function value, meaning we are done), or there exists an index $i \in [n]$ and an $(n - 1)$ -bit Boolean function $g : \{0, 1\}^{n-1} \rightarrow \{0, 1\}$ such that

$$f(x_1, \dots, x_n) = g(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \oplus x_i. \quad (5.57)$$

These two facts let us show the antecedent of the induction hypothesis via the following statement.

Lemma 5.13. *Let G_0 (resp. G_1) be the Gram matrix of the states corresponding to inputs of g that map to 0 (resp. 1), and let G' be the cross-Gram matrix. If f has the form in Eq. (5.51) and satisfies the Gram matrix relation in Eq. (5.35), then g also satisfies the Gram matrix relation, i.e., $G_0 G' = G' G_1$.*

Proof. We have that $\Gamma_0\Gamma' = \Gamma'\Gamma_1$. Order the elements of S_0 by first listing those with $x_i = 0$ and then those with $x_i = 1$. Analogously, order S_1 with $x_i = 1$ elements first and $x_i = 0$ second. With respect to this ordering, the Gram matrices have the block form

$$\Gamma_0 = \Gamma_1 = \begin{pmatrix} G_0 & sG' \\ sG'^* & G_1 \end{pmatrix}, \quad \Gamma' = \begin{pmatrix} sG_0 & G' \\ G'^* & sG_1 \end{pmatrix}, \quad (5.58)$$

where $s = \langle \psi_0 | \psi_1 \rangle$. We now compute both sides of Eq. (5.35). First,

$$\Gamma_0\Gamma' = \begin{pmatrix} sG_0^2 + sG'G'^* & G_0G' + s^2G'G_1 \\ s^2G'^*G_0 + G_1G'^* & sG'^*G' + sG_1^2 \end{pmatrix}. \quad (5.59)$$

Similarly,

$$\Gamma'\Gamma_1 = \begin{pmatrix} sG_0^2 + sG'G'^* & s^2G_0G' + G'G_1 \\ G'^*G_0 + s^2G_1G'^* & sG'^*G' + sG_1^2 \end{pmatrix}. \quad (5.60)$$

Equality of these two matrices implies equality of their $(1, 2)$ -blocks, hence

$$G_0G' + s^2G'G_1 = s^2G_0G' + G'G_1. \quad (5.61)$$

Rearranging gives

$$(1 - s^2)G_0G' = (1 - s^2)G'G_1. \quad (5.62)$$

Since $s \in (0, 1)$, we conclude

$$G_0G' = G'G_1, \quad (5.63)$$

as claimed. $\square$

Returning to the main proof: since we previously showed that f satisfies Eq. (5.35) and Eq. (5.51), the just-proven Lemma 5.13 implies that g also satisfies Eq. (5.35). By the

induction hypothesis, g is then affine, meaning it has the form

$$g(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) = b_0 \oplus \bigoplus_{j \neq i} b_j x_j. \quad (5.64)$$

Substituting into Eq. (5.51),

$$f(x_1, \dots, x_n) = \left(b_0 \oplus \bigoplus_{j \neq i} b_j x_j \right) \oplus x_i = b_0 \oplus \bigoplus_{j=1}^n b_j x_j, \quad (5.65)$$

where we set $b_i = 1$ and b_j for $j \neq i$ as in g . This is precisely the affine form in Eq. (5.7), completing the induction and the proof. $\square$

5.4 The Boolean hypercube graph

The structure of Boolean functions is intimately connected to the geometry of the n -dimensional hypercube. For $n \in \mathbb{N}$, the n -dimensional hypercube graph Q_n is defined as follows:

- The vertex set is $V(Q_n) = \{0, 1\}^n$, the set of all n -bit strings.
- Two vertices $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ are adjacent if and only if they differ in exactly one coordinate.

The *Hamming distance* $d(x, y)$ between two vertices x and y is the number of coordinates in which they differ, which equals the graph distance in Q_n :

$$d(x, y) = |\{i \in [n] : x_i \neq y_i\}|. \quad (5.66)$$

We denote adjacency by ' $\sim$ ', meaning $x \sim y$ if and only if $d(x, y) = 1$. The complete graph on two vertices is denoted as K_2 , i.e., the graph with vertex set $\{0, 1\}$ with a single edge

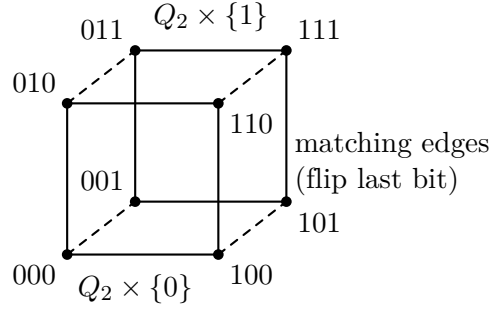


Figure 5.1: The three-dimensional hypercube $Q_3 \cong Q_2 \square K_2$, visualized as two copies of Q_2 with a perfect matching, shown by dashed edges, connecting corresponding vertices that differ in the last-bit coordinate.

connecting them. A perfect matching in a graph is a set of pairwise disjoint edges such that every vertex of the graph is incident to exactly one edge in the set.

An important property of the hypercube is its inductive structure via the Cartesian product. Recall that the Cartesian product $G \square H$ of two graphs has vertex set $V(G) \times V(H)$, where $(g, h) \sim (g', h')$ if either $g = g'$ and $h \sim h'$ in H , or $h = h'$ and $g \sim g'$ in G . The hypercube satisfies the recursive relation

$$Q_{n+1} \cong Q_n \square K_2, \quad (5.67)$$

meaning Q_{n+1} can be viewed as two copies of Q_n connected by a perfect matching. More precisely, if we partition $V(Q_{n+1})$ according to the last bit, we obtain two sub-cubes $Q_n \times \{0\}$ and $Q_n \times \{1\}$, each isomorphic to Q_n , with matching edges connecting vertices that differ only in the last coordinate. Figure 5.1 illustrates this structure for Q_3 .

The following lemma characterizes balanced partitions of the hypercube satisfying the path-counting symmetry (Eq. (5.49)) which are precisely those obtained by flipping a single coordinate.

Lemma 5.11 (Hypercube flip characterization). *Let Q_n be the n -dimensional hypercube graph, and let $A, B \subseteq V(Q_n)$ be a partition satisfying*

- (i) $A \cup B = V(Q_n)$ and $A \cap B = \emptyset$;
- (ii) $|A| = |B| = 2^{n-1}$ (balanced partition);
- (iii) for all $L \in \mathbb{N}$ and for all $x \in A, y \in B$,

$$|\{u \in A : d(x, u) + d(u, y) = L\}| = |\{v \in B : d(x, v) + d(v, y) = L\}| \quad (5.49)$$

Then there exists a unique index $i \in [n]$ such that for every $x = (x_1, \dots, x_n) \in \{0, 1\}^n$,

$$x \in A \iff x^{(i)} \in B,$$

where $x^{(i)} = (x_1, \dots, x_{i-1}, \overline{x_i}, x_{i+1}, \dots, x_n)$ denotes the bit flip of x in the i -th coordinate.

To prove Lemma 5.11, we need the following lemma which establishes that the two parts must have adjacent vertices.

Lemma 5.14. *Let $A, B \subseteq V(Q_n)$ satisfy the hypotheses of Lemma 5.11. Then*

$$\min\{d(x, y) : x \in A, y \in B\} = 1. \quad (5.68)$$

Proof. Let $k = \min\{d(x, y) : x \in A, y \in B\}$. Assume for contradiction that $k > 1$. Choose $x \in A$ and $y \in B$ with $d(x, y) = k$, and let $x-x_1-\dots-x_{k-1}-y$ be a shortest path from x to y in Q_n .

Then, $d(x, x_1) = 1$ and $d(x_1, y) = k - 1 < k$. Since A and B form a partition, for any $x_1 \in V(Q_n)$ we either have:

- $x_1 \in A$, meaning $d(x_1, y) < k$ with $x_1 \in A$ and $y \in B$, contradicting minimality of k ,
- or

- $x_1 \in B$, meaning $d(x, x_1) = 1 < k$ with $x \in A$ and $x_1 \in B$, again contradicting minimality.

Hence, $k = 1$. □

Proof of Lemma 5.11. The case when $n = 1$ is trivial. For illustrative purposes, we additionally show the case when $n = 2$.

The vertex set of the two-dimensional hypercube is $V(Q_2) = \{00, 01, 10, 11\}$. By Lemma 5.14, there exist vertices $x \in A$ and $y \in B$ such that $d(x, y) = 1$. Without loss of generality, choose $x = 00$ and $y = 01$. Since $|A| = |B| = 2$, the only possible balanced partitions with $00 \in A$ and $01 \in B$ are

$$A = \{00, 10\}, B = \{01, 11\} \quad \text{or} \quad A = \{00, 11\}, B = \{01, 10\}. \quad (5.69)$$

In both cases, B is obtained from A by flipping the second bit. Thus, the claim holds for $n = 2$.

We next prove the statement for an arbitrary positive integer n . By Lemma 5.14, choose vertices $x \in A$ and $y \in B$ such that $d(x, y) = 1$. Let $i \in [n]$ be the unique coordinate where x and y differ, so $y = x^{(i)}$. We now show that coordinate i implements the flip globally.

Take any coordinate $k \neq i$. The four vertices $x, x^{(i)}, x^{(i,k)} = (x^{(i)})^{(k)}, x^{(k)}$ form a 4-cycle in Q_n :

$$x - x^{(i)} - x^{(i,k)} - x^{(k)} - x. \quad (5.70)$$

We now consider two cases based on whether $x^{(i,k)}$ belongs to A or B .

Case 1: $x^{(i,k)} \in B$. Apply Eq. (5.49) at $L = 2$ to the antipodal pair $(x, x^{(i,k)})$. The only

two length-2 paths between them pass through $x^{(i)}$ and $x^{(k)}$ respectively. Since $x^{(i,k)} \in B$, balancing the count forces $x^{(k)} \in A$. Thus, we have

$$x \in A, \quad x^{(i)} \in B, \quad x^{(i,k)} \in B, \quad x^{(k)} \in A. \quad (5.71)$$

The pair $(x^{(i,k)}, x^{(k)})$ differs only in coordinate i , just as $(x, x^{(i)})$ does.

Case 2: $x^{(i,k)} \in A$. If $x^{(k)} \in A$, then it belongs to a different partition than $x^{(i)}$, and we may apply Eq. (5.49) with $L = 2$ to the antipodal pair $(x^{(i)}, x^{(k)})$. The only two length-2 paths between them pass through x and $x^{(i,k)}$, both in A , giving two A -intermediates and zero B -intermediates, which leads to a contradiction. Hence, we must have $x^{(k)} \in B$, meaning

$$x \in A, \quad x^{(i)} \in B, \quad x^{(i,k)} \in A, \quad x^{(k)} \in B. \quad (5.72)$$

Again, the pair $(x^{(i,k)}, x^{(k)})$ differs only in coordinate i , just as $(x, x^{(i)})$ does.

We have established the following propagation property: if $x \in A$ and $x^{(i)} \in B$, then for every coordinate $k \neq i$, the vertices $x^{(k)}$ and $x^{(i,k)}$ belong to distinct partitions. In other words, flipping any non- i coordinate sends a split pair to a split pair, preserving the same flip coordinate i .

Since every vertex pair $z, z^{(i)} \in V(Q_n)$ is reachable from the pair x, y by a finite sequence of non- i coordinate flips, we conclude that

$$\forall x \in V(Q_n), \quad x \in A \iff x^{(i)} \in B, \quad (5.73)$$

ending the proof. □

5.5 Greedy strategy performance for non-affine functions

In this section, we examine two important non-affine functions: the AND function and the majority (MAJ) function. For these functions, the greedy strategy is suboptimal, but understanding the gap between greedy and global performance provides insight into when adaptive or joint measurements offer practical advantages.

5.5.1 The AND function

The AND function on n bits is defined by

$$\text{AND}(x_1, \dots, x_n) = x_1 \wedge x_2 \wedge \dots \wedge x_n = \begin{cases} 1 & \text{if } x_i = 1 \text{ for all } i, \\ 0 & \text{otherwise.} \end{cases} \quad (5.74)$$

This function is highly unbalanced: $S_1 = \{11 \dots 1\}$ contains only the all-ones string, while S_0 contains all other $2^n - 1$ strings.

Greedy success probability

For the greedy strategy, we measure each qubit independently using the optimal single-qubit measurement, which succeeds with probability

$$p = \frac{1}{2} + \frac{1}{4} \left\| |\psi_0\rangle\langle\psi_0| - |\psi_1\rangle\langle\psi_1| \right\|_1 = \frac{1}{2}(1 + \sqrt{1 - s^2})$$

in distinguishing $|\psi_0\rangle$ from $|\psi_1\rangle$ with equal priors. The greedy strategy fails in evaluating AND when

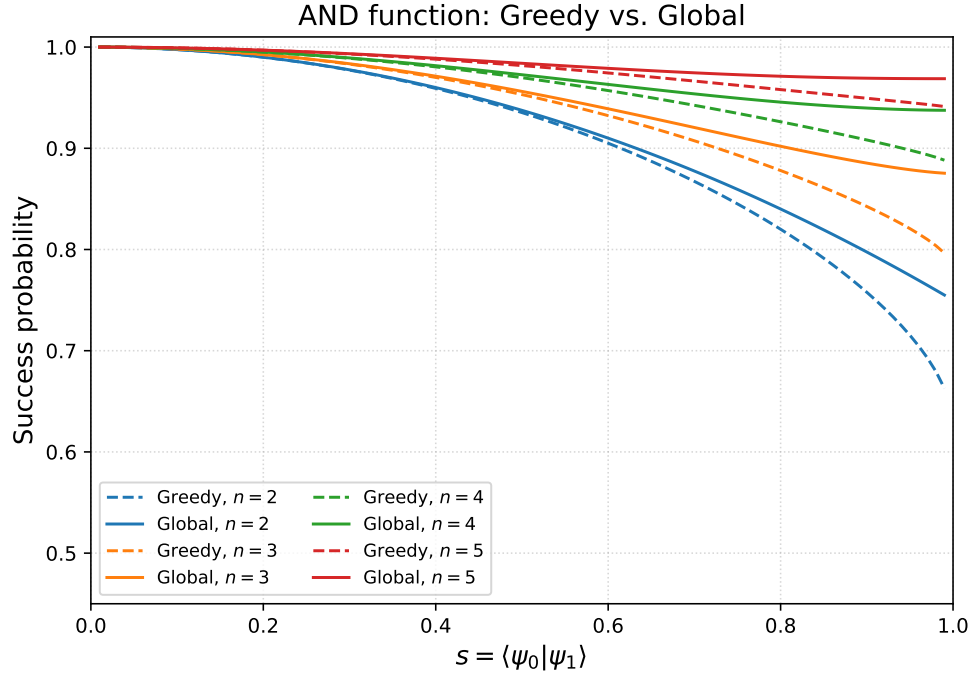


Figure 5.2: Greedy and global success probabilities for the AND function. As n increases, both strategies approach perfect success exponentially quickly, and the gap between them decreases rapidly.

- $x = 11 \dots 1$ but we guess $y \neq 11 \dots 1$, which happens with probability $(1 - p^n)/2^n$.
- $x \neq 11 \dots 1$ but we guess $y = 11 \dots 1$, which happens with probability

$$\sum_{x \neq 11 \dots 1} \frac{1}{2^n} p^{w(x)} (1 - p)^{n-w(x)} = \frac{1}{2^n} \sum_{k=0}^{n-1} \binom{n}{k} p^k (1 - p)^{n-k} = \frac{1}{2^n} (1 - p^n), \quad (5.75)$$

where $w(x)$ is the Hamming weight of x . The greedy success probability is therefore

$$P(n, \text{AND}, s, \text{greedy}) = 1 - \frac{1 - p^n}{2^{n-1}}. \quad (5.76)$$

Asymptotic behavior

As $n \rightarrow \infty$ with p fixed (corresponding to fixed non-orthogonal states $|\psi_0\rangle$ and $|\psi_1\rangle$), we have $P(n, \text{AND}, s, \text{greedy}) \rightarrow 1$ since $1/2^{n-1} \rightarrow 0$ exponentially fast.

This limiting behavior is intuitive: with overwhelming probability $(2^n - 1)/2^n \rightarrow 1$, the input contains at least one zero. The probability of accidentally guessing all ones when the input is not all ones vanishes exponentially. Thus, the greedy strategy succeeds with probability approaching 1. The global optimal strategy achieves slightly better performance by leveraging correlations across qubits, but this advantage becomes negligible as n increases. For large n , the extreme imbalance of the AND function dominates, and both strategies approach perfect success.

5.5.2 The majority function

For odd $n = 2k + 1$, the majority function is defined by

$$\text{MAJ}(x_1, \dots, x_{2k+1}) = \begin{cases} 1 & \text{if } \sum_{i=1}^{2k+1} x_i > k, \\ 0 & \text{if } \sum_{i=1}^{2k+1} x_i \leq k. \end{cases} \quad (5.77)$$

Unlike the AND function, majority is balanced: $|S_0| = |S_1| = 2^{2k}$.

Greedy success probability

We now compute the greedy success probability for the majority function. The greedy strategy measures each qubit independently with the optimal single-qubit measurement, which correctly identifies the bit value with probability p and incorrectly identifies it with probability $1 - p$.

Given an input string x with Hamming weight $w(x) = j$ (i.e., j ones and $2k + 1 - j$ zeros), the measurement produces an output string y whose Hamming weight $w(y)$ is a random variable. We can decompose $w(y)$ as

$$w(y) = Y_1 + Y_2, \quad (5.78)$$

where

- Y_1 is the number of ones in y that came from ones in x (correctly identified ones), with $Y_1 \sim \text{Binomial}(j, p)$;
- Y_2 is the number of ones in y that came from zeros in x (incorrectly flipped zeros), with $Y_2 \sim \text{Binomial}(2k + 1 - j, 1 - p)$.

Since different qubits are measured independently, Y_1 and Y_2 are independent random variables.

The greedy strategy succeeds when $\text{MAJ}(x) = \text{MAJ}(y)$, which occurs when:

- If $j \leq k$ (so $f(x) = 0$): we need $w(y) \leq k$, i.e., $Y_1 + Y_2 \leq k$.
- If $j > k$ (so $f(x) = 1$): we need $w(y) > k$, i.e., $Y_1 + Y_2 > k$.

Let γ_j denote the conditional success probability given that $w(x) = j$. For $j \leq k$:

$$\gamma_j = P(Y_1 + Y_2 \leq k) = \sum_{a=0}^j \sum_{b=0}^{\min(2k+1-j, k-a)} \binom{j}{a} p^a (1-p)^{j-a} \binom{2k+1-j}{b} (1-p)^b p^{2k+1-j-b}. \quad (5.79)$$

For $j > k$ (equivalently, $j \geq k + 1$):

$$\gamma_j = \mathbb{P}(Y_1 + Y_2 > k) = \sum_{a=0}^j \sum_{b=\max(0, k+1-a)}^{2k+1-j} \binom{j}{a} p^a (1-p)^{j-a} \binom{2k+1-j}{b} (1-p)^b p^{2k+1-j-b}. \quad (5.80)$$

By symmetry of the majority function, we have $\gamma_j = \gamma_{2k+1-j}$ for all j . To see this, note that if we flip all bits in both x and y , the majority value also flips, and the distribution of measurement outcomes remains the same (with the roles of 0 and 1 interchanged). Therefore, the conditional success probability for j ones is the same as for j zeros (i.e., $2k+1-j$ ones).

The greedy success probability is:

$$\mathbb{P}(n = 2k + 1, \text{MAJ}, s, \text{greedy}) = \sum_{j=0}^{2k+1} \frac{1}{2^{2k+1}} \binom{2k+1}{j} \gamma_j = \sum_{j=0}^k \frac{1}{2^{2k}} \binom{2k+1}{j} \gamma_j, \quad (5.81)$$

where we used the symmetry $\gamma_j = \gamma_{2k+1-j}$ to combine terms.

Asymptotic behavior

The asymptotic behavior of the majority function is governed by its noise sensitivity. Unlike the AND function, the failure probability for majority does not vanish as $n \rightarrow \infty$.

To analyze this, we relate our setting to the noise sensitivity framework. When we measure each qubit with the optimal single-qubit measurement, we correctly identify each bit with probability $p = \frac{1}{2}(1 + \sqrt{1 - s^2})$ and incorrectly with probability $\epsilon := 1 - p$. This is equivalent to applying independent random noise to each bit with noise rate ϵ .

The noise sensitivity of the majority function has been precisely characterized [MO03]. For the majority function on $n = 2k + 1$ bits with noise parameter ϵ , the probability that the

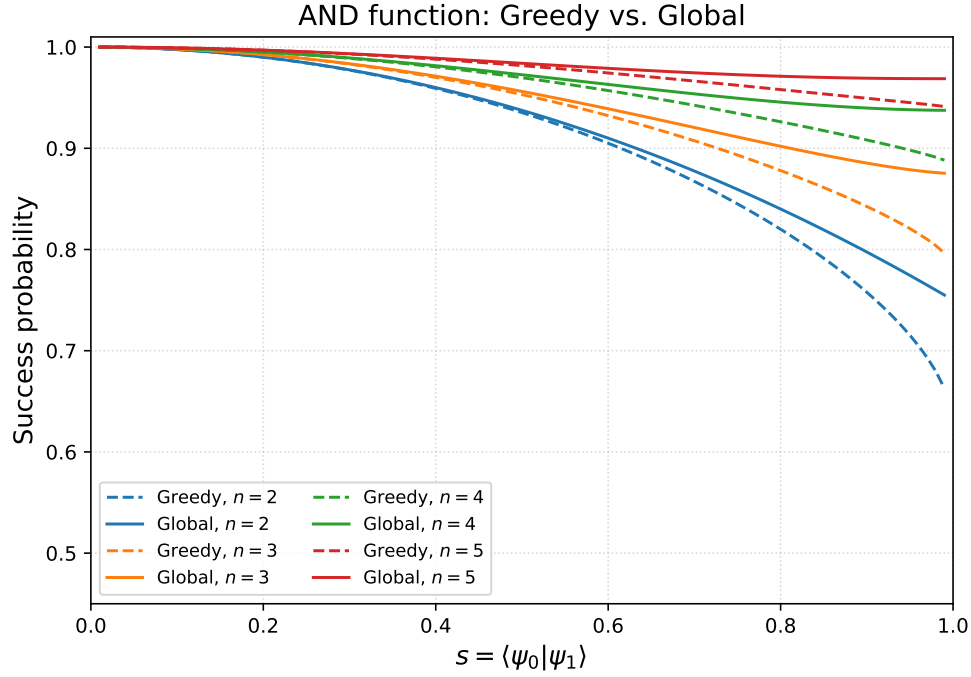


Figure 5.3: Greedy and global success probabilities for the AND function. As n increases, both strategies approach perfect success exponentially quickly, and the gap between them decreases rapidly.

function value changes (i.e., the failure probability) is given by

$$P_{\text{fail}}^{\text{MAJ}} = \frac{1}{2} - \frac{1}{\pi} \arcsin(1 - 2\epsilon) + O\left(\frac{1}{\sqrt{n\epsilon}}\right). \quad (5.82)$$

In the limit as $n \rightarrow \infty$ with small ϵ , this approaches

$$P_{\text{fail}}^{\text{MAJ}} \rightarrow \frac{2}{\pi} \sqrt{\epsilon} = \frac{2}{\pi} \sqrt{1 - p}. \quad (5.83)$$

Therefore, the local success probability approaches

$$P(n, \text{MAJ}, s, \text{greedy}) \rightarrow \frac{1}{2} + \frac{1}{\pi} \arcsin(\sqrt{1 - s^2}) \quad (5.84)$$

as $n \rightarrow \infty$. This limiting value is strictly less than 1 for any $p < 1$ (i.e., for non-orthogonal states).

5.5.3 Comparison and discussion

The contrasting behaviors of the AND and majority functions illustrate two distinct regimes:

Highly unbalanced functions (AND): When $|S_0| \gg |S_1|$ or vice versa, the greedy strategy performs nearly optimally for large n because the overwhelming size of one pre-image dominates the success probability. Both greedy and global strategies converge to perfect success exponentially fast as $n \rightarrow \infty$, and the advantage from joint measurements becomes negligible in this regime.

Balanced non-affine functions (majority): For balanced functions that are not affine, the situation is fundamentally different. The majority function's noise sensitivity ensures that the greedy strategy converges to a limiting success probability strictly bounded away from 1. Specifically, $P(n, \text{MAJ}, s, \text{greedy}) \rightarrow 1 - \frac{2}{\pi} \arcsin(\sqrt{1-p})$ as $n \rightarrow \infty$, which remains bounded below 1 for any non-orthogonal states ($p < 1$), while the global optimal strategy achieves better performance. This gap between greedy and global strategies persists even asymptotically, demonstrating a persistent advantage of joint measurements that does not vanish as the problem size increases.

These examples confirm that our characterization theorem captures a fundamental dichotomy: affine functions are the unique class where local memoryless measurements are always optimal, while all other functions require more complex strategies. Crucially, the practical significance of this advantage is intimately tied to the function's noise sensitivity. Low noise sensitivity functions (like AND for large n) exhibit vanishing advantage from joint mea-

surements, while high noise sensitivity functions (like majority) maintain advantage from joint-measurements even for arbitrarily large input dimensions. This connection between noise sensitivity and joint-measurement-advantage suggests deeper structural relationships between classical Boolean function complexity and quantum information-theoretic resources.

Chapter 6

Conclusion

In this thesis, we have investigated several fundamental problems in quantum information theory, primarily focusing on computing and/or approximating the value of large optimization problems. By exploiting the inherent structure of the problem, we have developed algorithms and theoretical frameworks that address the computational bottlenecks associated with the high-dimensional spaces.

The first part of this work addressed the separability problem. By introducing a hybrid quantum-classical heuristic, we demonstrated that the effective dimension of the optimization problem can be significantly reduced while maintaining feasibility guarantees. Our results for the one-dimensional Ising Hamiltonian on up to 28 qubits illustrate that this approach can effectively approximate the separable ground energy, providing a scalable tool for characterizing entanglement in many-body systems.

In the second part, we extended the concept of dimension reduction to quantum state discrimination. We showed that the search for optimal measurements for state discrimination problems can be reduced from a problem scaling with the dimension d of the space to one

scaling with the number of candidate states N . This NL reduction framework is general, accommodating various figures of merit including minimum-error, unambiguous discrimination, and the state exclusion problem. We first characterized the optimal identifications for quantum changepoint problems under several reward structures, including multiple-changepoint settings that were previously computationally inaccessible. Second, we considered a quantum error classification problem and showed how our reduction makes it tractable for systems of hundreds of qubits.

Finally, we explored structural measurement strategies for the state exclusion problem. We introduced the pretty bad measurement as a natural analogue to the pretty good measurement, proving its efficacy relative to blind guessing. Furthermore, our characterization of Boolean property learning from quantum sequences established the global optimality of memoryless “greedy” strategies for affine functions. This provides a clear boundary for when local measurements suffice and when joint measurements are fundamentally required to extract coarse-grained information from quantum data.

In summary, the results presented here suggest that many “hard” problems in quantum information possess a hidden structure that can be exploited for efficient computation. As quantum hardware continues to evolve, the hybrid workflows and dimension-reduction techniques developed in this thesis could play a role in bridging the gap between theoretical protocols and practical implementation on near-term devices.

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